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# Data Communication CS601 Lecture No 1

Lecture Prepared By

Muhammad Abid Ali

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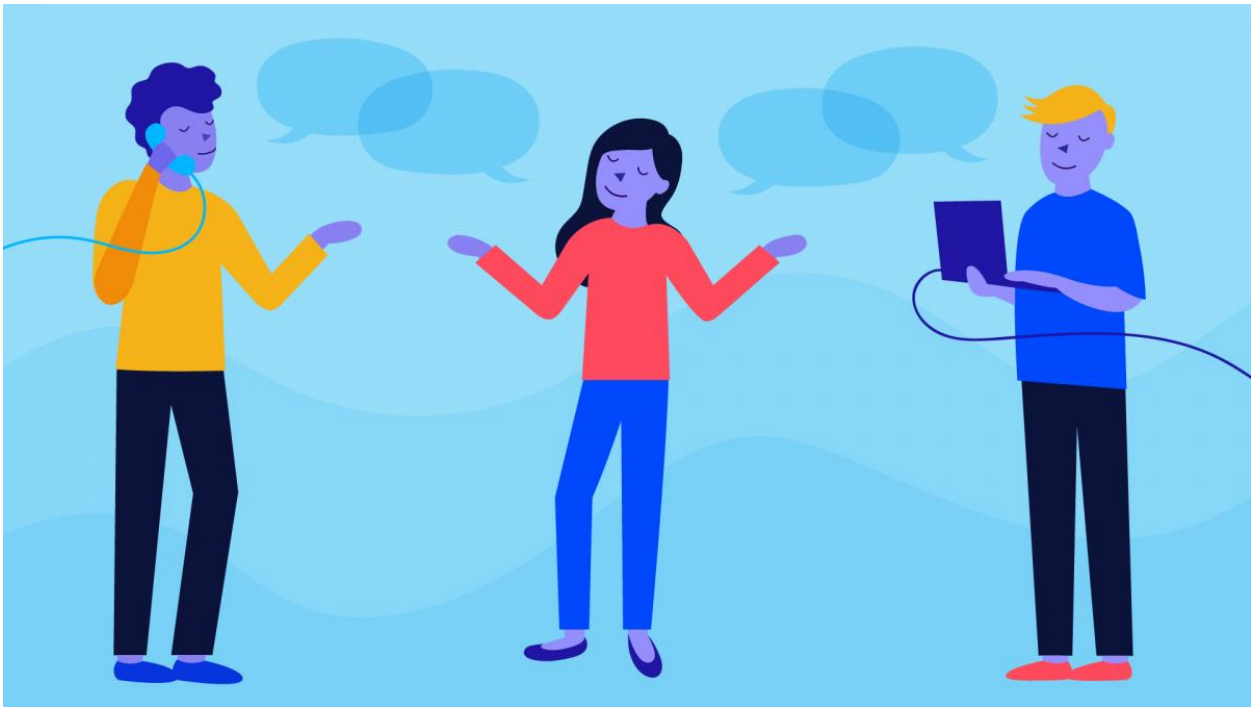
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## Communication

**A process by which information is exchanged. Communication is simply the act of transferring information from one place, person or group to another.**



## Telecommunications

Telecommunications are **the means of electronic transmission of information over distances**. The information may be in the form of voice telephone calls, data, text, images, or video. Today, telecommunications are used to organize more or less remote computer systems into telecommunications networks.

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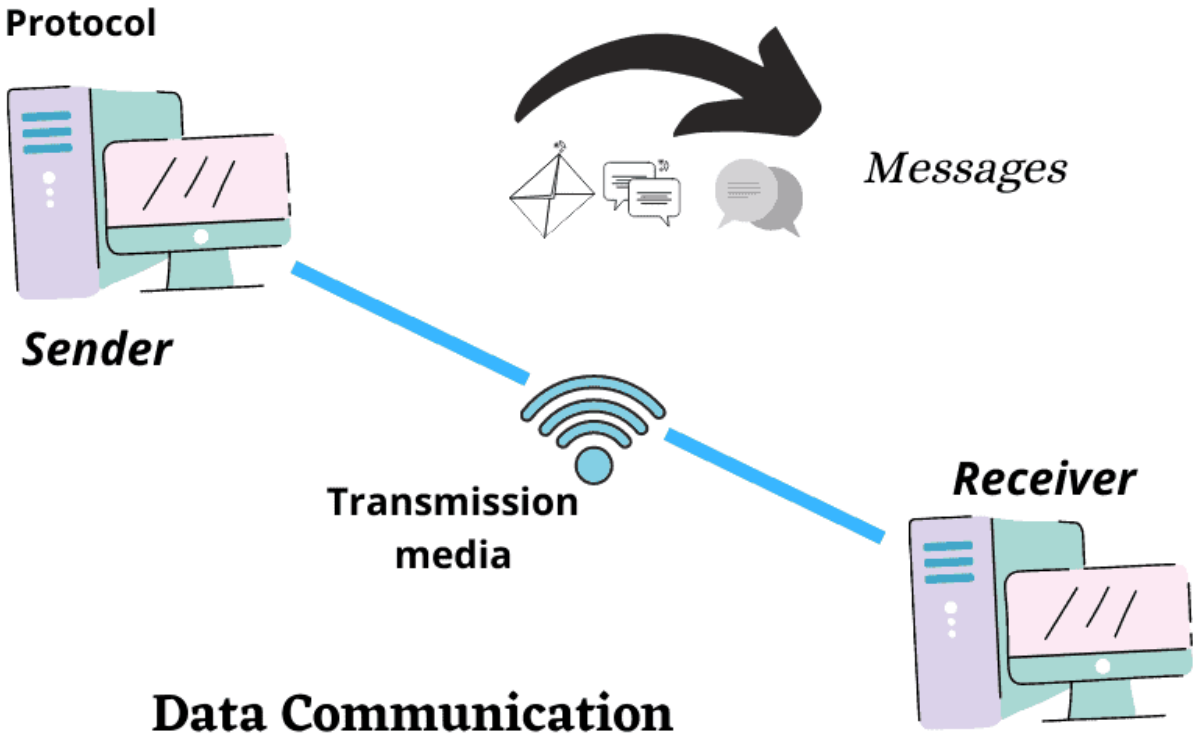
## Data communications

Exchange of data between two devices via some form of transmission media. Data communications (DC) is **the process of using computing and communication technologies to transfer data from one place to another, or between participating parties**.

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## Characteristics of a Data Communication System

- **Delivery**
- **Accuracy**
- **Timeliness**
- **Jitter**

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- **Delivery** means sending the data to the correct destination.
- **Accuracy** means reception of data with no errors.
- **Timeliness** means delivering/transmitting the data within a time interval.
- **Jitter** refers to the variation in the packet arrival of time.

I. Delivery- The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

2. Accuracy- The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

3. Timeliness- The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

4. Jitter- Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

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## Forms of Information

- **Text**
- **Numbers**
- **Images**
- **Audio**
- **Video**

♦ Text: ASCII: 7-bit pattern (128 different symbols) Extended ASCII: 8-bit pattern Unicode: 32 bits pattern, which is definitely enough to represent any symbol in the world.

♦ Numbers: represented by bit pattern (binary number)

♦ Images: represented by matrix of pixels (picture element), small dot. The size of pixel represents the resolution.

♦ Audio: represent sound by continuous (analog) signal

♦ Video: can be analog or digital signal

## 1. Text-

In data communications, text is represented as a bit pattern, a sequence of bits (Os or Is). Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding.

## 2. Numbers-

Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations.

## 3. Images-

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Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the resolution. For example, an image can be divided into 1000 pixels or 10,000 pixels.

## 4. Audio-

Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete. Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.

## 5. Video-

Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion. Again, we can change video to a digital or an analog signal.

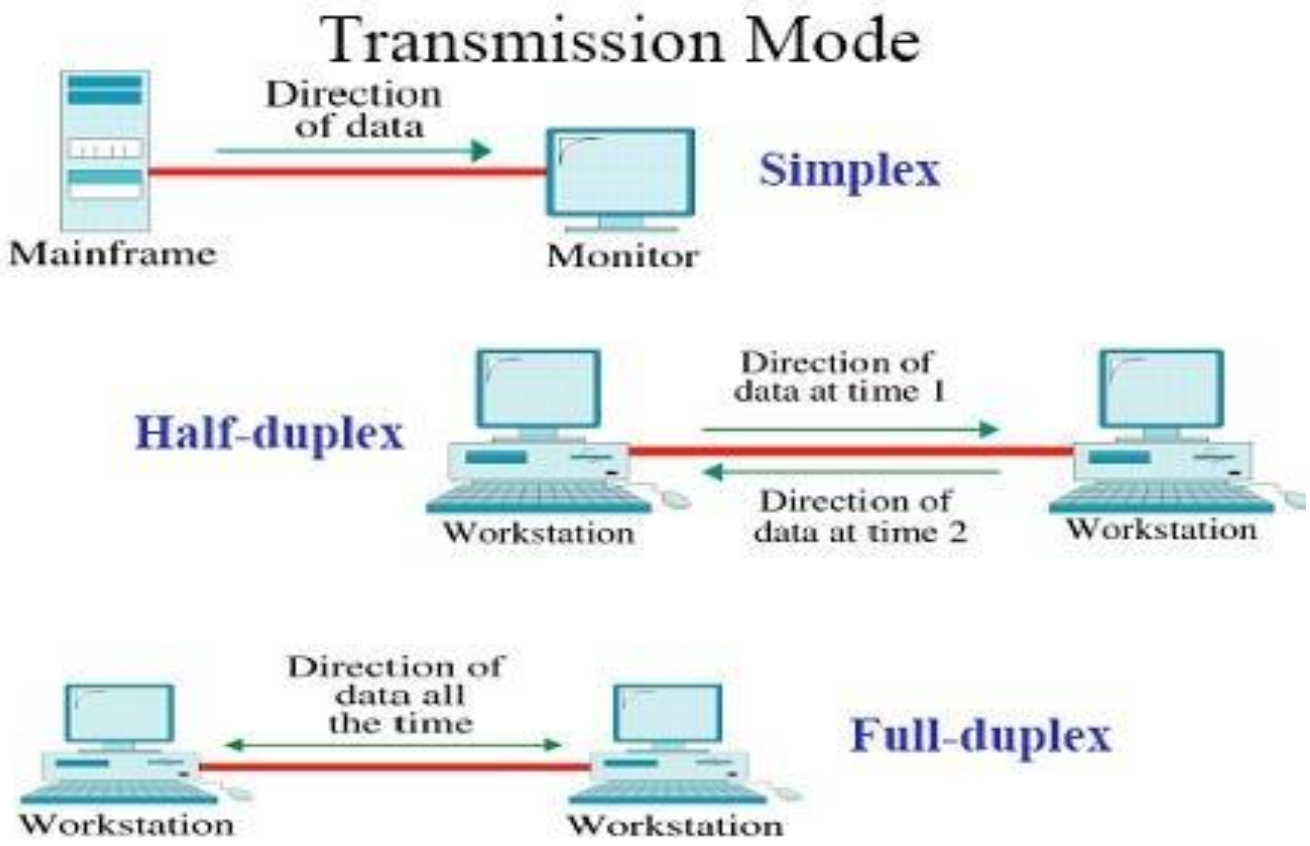
## Data Flow between two devices:

- **Simplex**
- **Half-Duplex**
- **Full-Duplex**

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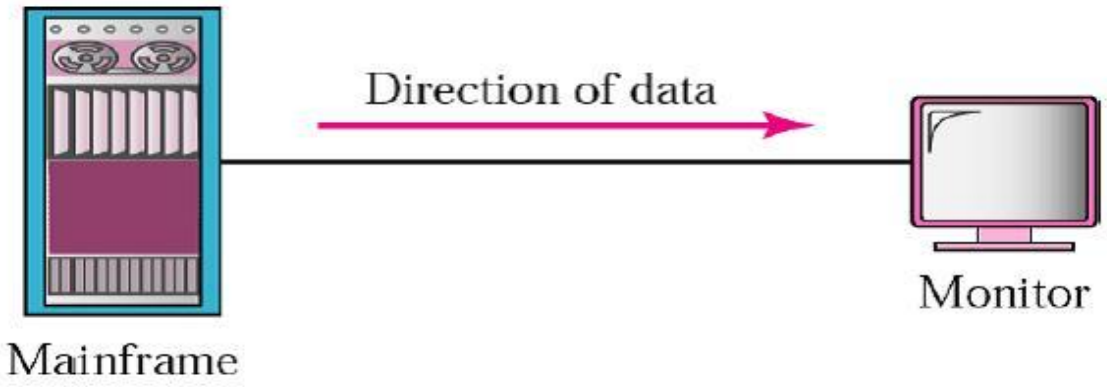

## 1. Simplex:

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; Keyboards and traditional monitors are examples of simplex devices. the other can only receive which can be represented in the following figure.

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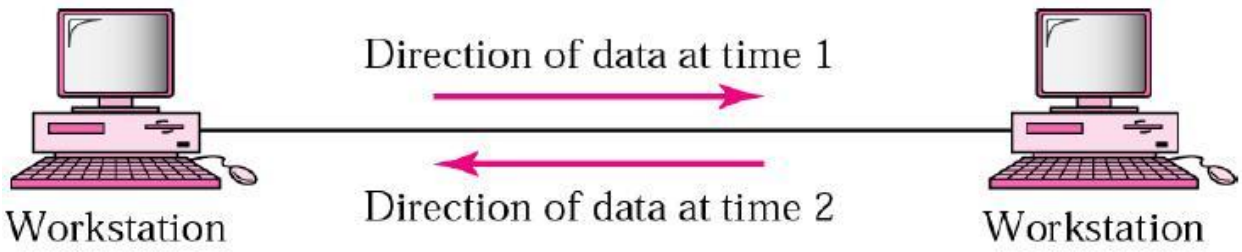
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## 2. Half-Duplex:

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa which will represent in the following figure.




he half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait.

## 3. Full-Duplex:

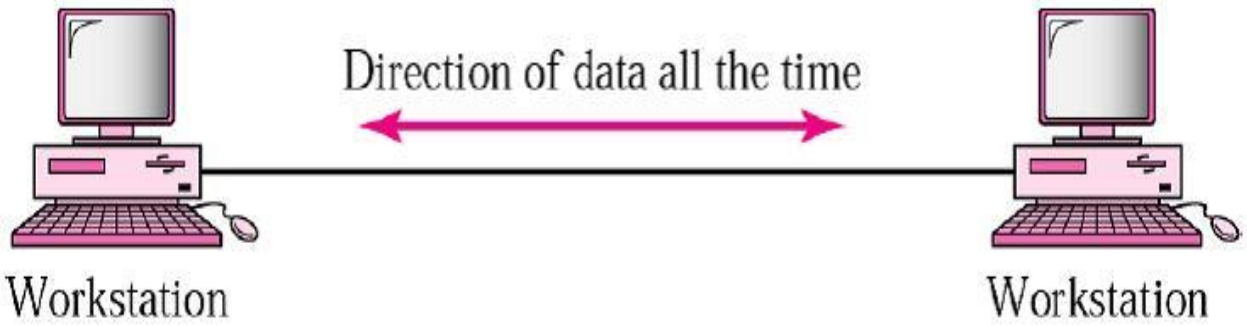
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In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously as shown in the following figure.




## Network:

Interconnection of a set of devices capable of communication

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A data network is **a system designed to transfer data from one network access point to one other or more network access points via data switching, transmission lines, and system controls**. Data networks consist of communication systems such as circuit switches, leased lines, and packet switching networks.

## Throughput

Throughput is **the actual amount of data that is successfully sent/received over the communication link**. Throughput is presented as kbps, Mbps or Gbps, and can differ from bandwidth due to a range of technical issues, including latency, packet loss, jitter and more.

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## Delay

Network delay is a design and performance characteristic of a telecommunications network. It **specifies the latency for a bit of data to travel across the network from one communication endpoint to another**.

## Reliability

In computer networking, a reliable protocol is **a communication protocol that notifies the sender whether or not the delivery of data to intended recipients was successful**.

## Physical Network Attributes

- **Link**
- **Type of Connection**
  - ✓ **Point-to-Point**
  - ✓ **Multipoint**




a. Point-to-point

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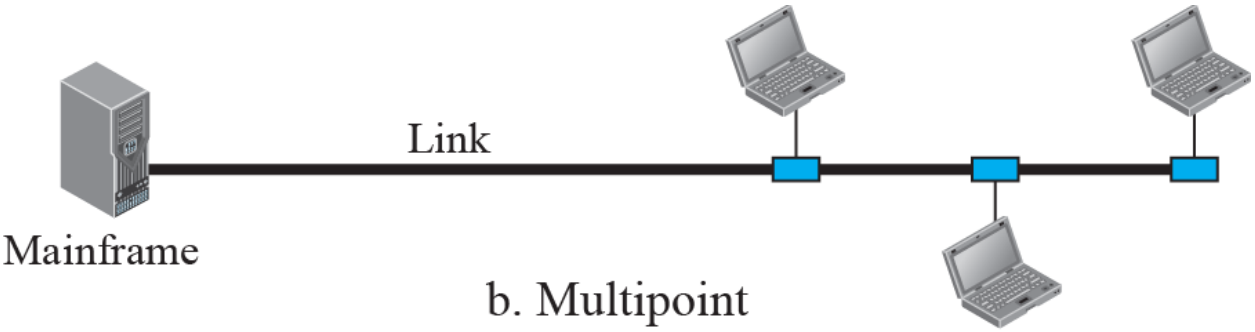





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b. Multipoint

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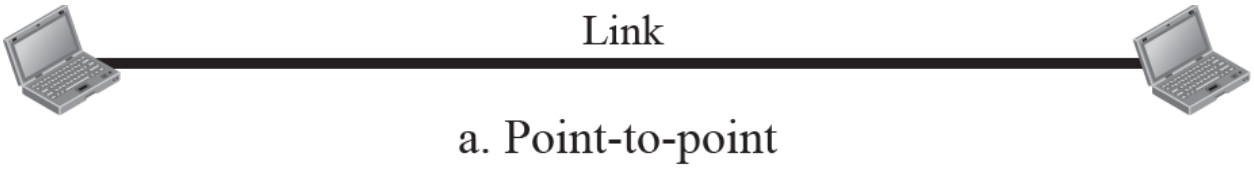
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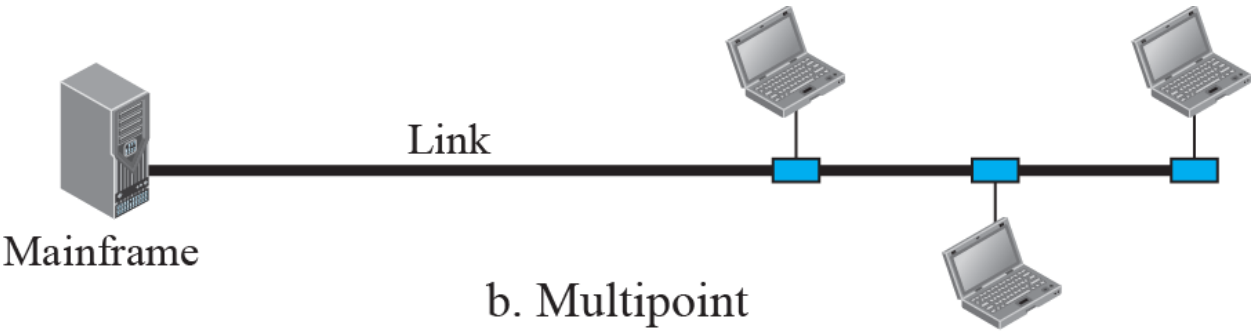
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## Physical Network Attributes

- **Link**
- **Type of Connection**
  - ✓ **Point-to-Point**
  - ✓ **Multipoint**




a. Point-to-point




b. Multipoint

## Physical Topologies

- **Physical Layout of Network**
- **Links + Nodes = Topology**
- **Physical Topologies:**
  - ✓ **Mesh**

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- ✓ **Star**
- ✓ **Bus**
- ✓ **Ring**

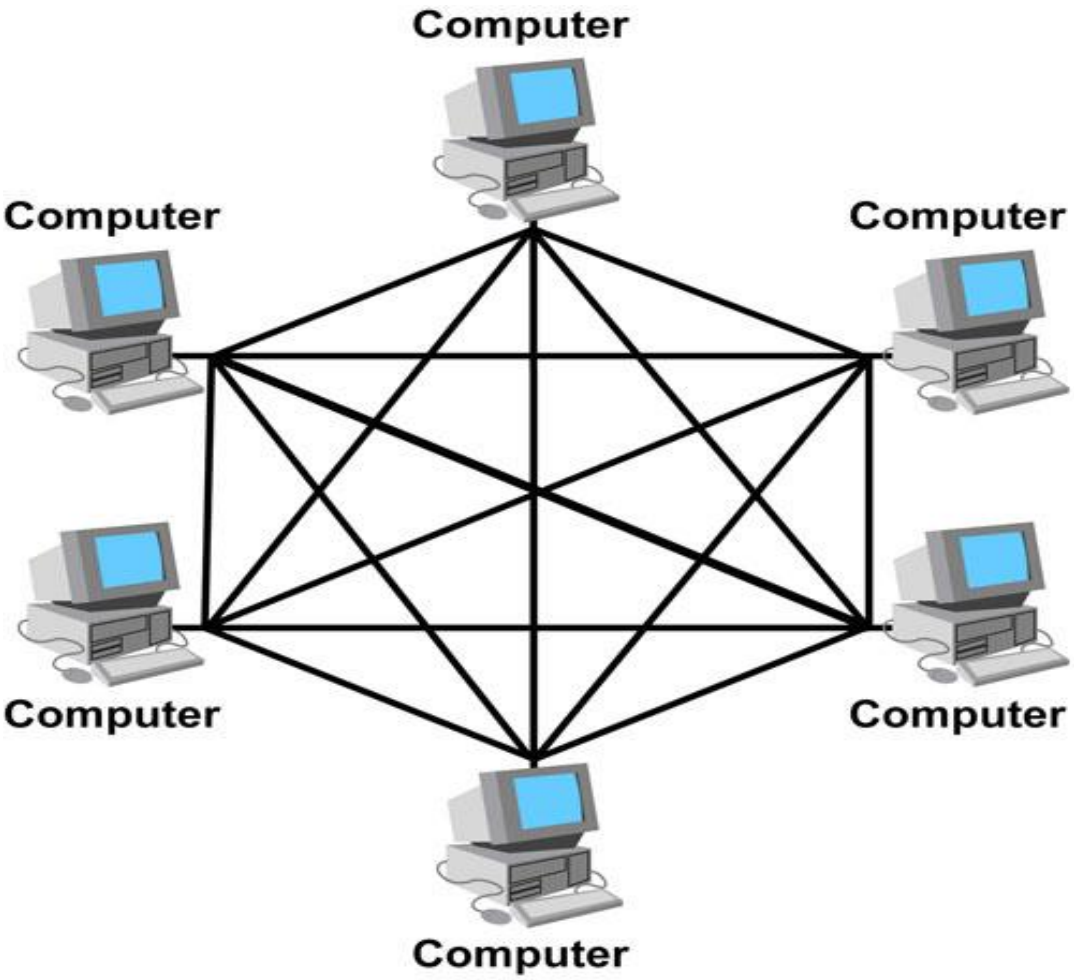
## Mesh topology

In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. A fully connected mesh network therefore has [n (n-1) / 2] physical channels to link it devices.

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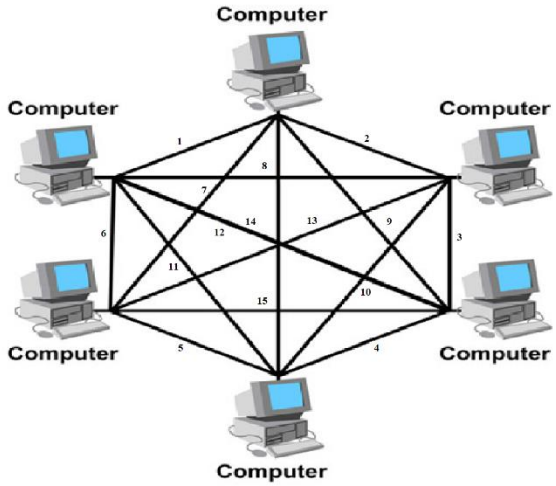
**A fully connected mesh network therefore has [n (n-1) / 2] physical channels to link it devices.**

**n (n-1) / 2**

**6 (6-1) / 2**

**6 (5) / 2**

**30 / 2 = 15 Links**




## A mesh offers several advantages over other network topologies:

1. The use of dedicated links guarantees that each connection can carry its data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.

2. A mesh topology is robust; if one link becomes unusable, it does not affect the entire system.

3. Security where every message sent travels along a dedicated line, only the intended recipient sees it.

4. Point-to-point links make fault identification and fault isolation easy.

## The main disadvantages of a mesh topology are:

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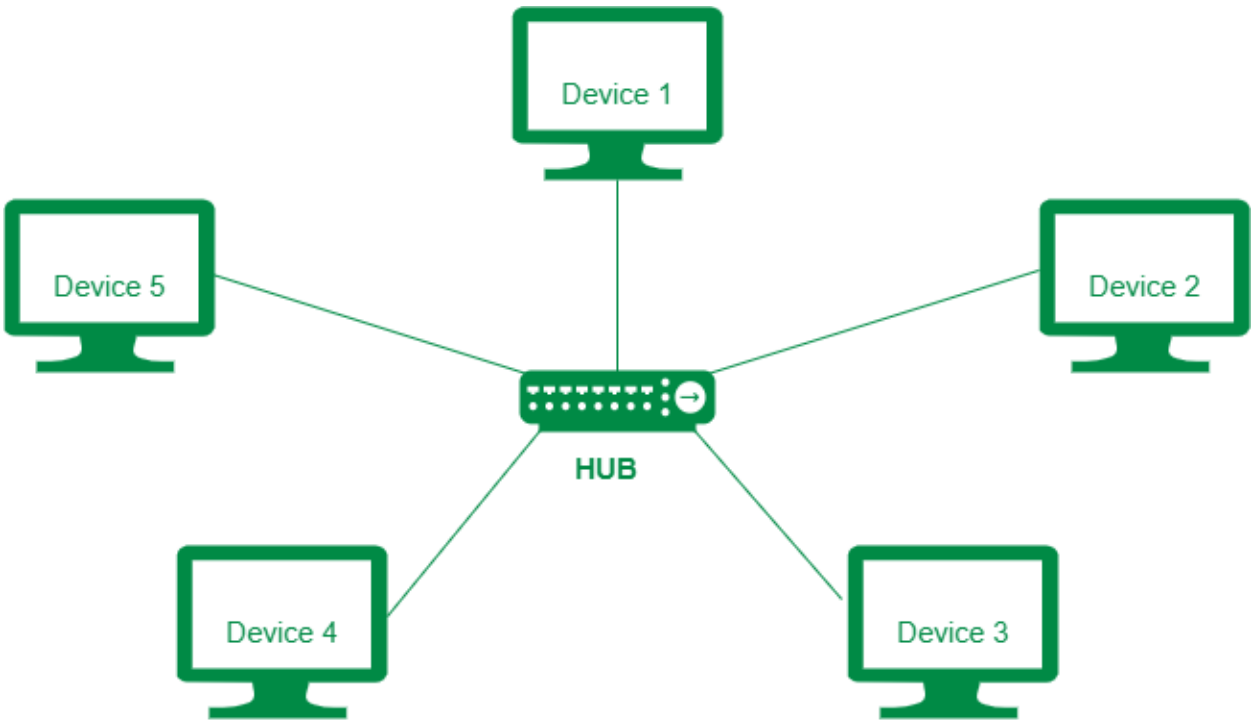
1. Difficult installation and reconfiguration because every device must be connected to every other device.

2. The bulk of the wiring can be greater than the available space can accommodate.

3. The hardware required to connect each link can be expensive.

## Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not linked to each other. Unlike a mesh topology, a star topology does not allow direct traffic between devices.




Star Topology

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## The advantages of star topology are:

1. Less expensive than a mesh topology.

2. Easy to install.

3. Robustness. If one link fails, only that link is affected.

4. Easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

## The disadvantages of the star topology are:

1. Any fault occur to the central controller will drop down the entire network.

2. Although the star needs less cabling than mesh, it still needs more cables than ring and tree topology.

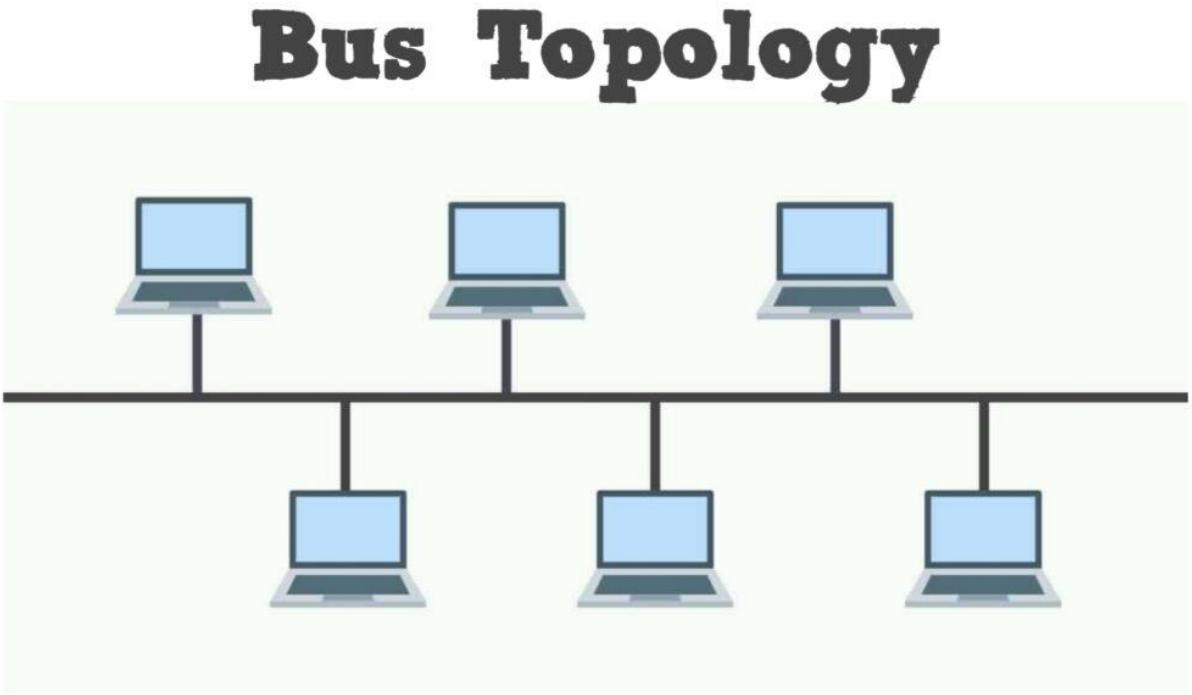
## Bus Topology

The preceding examples all describe point-to-point configurations A bus topology on the other hand, is multipoint. One long cable act as a backbone to link all the devices in the network

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## The advantages of a bus topology are:

1. Ease of installation.

2. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.

3. Use less cabling than mesh, star, and tree topologies. Terminator Terminator Tap Tap 11

## While the disadvantages are:

1. Difficult reconfiguration and fault isolation.

2. A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

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3. Adding new devices may therefore require modification or replacement of the backbone.

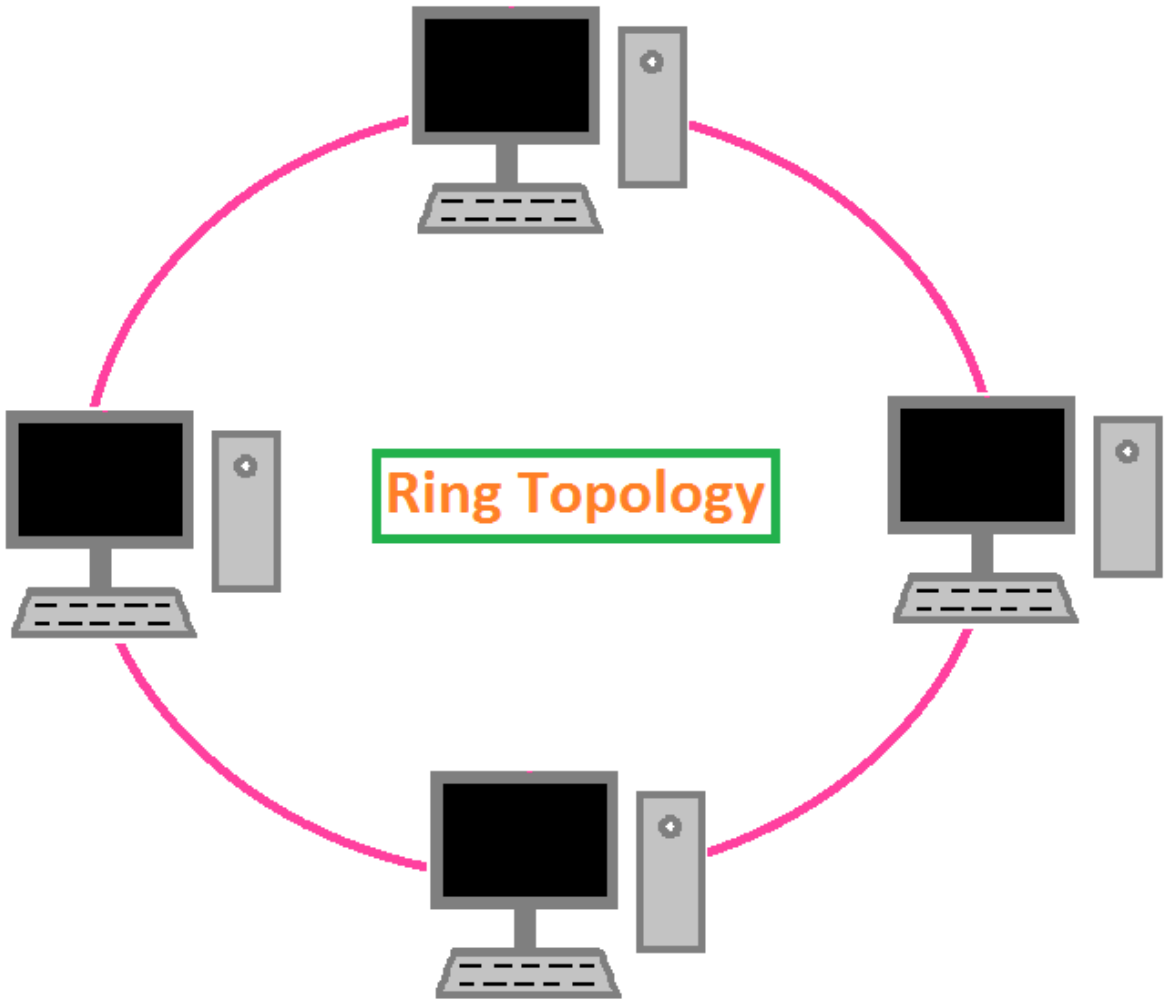
## Ring Topology

In a ring topology, each device has a dedicated point-to-point line configuration only with the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.

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## The advantages of ring topology are:

1. Relatively easy to install and reconfigure.

2. Addition or deletion of a device requires moving only two connections.

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3. Fault isolation is simplified.

The disadvantage of the ring topology is that a break in the ring (such as a disabled station) can disable the entire network, and it is one way data flow.

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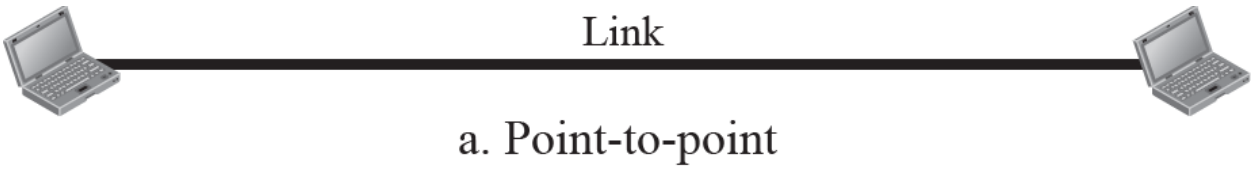
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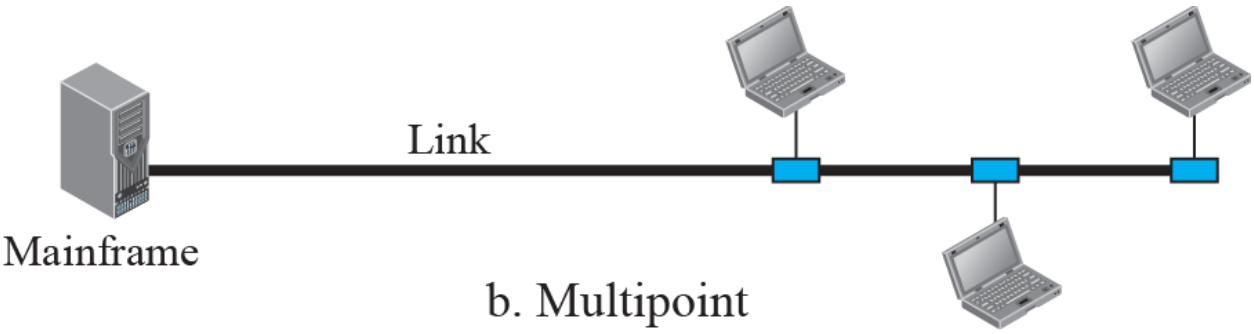
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## Physical Network Attributes

- **Link**
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  - ✓ **Point-to-Point**
  - ✓ **Multipoint**




a. Point-to-point




b. Multipoint

## Physical Topologies

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- **Physical Topologies:**
  - ✓ **Mesh**
  - ✓ **Star**
  - ✓ **Bus**
  - ✓ **Ring**

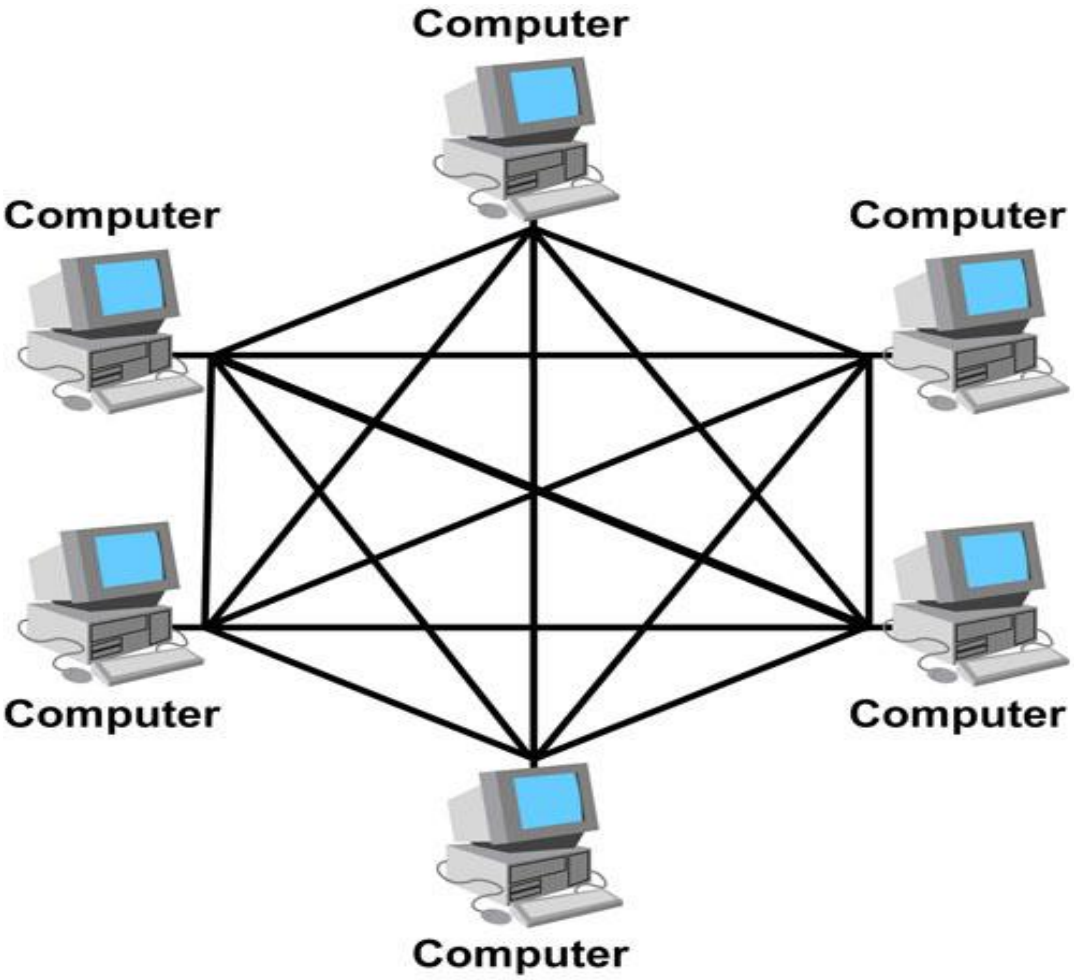
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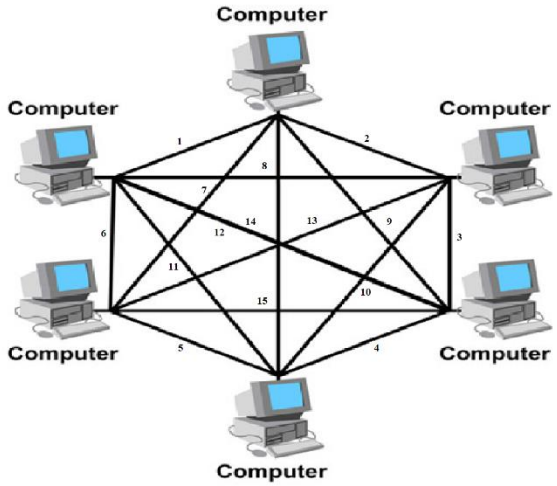
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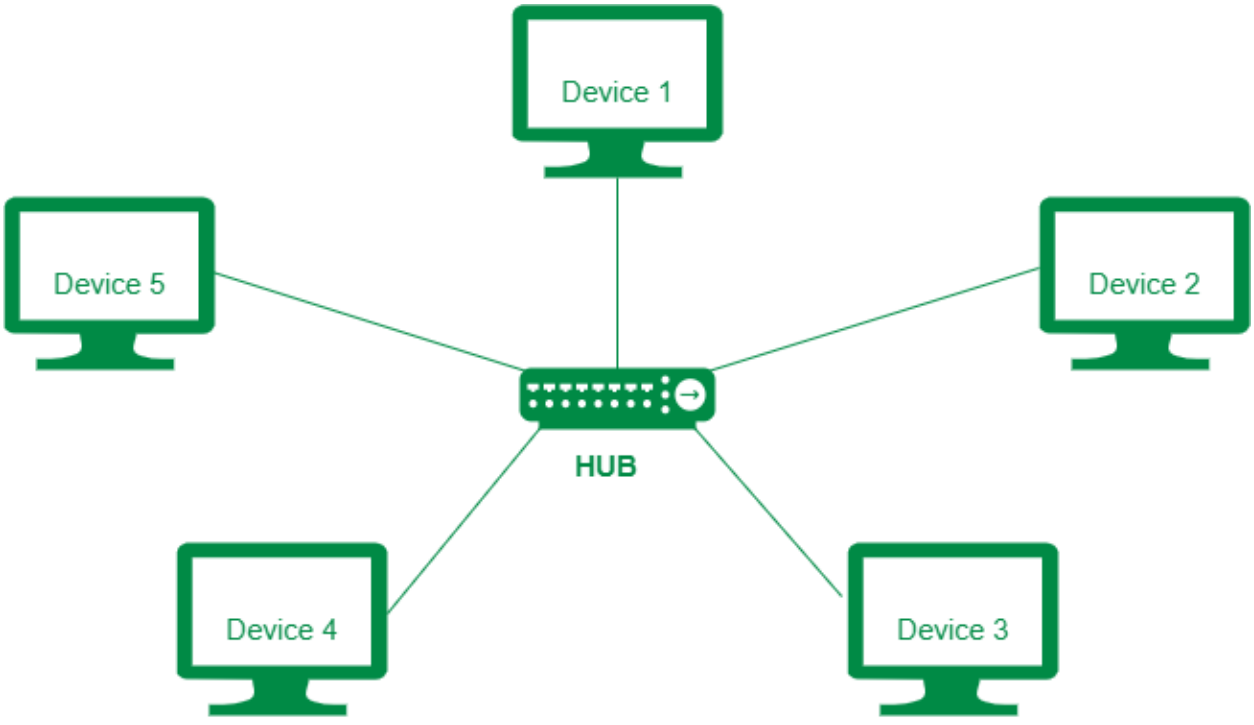
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Star Topology

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3. Robustness. If one link fails, only that link is affected.

4. Easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

## The disadvantages of the star topology are:

1. Any fault occur to the central controller will drop down the entire network.

2. Although the star needs less cabling than mesh, it still needs more cables than ring and tree topology.

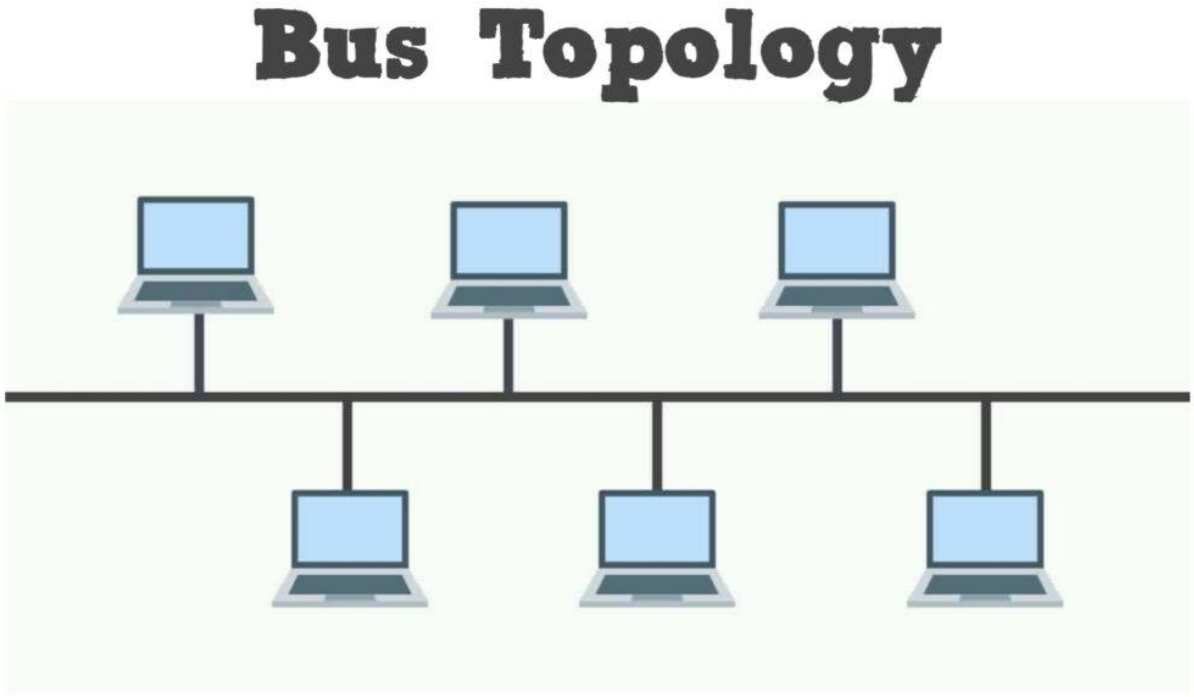
## Bus Topology

The preceding examples all describe point-to-point configurations A bus topology on the other hand, is multipoint. One long cable act as a backbone to link all the devices in the network

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## The advantages of a bus topology are:

1. Ease of installation.

2. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.

3. Use less cabling than mesh, star, and tree topologies. Terminator Terminator Tap Tap 11

## While the disadvantages are:

1. Difficult reconfiguration and fault isolation.

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2. A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

3. Adding new devices may therefore require modification or replacement of the backbone.

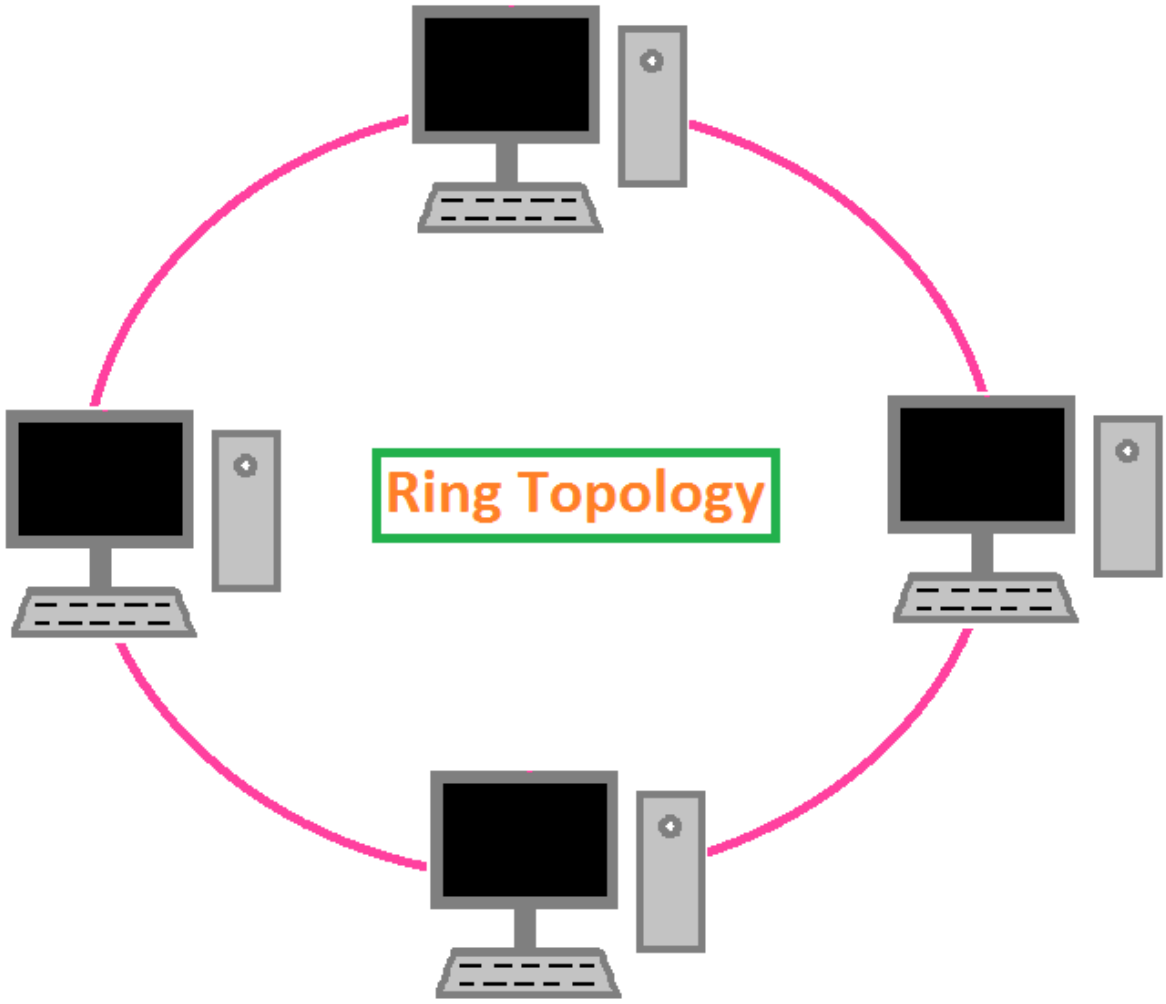
## Ring Topology

In a ring topology, each device has a dedicated point-to-point line configuration only with the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.

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## The advantages of ring topology are:

1. Relatively easy to install and reconfigure.

2. Addition or deletion of a device requires moving only two connections.

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3. Fault isolation is simplified.

The disadvantage of the ring topology is that a break in the ring (such as a disabled station) can disable the entire network, and it is one way data flow.

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## Networks Types

- **Network classification:**
  - ✓ **Size**
  - ✓ **Geographical Coverage**
  - ✓ **Ownership**

- Local Area Networks

(LANs)

- Wide Area Networks (WANs)

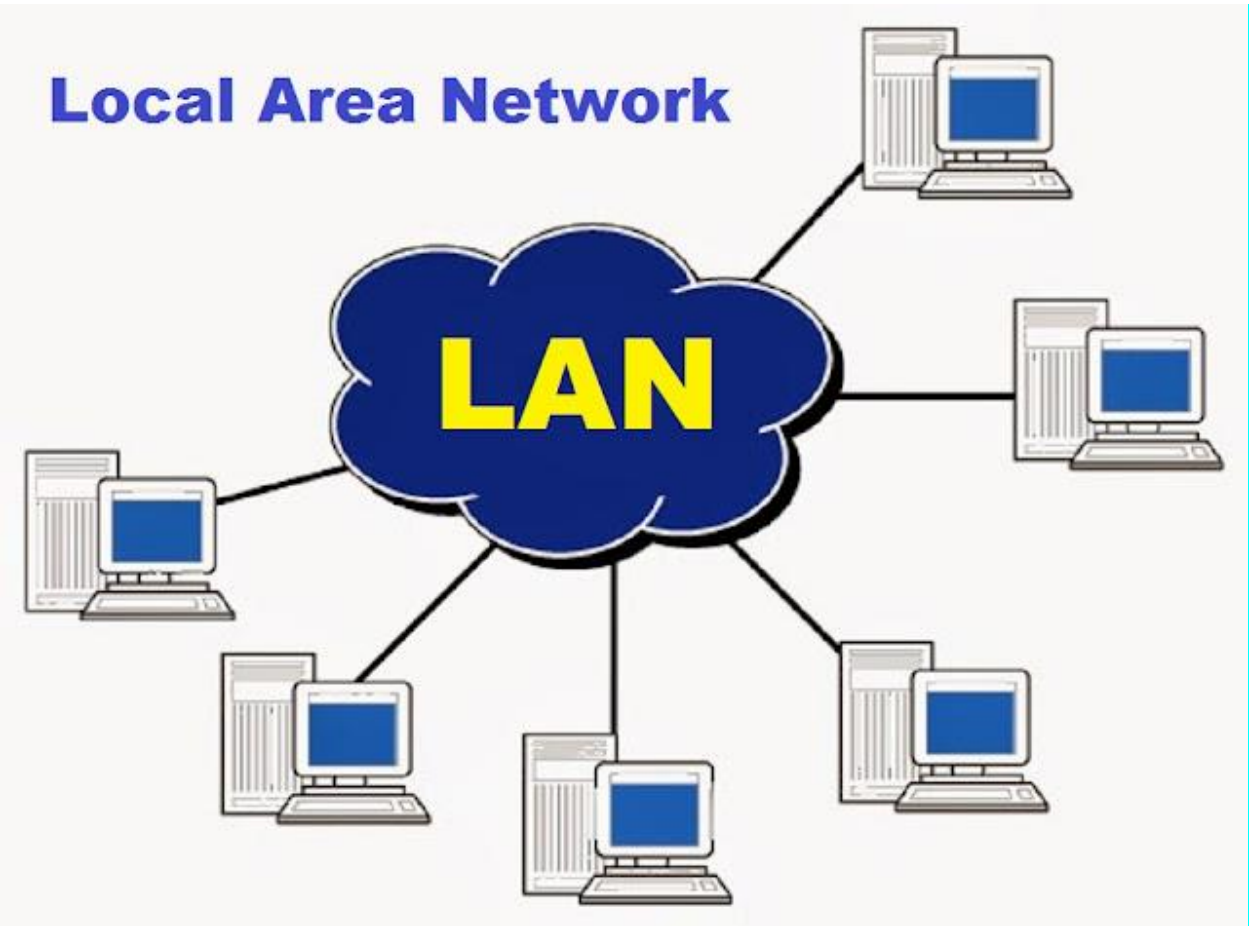
## Local Area Networks

- **Usually Privately owned**
- **Connects some hosts in a single office, building, or campus**
- **Can be as simple as two PCs and a printer in someone's home office**
- **Can extend throughout a company**
- **Host Address**

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## Wide Area Network

- **Wider geographical span than a LAN**
- **Spans a town, a state, a country, or even the world**
- **Interconnects connecting devices such as switches, routers, or modems**
- **Normally created and run by communication companies**

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- **Point-to-Point WAN**
- **Switched WAN**
- **Internetwork**

## Point-to-Point WANs

Point-to-point connections are **typically used to connect two systems together over a wide area network (WAN)**. You can use a point-to-point connection to get data from your local system to a remote system or to get data from a local network to a remote network.




## Switched WANs

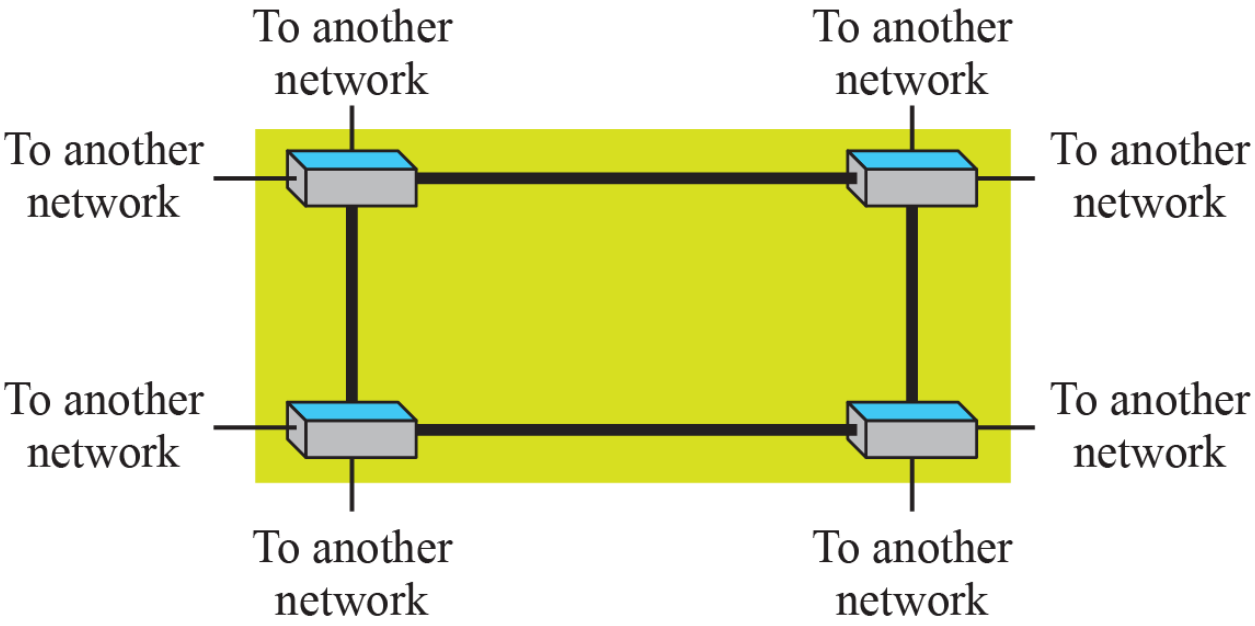
A switched WAN network is **used to connect multiple end nodes through a common WAN network**, example is Asynchronous Transfer Mode (ATM) network while Point-to-point WAN is one which consists of two end nodes connected by a leased line, example is a dial-up line that connects a home computer to the Internet.

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## Internetwork

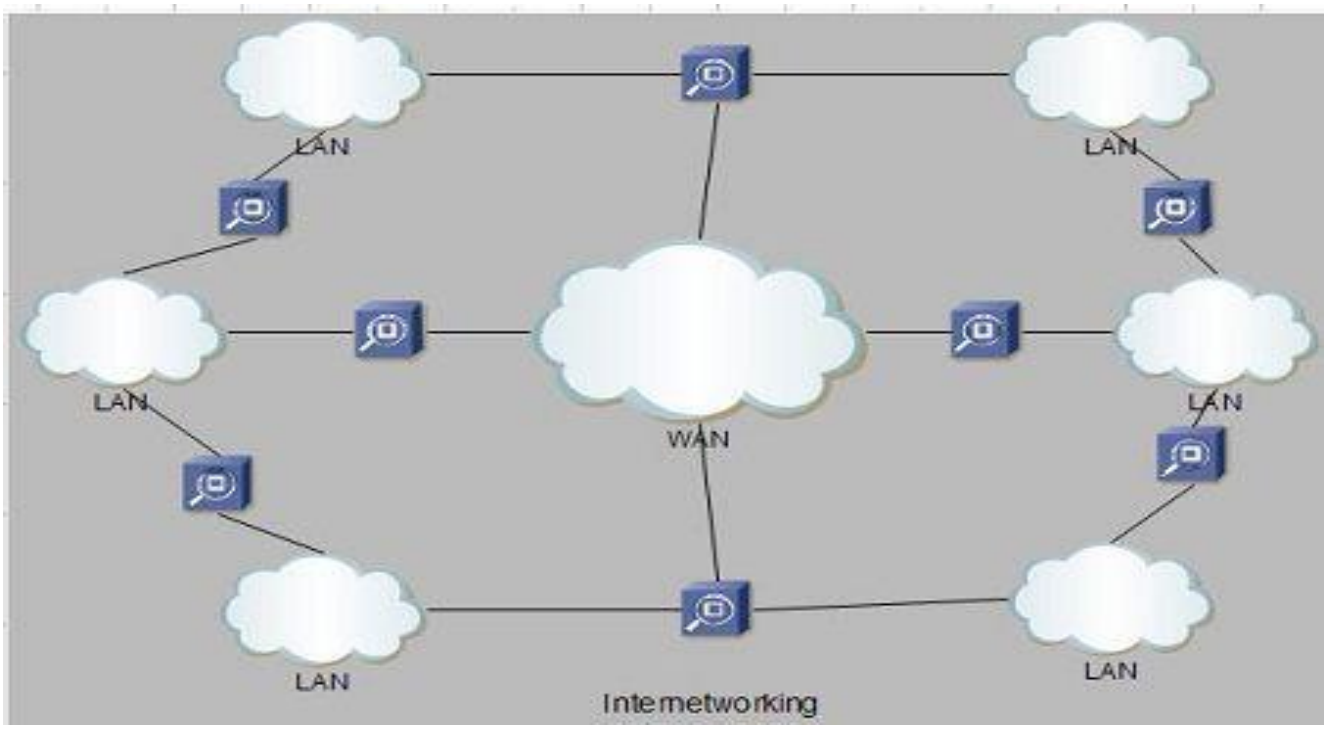
Internetworking is **the process or technique of connecting different networks by using intermediary devices such as routers or gateway devices**. Internetworking ensures data communication among networks owned and operated by different entities using a common data communication and the Internet Routing Protocol.

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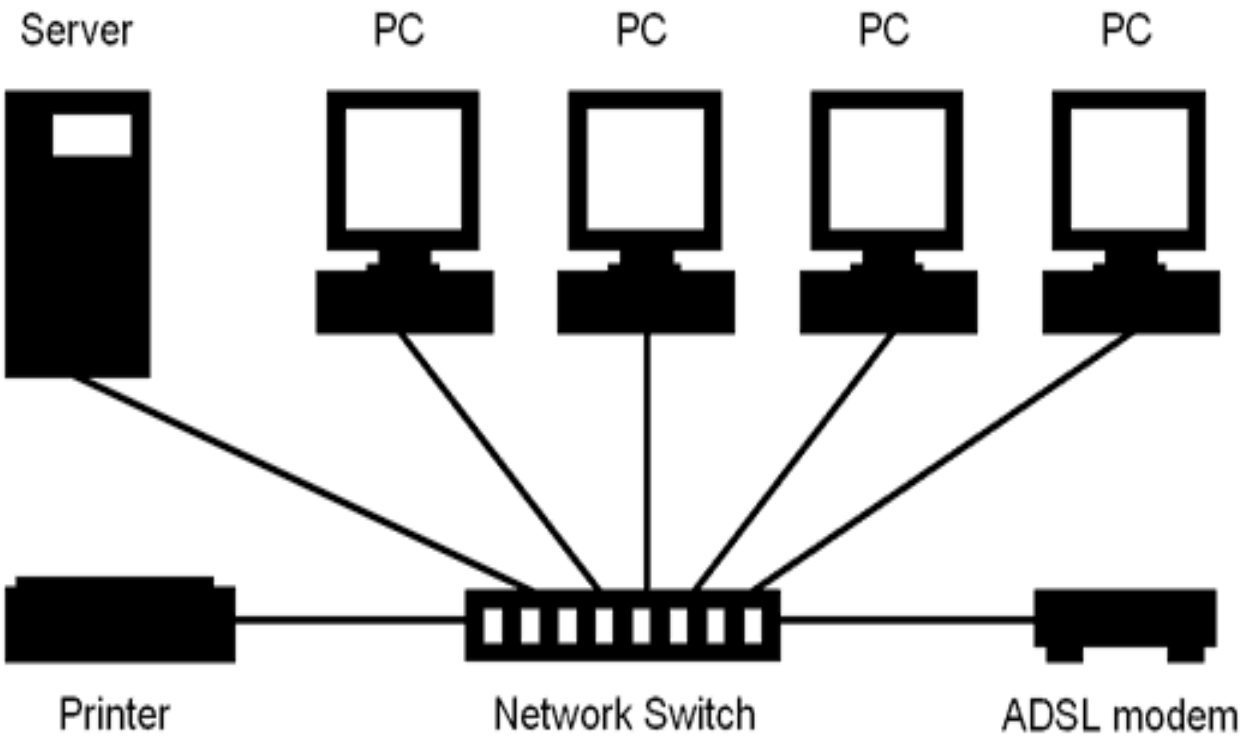

## Switching

Switching in a computer network is **achieved by using switches**. A switch is a small hardware device which is used to join multiple computers together with one local area network (LAN). Network switches operate at layer 2 (Data link layer) in the OSI model.

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- **Switching**
  - ✓ **Circuit-Switched Network**
  - ✓ **Packet- Switched Network**

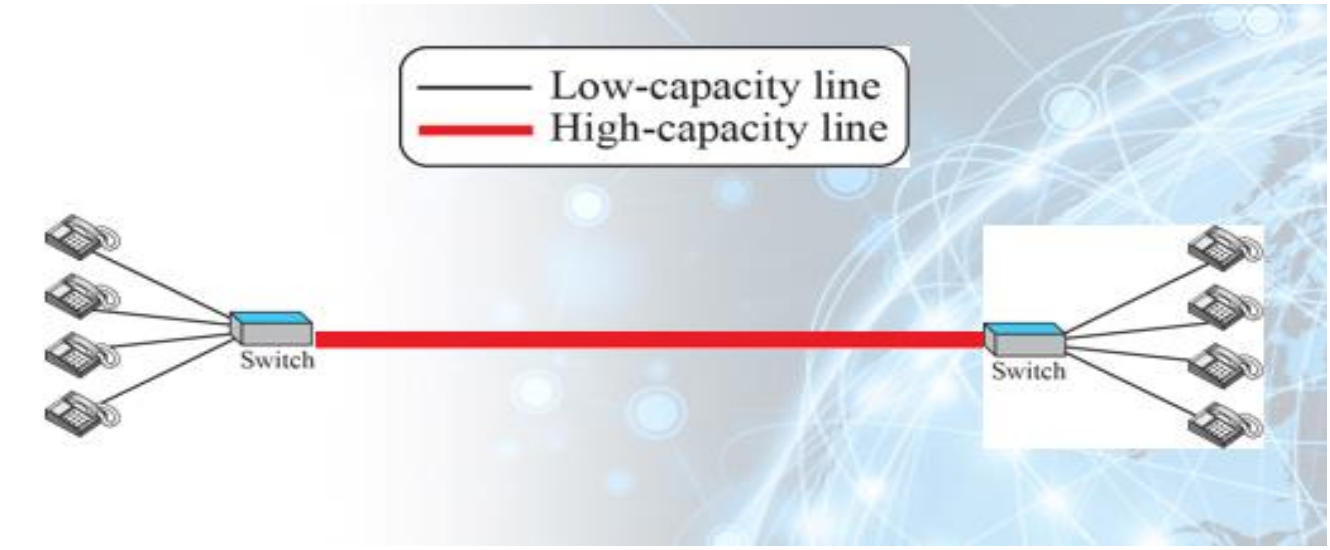
## Circuit Switched Network

Circuit switching is **a type of network configuration in which a physical path is obtained and dedicated to a single connection between two endpoints in the network for the duration of a dedicated connection**.

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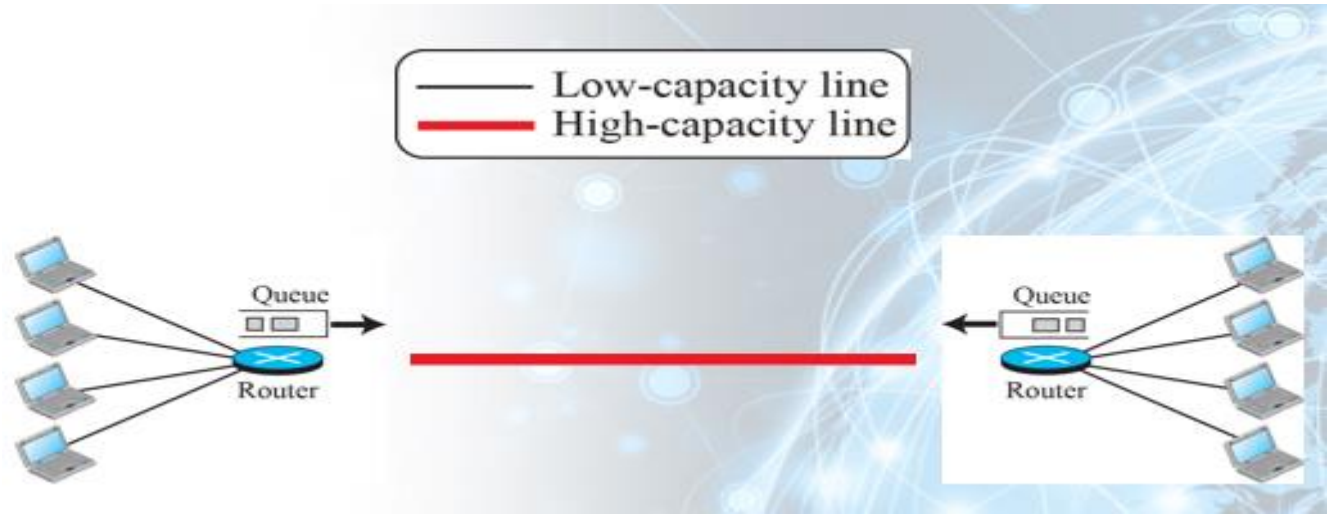
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## Packet Switched Network

A packet switched network (PSN) is **a type of computer communications network that groups and sends data in the form of small packets**. It enables the sending of data or network packets between a source and destination node over a network channel that is shared between multiple users and/or applications.



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**The Internet**

- **An internet (note the lowercase i) is two or more networks that can communicate with each other**

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- **The Internet (uppercase I), and is composed of thousands of interconnected networks.**

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## Internet History

- **Telegraph and Telephone networks, before 1960:**
  - ✓ **Constant-rate communication only**
  - ✓ **ARPANET- Packet Switched**
- **Birth of the Internet &TCP/IP**
- **MILNET**
- **CSNET**
- **NSFNET**
- **Internet Today**

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## What is ARPANET and why is it important?

The Advanced Research Projects Agency (ARPA), an arm of the U.S. Defense Department, funded the development of the Advanced Research Projects Agency Network (ARPANET) in the late 1960s. Its initial purpose was **to link computers at Pentagon-funded research institutions over telephone lines**.

## Birth of the Internet &TCP/IP

**January 1, 1983** is considered the official birthday of the Internet. Prior to this, the various computer networks did not have a standard way to communicate with each other. A new communications protocol was established called Transfer Control Protocol/Internetwork Protocol (TCP/IP).

## MILNET

MILNET was **the name given to a network that split off from ARPANET in 1983 to create an internetwork designated for use by the U.S. Department of Defense**. MILNET later became part of the DoD Defense Network (DDN). MILNET is also known as military net.

## Who developed Csnet?

Short for Computer Science Network, CSNET is a computer network that was developed by the **U.S. National Science Foundation** in the early 1980s.

## NSFNET

**The National Science Foundation Network** (NSFNET) was a program of coordinated, evolving projects sponsored by the National Science Foundation (NSF) from 1985 to 1995 to promote advanced research and education networking in the United States.

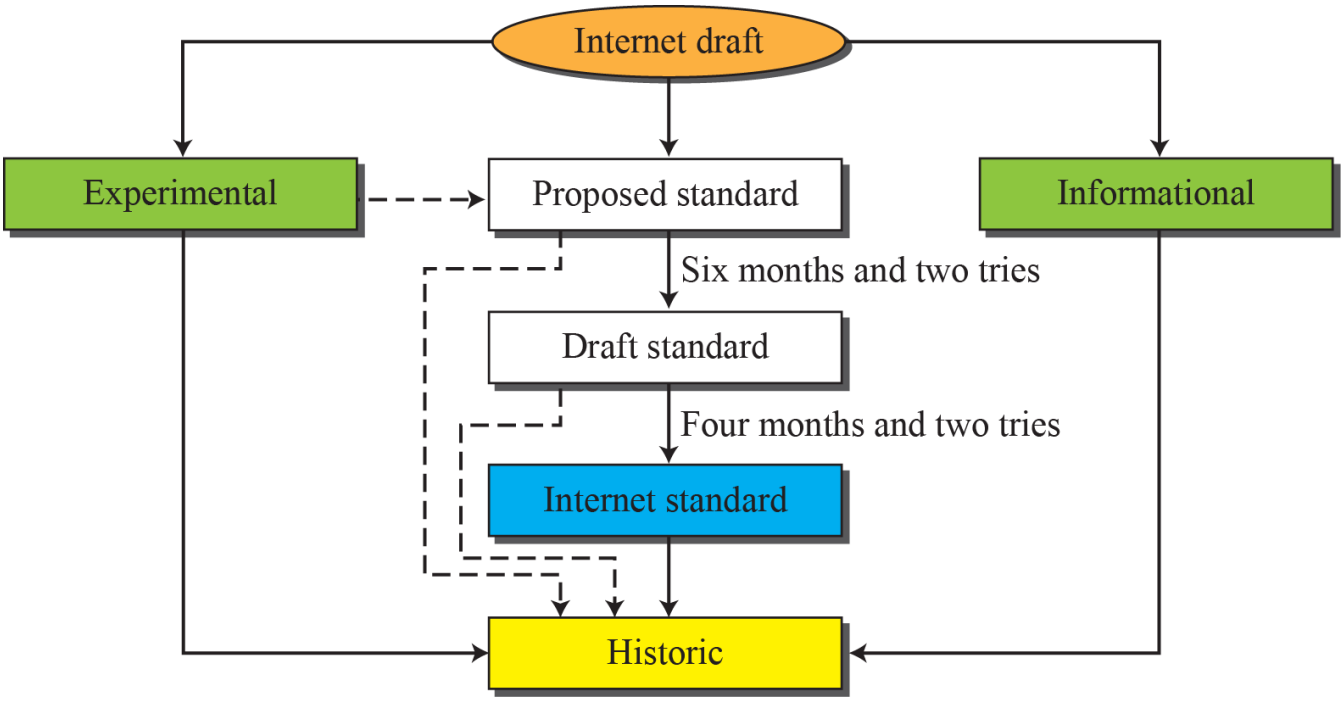
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## Internet Standards

- **Internet draft**
- **Request for Comments (RFC)**
  - ✓ **Proposed Standard**
  - ✓ **Draft Standard**
  - ✓ **Internet Standard**
  - ✓ **Historic**
  - ✓ **Experimental**
  - ✓ **Informational**




## Protocol Layering - Introduction

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- **Protocol**

Rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively

- **Protocol Layering**

Simple Communication: only one simple protocol

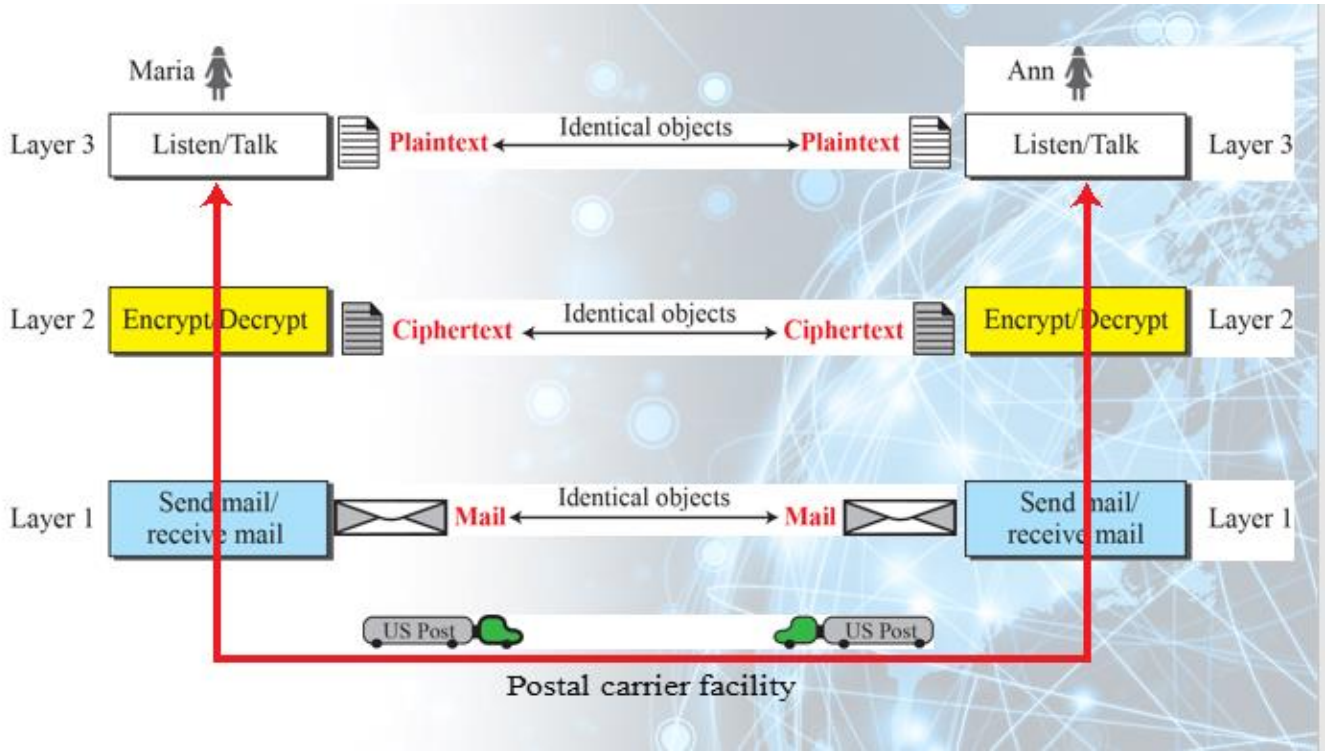



Complex Communication, we need a protocol at each layer, or Protocol Layering

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## Protocol Layering - Advantages and Disadvantages

### Advantages

- ✓ Modularity
- ✓ Separation of Service & Implementation
- ✓ Reduced Complexity & Cost

### Disadvantages

- None Really!

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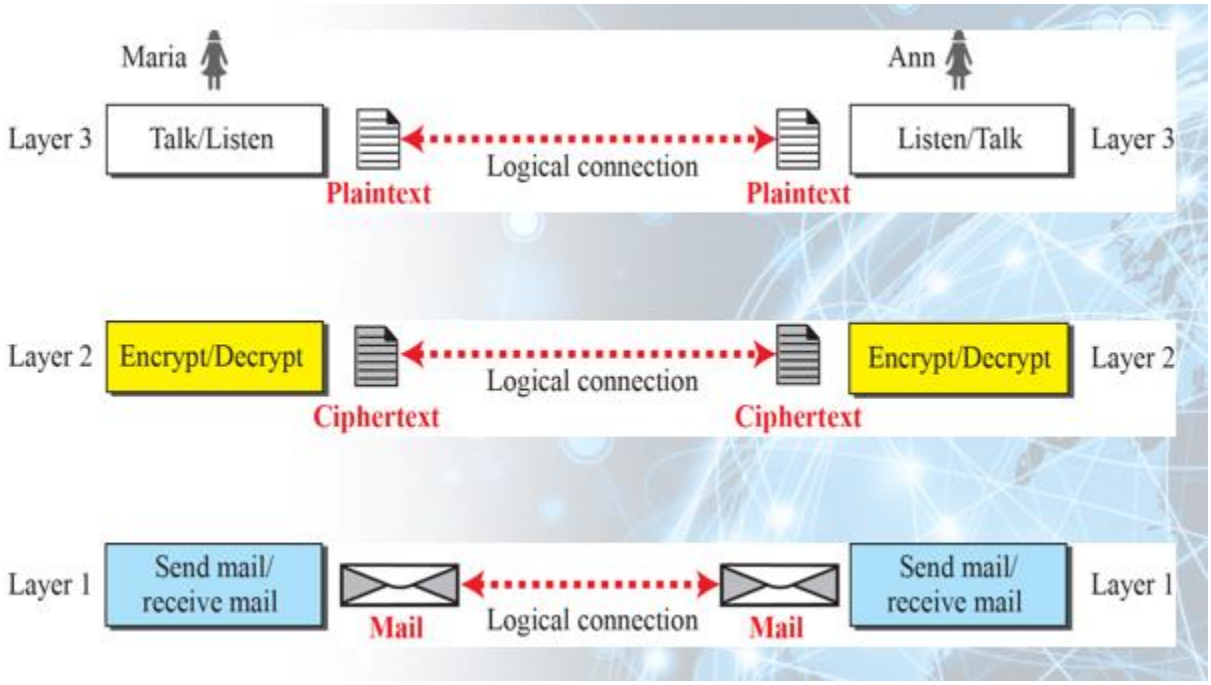
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## Protocol Layering - Principles

- **Two Principles**
  - ✓ Bidirectional Communication → Each Layer performs two opposite tasks in each direction
  - ✓ Two objects under each layer at both sites should be identical

## Protocol Layering - Logical Connections

- **Logical Connections**
  - ✓ Imaginary connection between each layer



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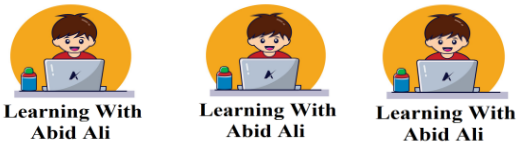
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## TCP/IP Protocol Suite

TCP/IP stands for Transmission Control Protocol/Internet Protocol and is **a suite of communication protocols used to interconnect network devices on the internet**. TCP/IP is also used as a communications protocol in a private computer network (an intranet or extranet).

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# TCP/IP Protocol Suite

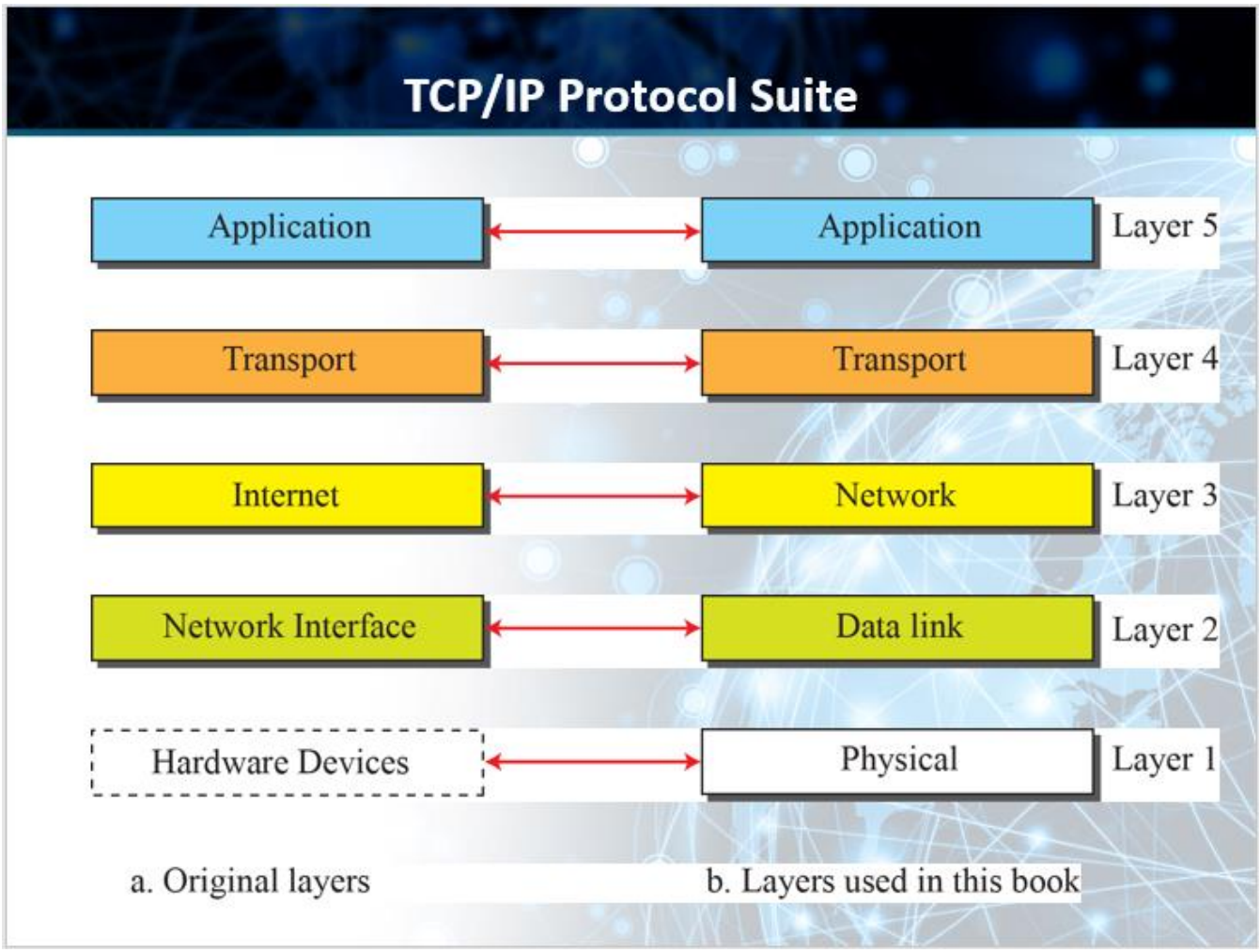
## History

- The Internet Protocol Suite (commonly known as TCP/IP Suite or the TCP/IP Model) is the set of computer network communications protocols and a description framework used for the Internet and other similar networks.
- The TCP/IP Model was created in the 1970s by DARPA, an agency of the United States Department of Defense (DOD). It evolved from ARPANET, which was the world's first wide area network and a predecessor of the Internet.

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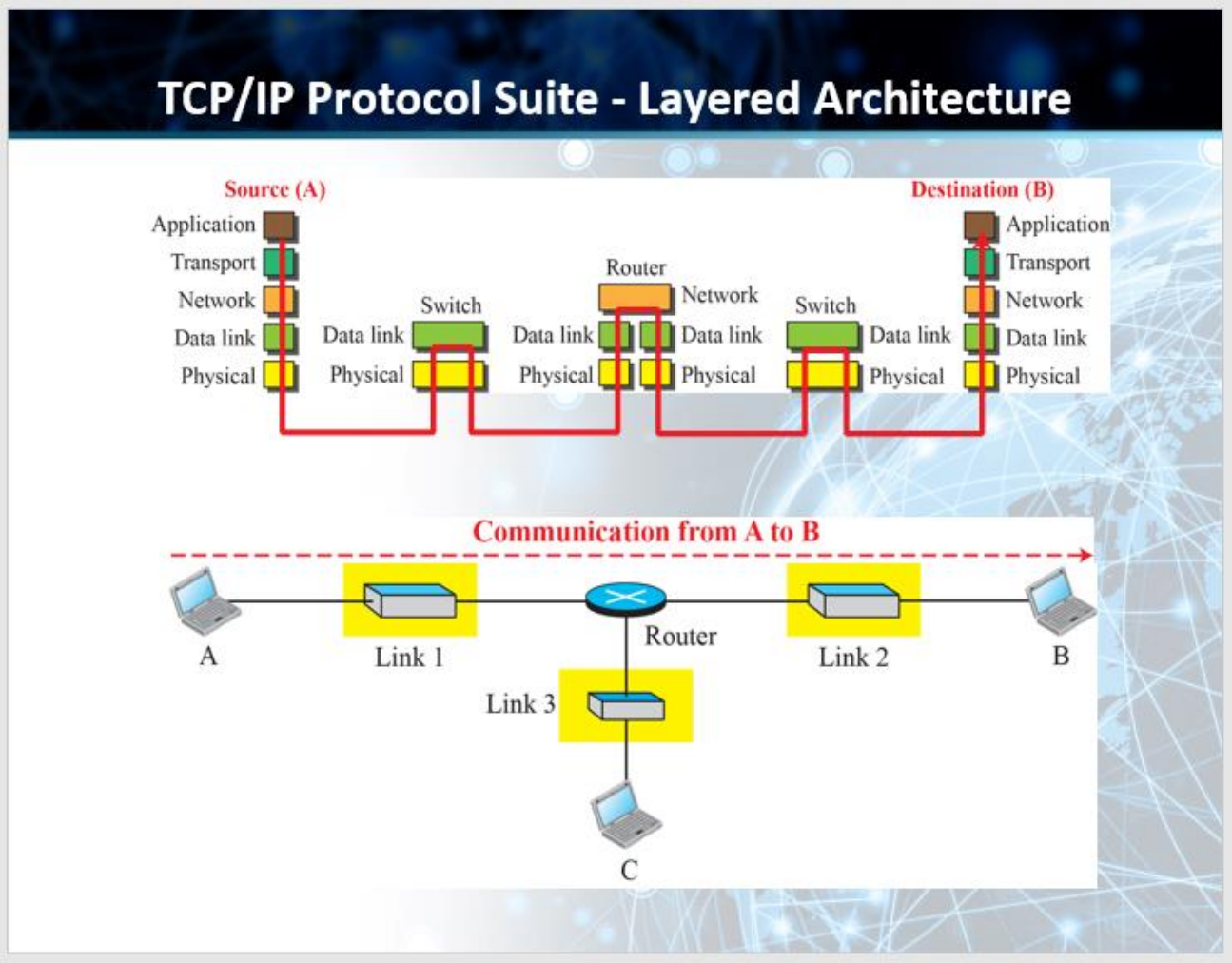
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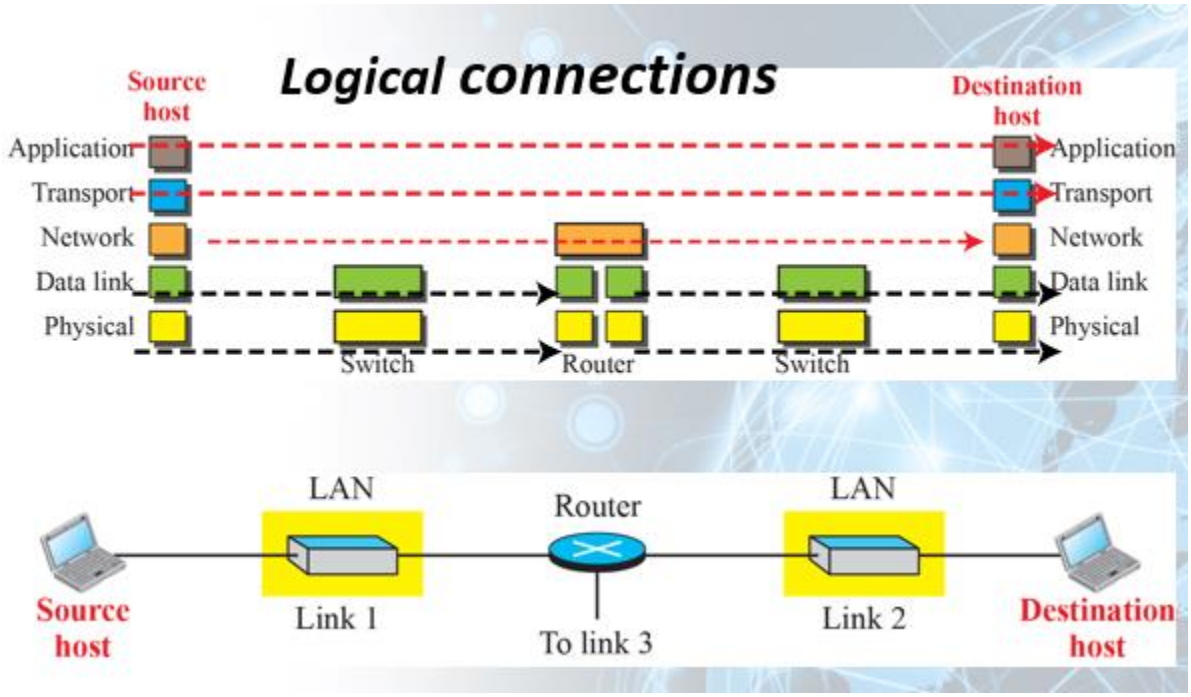


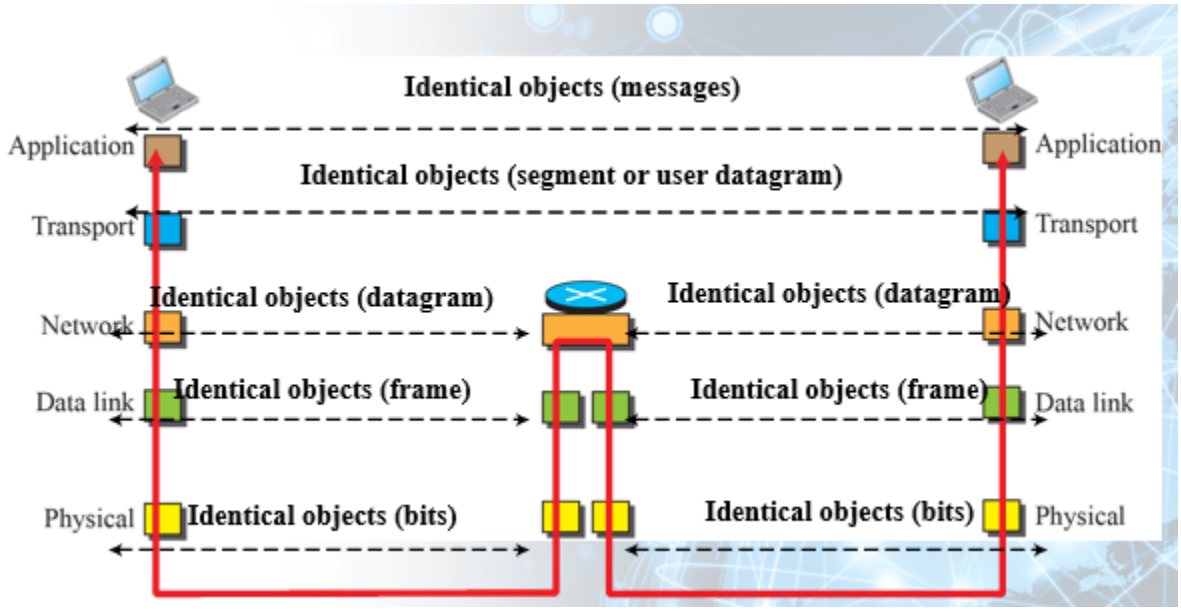

- **Logical Connections between TCP/IP Layers**

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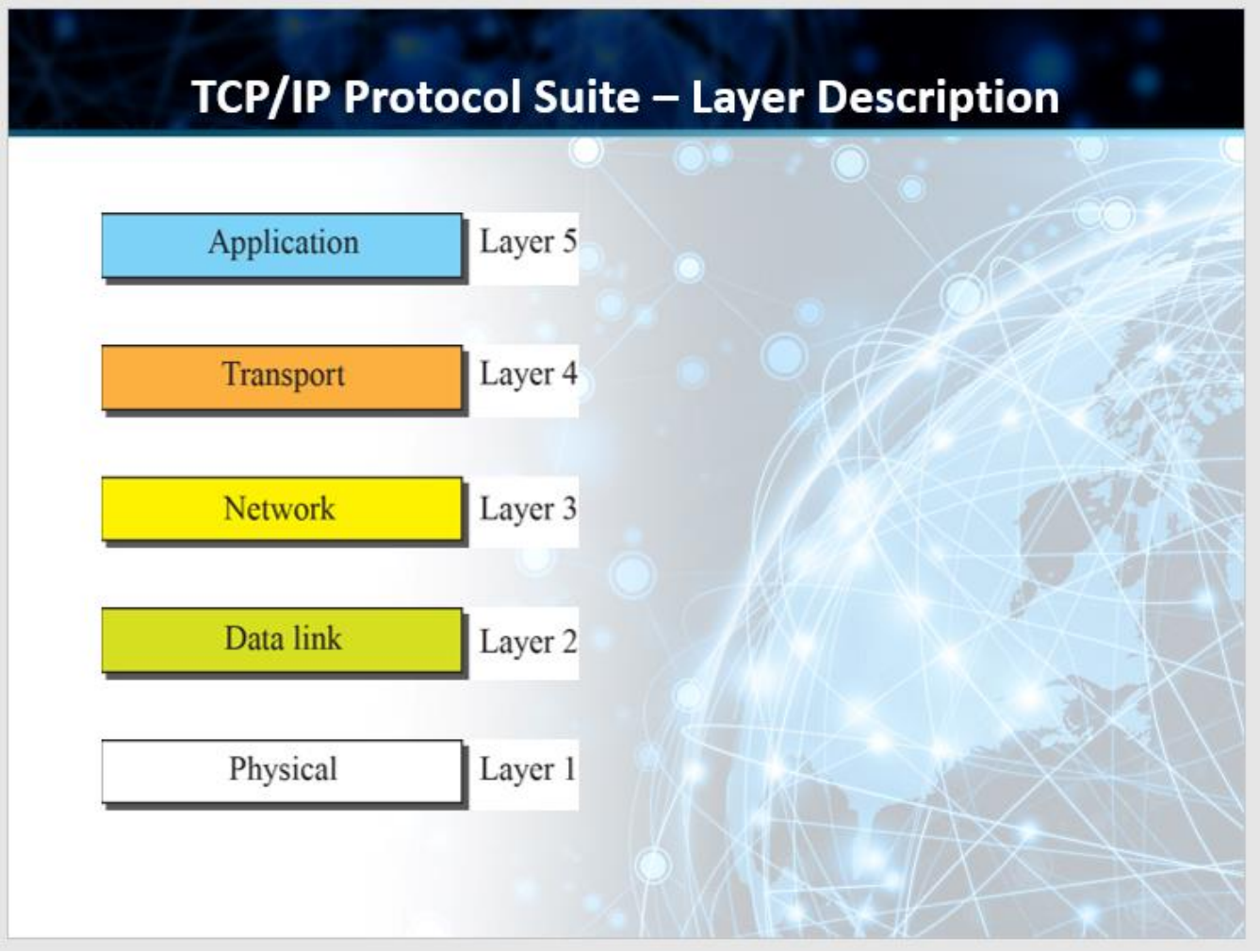


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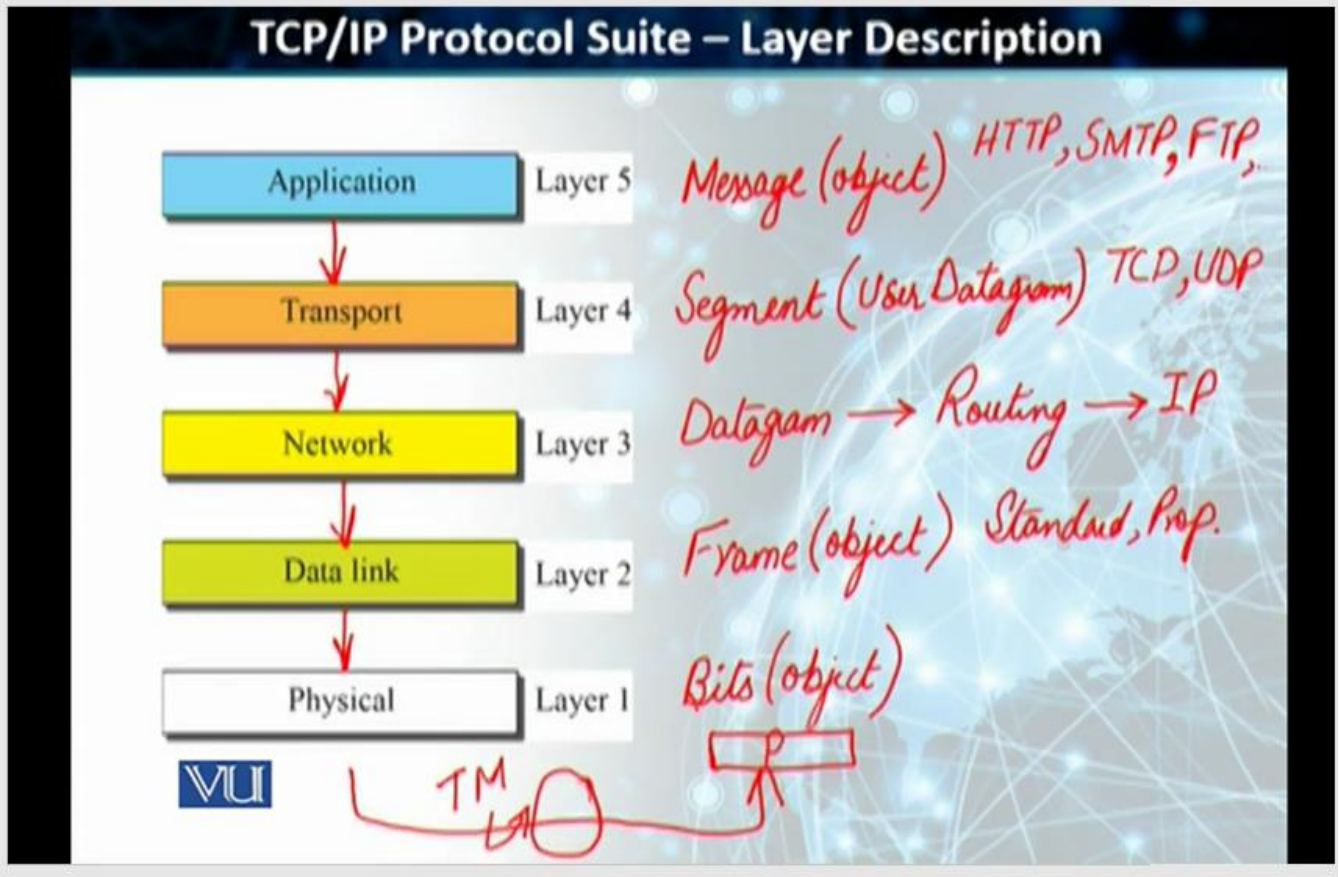
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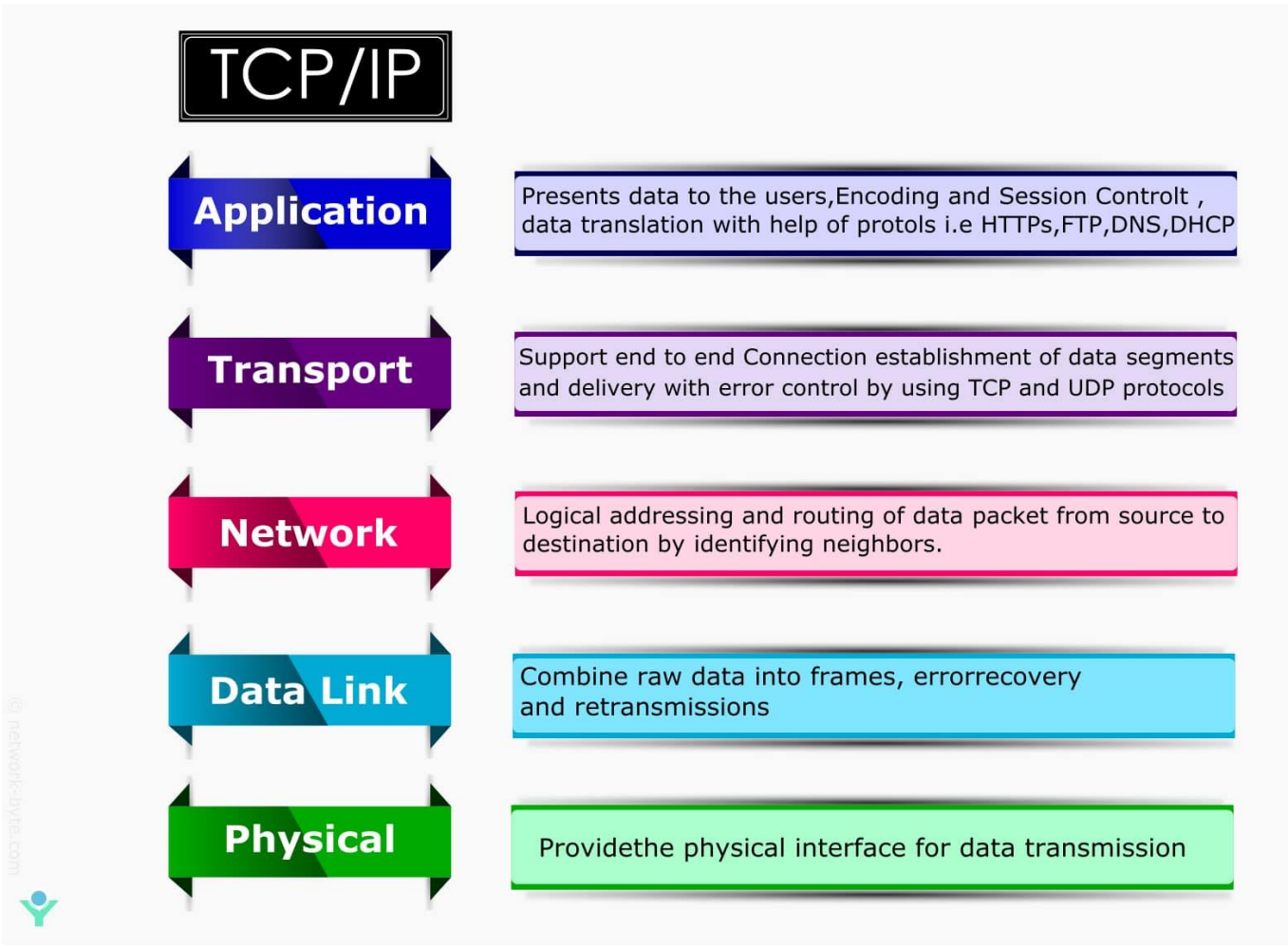
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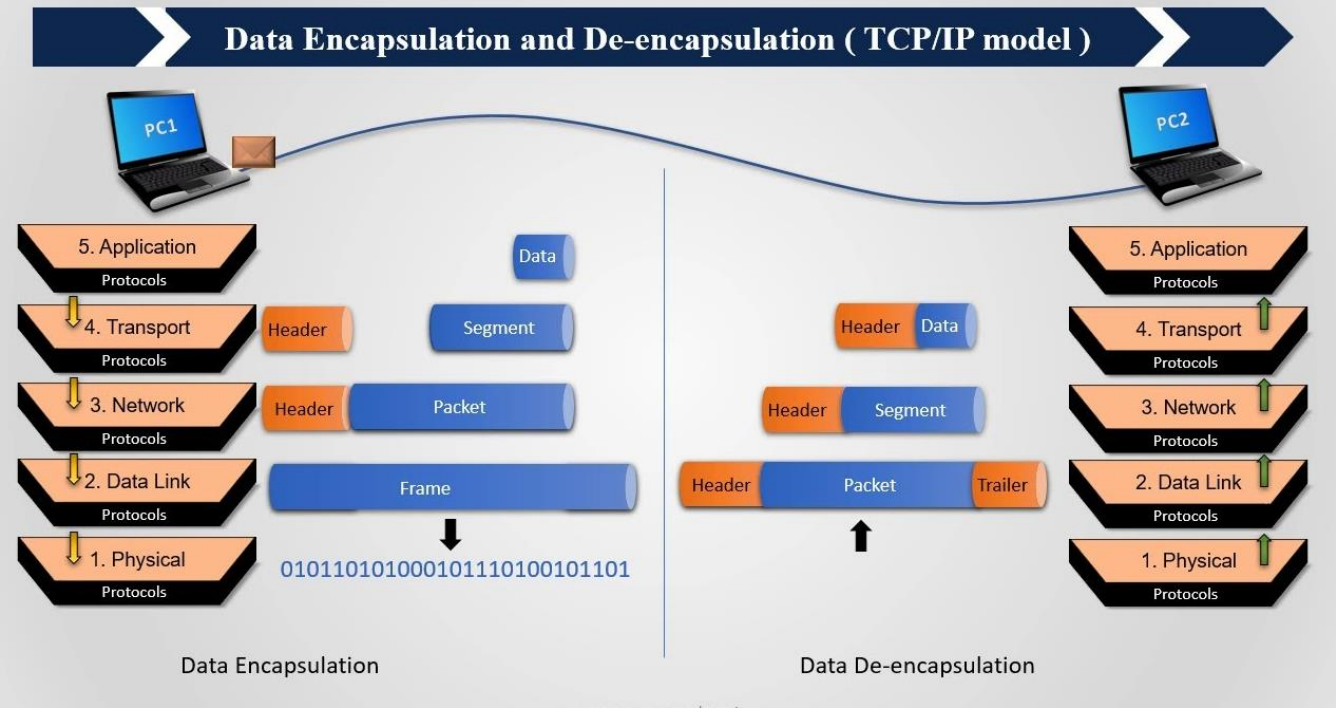

## What is encapsulation and Decapsulation?

**Encapsulation adds information to a packet as it travels to its destination. Decapsulation reverses the process by removing the info, so a destination device can read the original data**.

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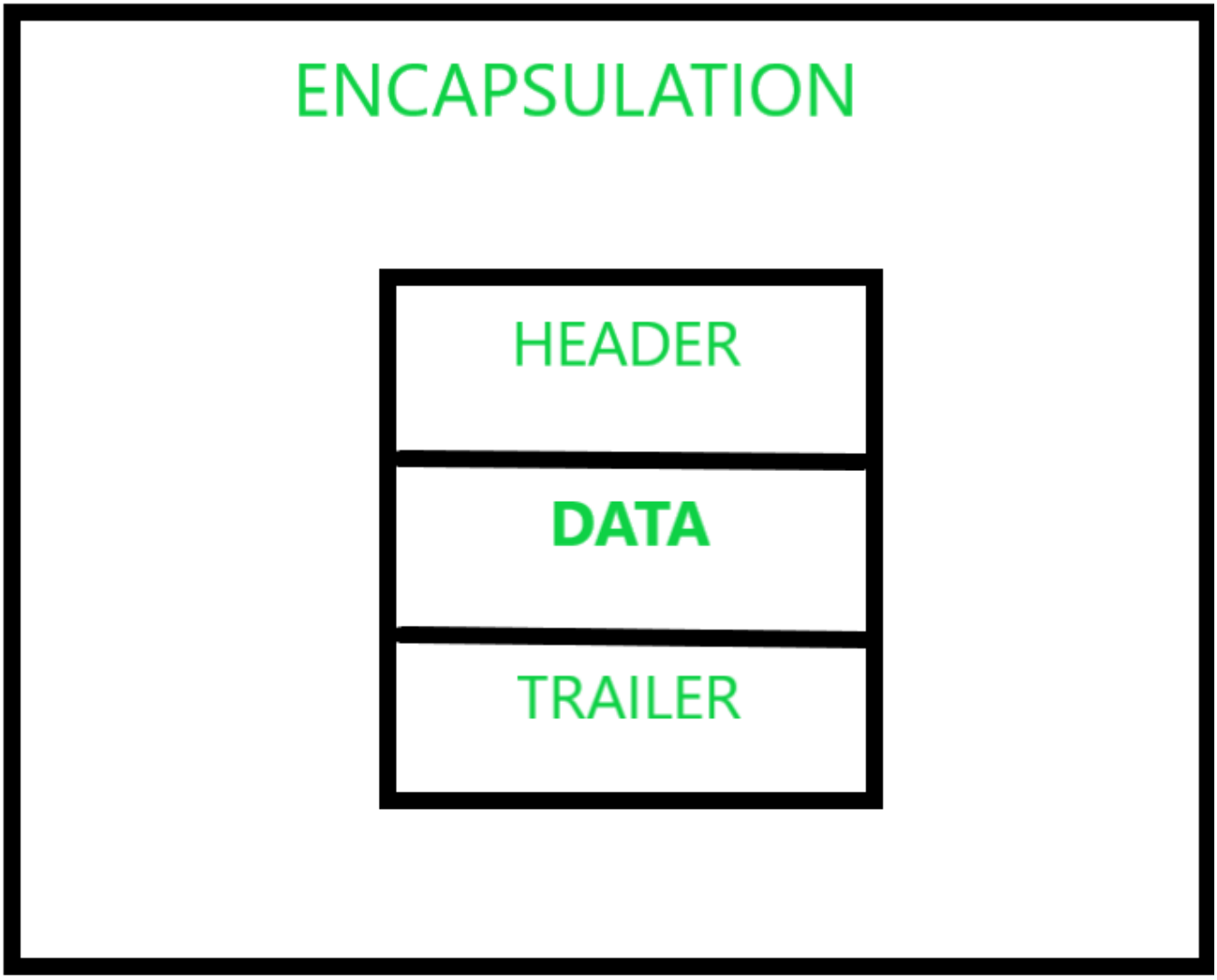

**Encapsulation:**

Encapsulation refers to attaching new information in the Application Layer data as it is passed onto next layers in the TCP/IP model. This additional information basically divided into two parts, Header and Trailer. These are elements attached in order to make the transmission smoother, on each layer a PDU (Protocol Data Unit) is generated. The concept of Encapsulations can be summarized in the screenshot attached ahead.

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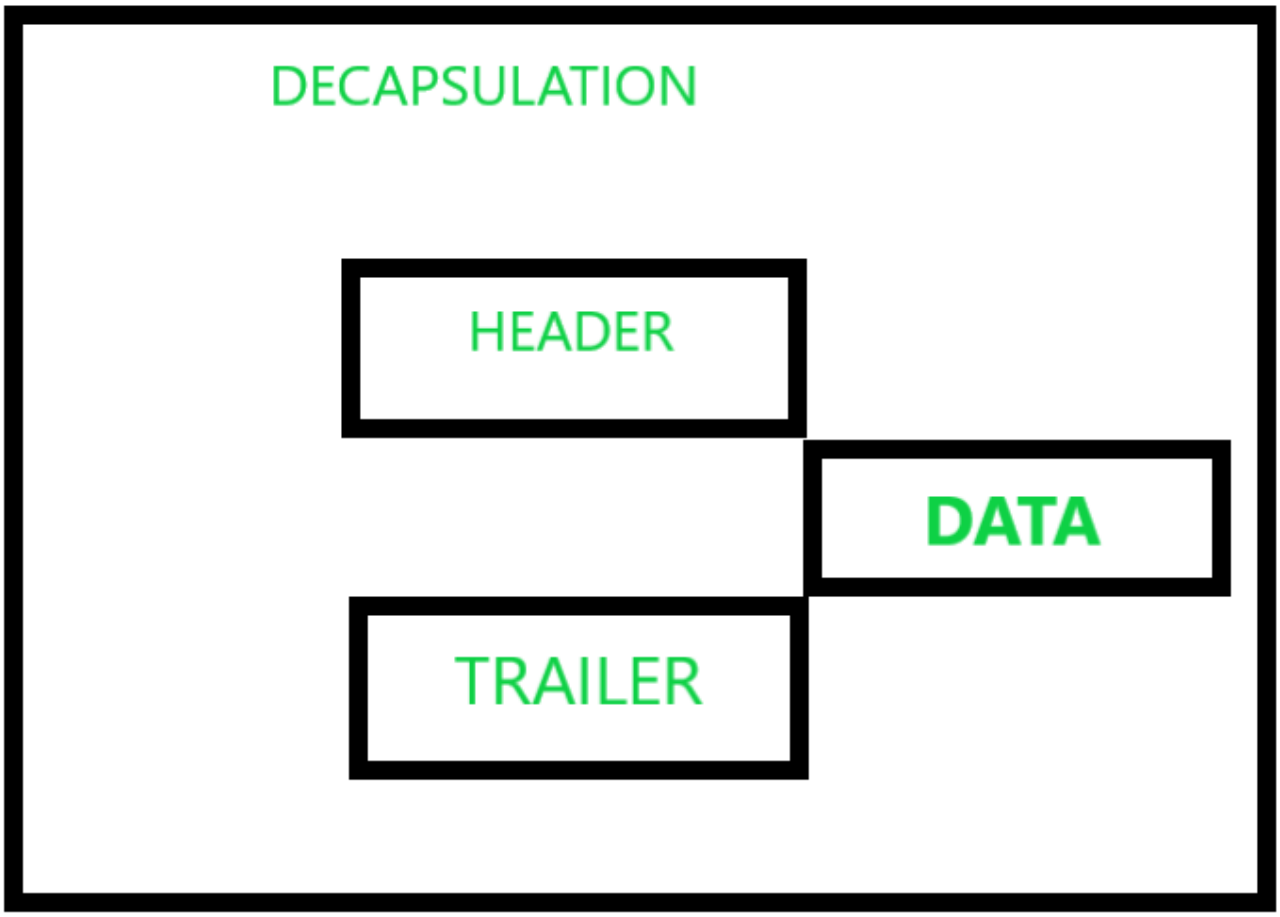

**Decapsulation:**

Decapsulation refers to the removal of all these additional information and extraction of originally existing data, and this process continues till the last layer i.e., the Application Layer. This process removes, fragments of distinct information in each layer as it approaches that layer. Here is the pictorial representation of the whole process.

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## The Open System Interconnection (OSI) Model

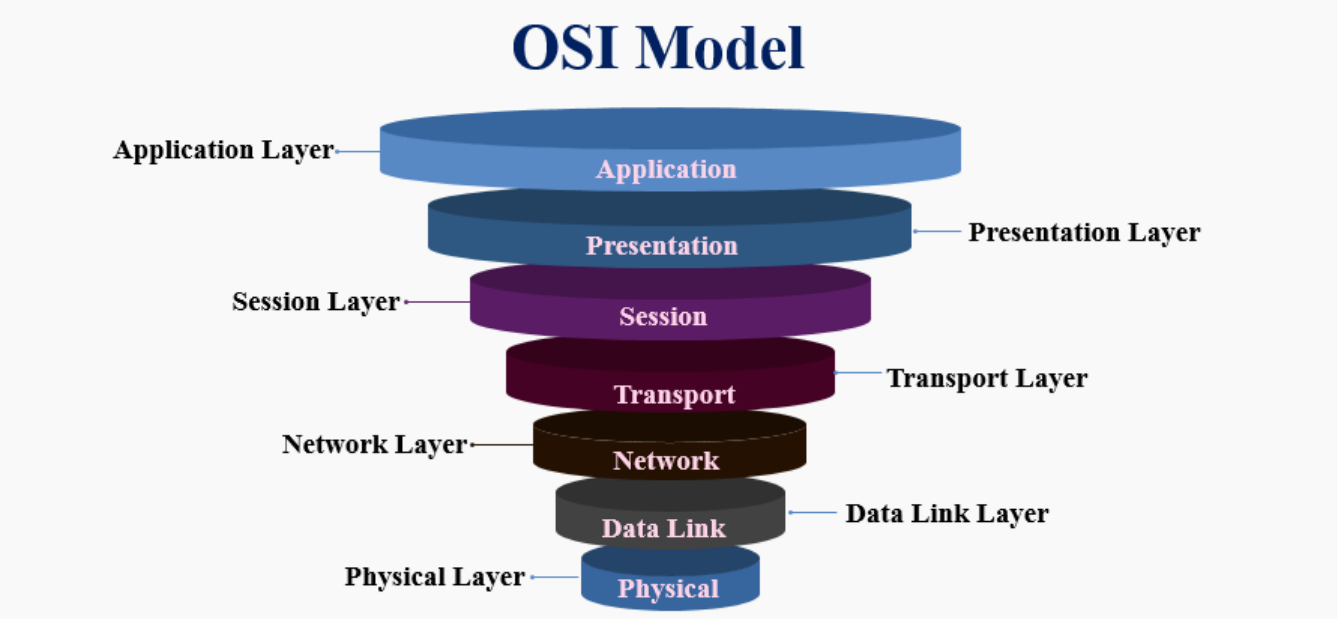
Open Systems Interconnection (OSI) – a framework for transmitting messages between any two entities in a network. It defines seven layers of functions: Physical Layer – is at the electrical and mechanical level; referring to the hardware and physical wires/devices required to transport the data.

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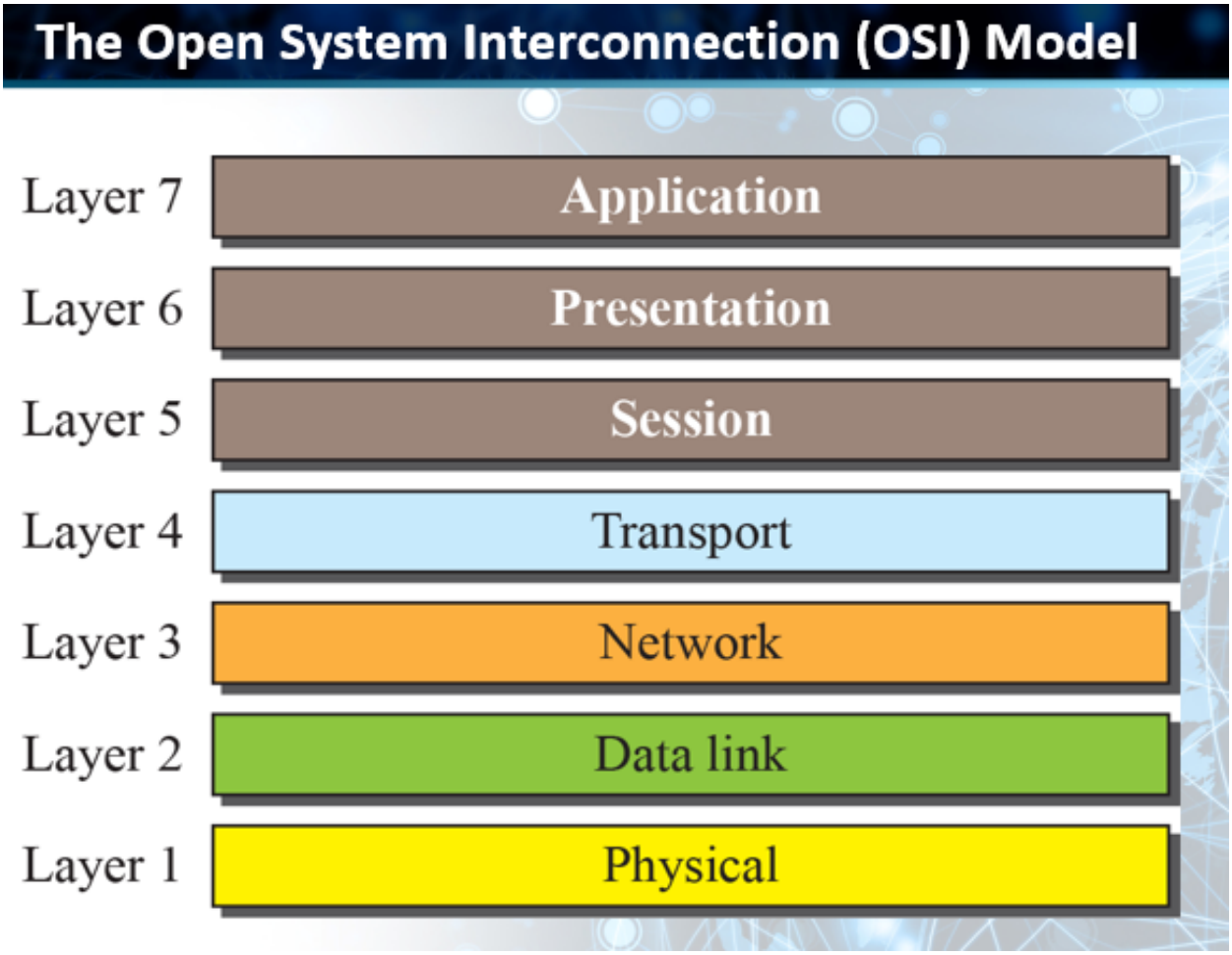
- **International Organization for Standardization (ISO)**
- **ISO established in 1947**
- **Close to three-fourths of countries represented**
- **Introduced OSI Model in late 1970s**
- **OSI: a 7-Layer Model**



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## Lack of OSI Model's Success

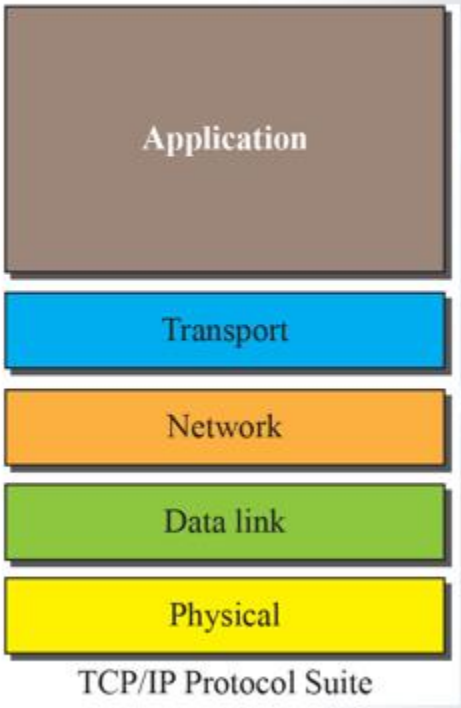
- **Three reasons OSI did not replace TCP/IP:**
  - ✓ **OSI was completed when TCP/IP was fully in place**
  - ✓ **Some layers in OSI not fully defined**
  - ✓ **Performance of TCP/IP better than that of OSI**

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## Data Communication versus Computer Networks




- Analog & Digital Transmission
- Transmission Media
- Switching
- Error Detection and Correction
- Media Access and Data Link Control
- Wired and Wireless LANs

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## Physical Layers

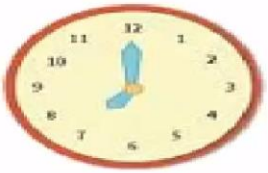
The data is travel from one network to another network through Physical Layer in the Form of Analog & Digital Data.

**Analog Data → Continuous**

Analog Clock, Human Voice



Analog Signal



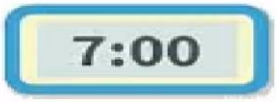
Analog clock

**Digital Data → Discrete**

Digital Clock, Data Store in Computer



Digital Signal

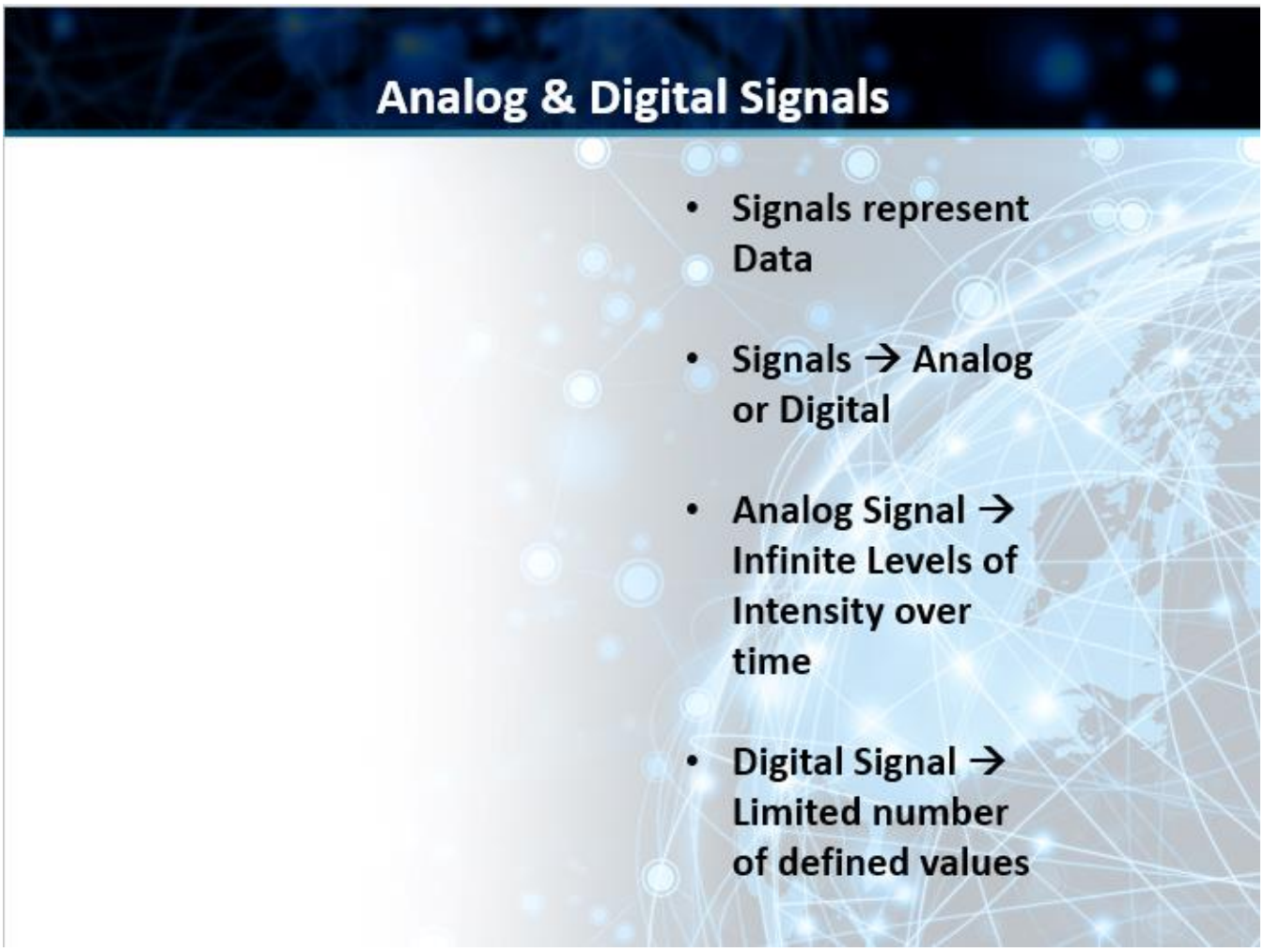



Digital clock

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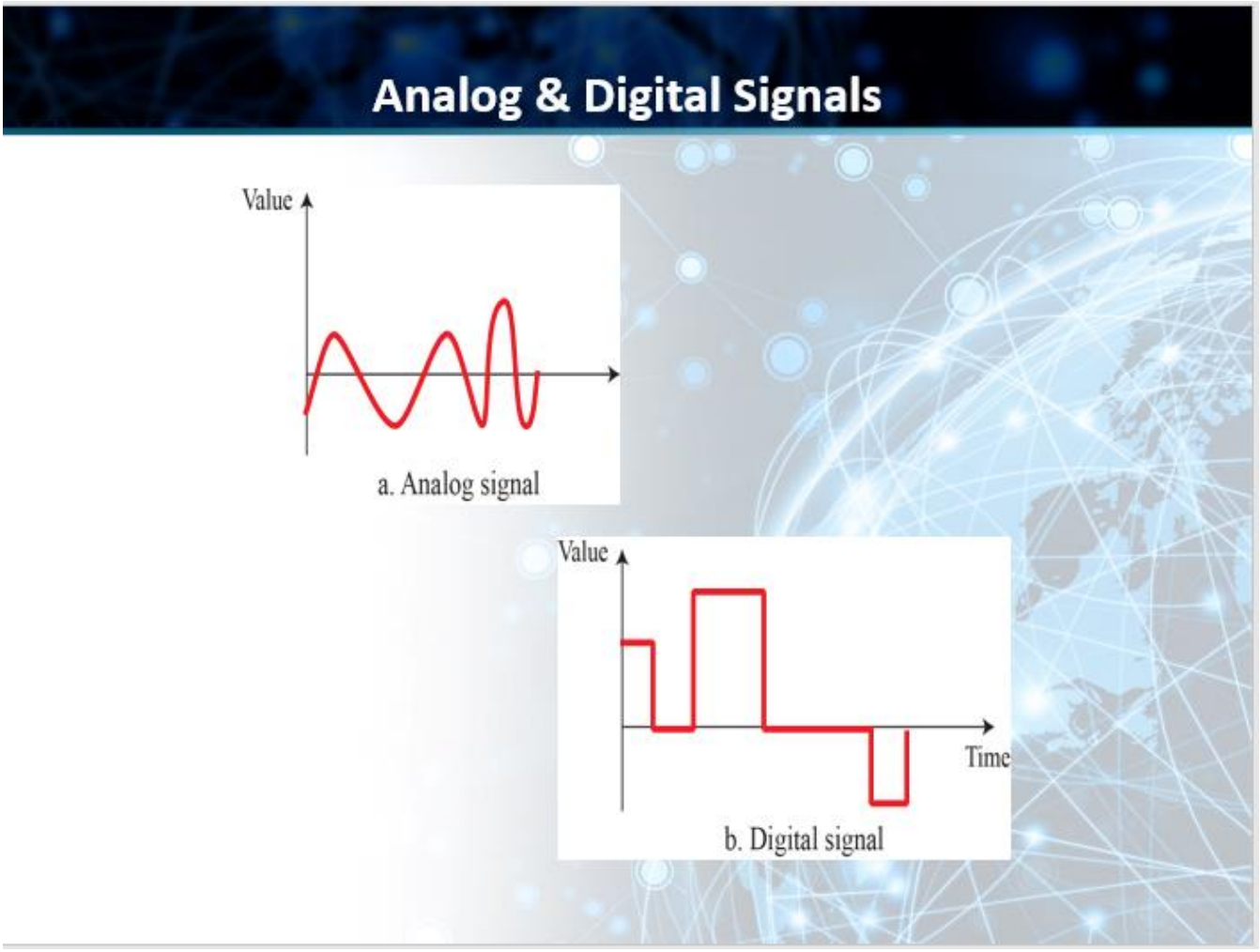
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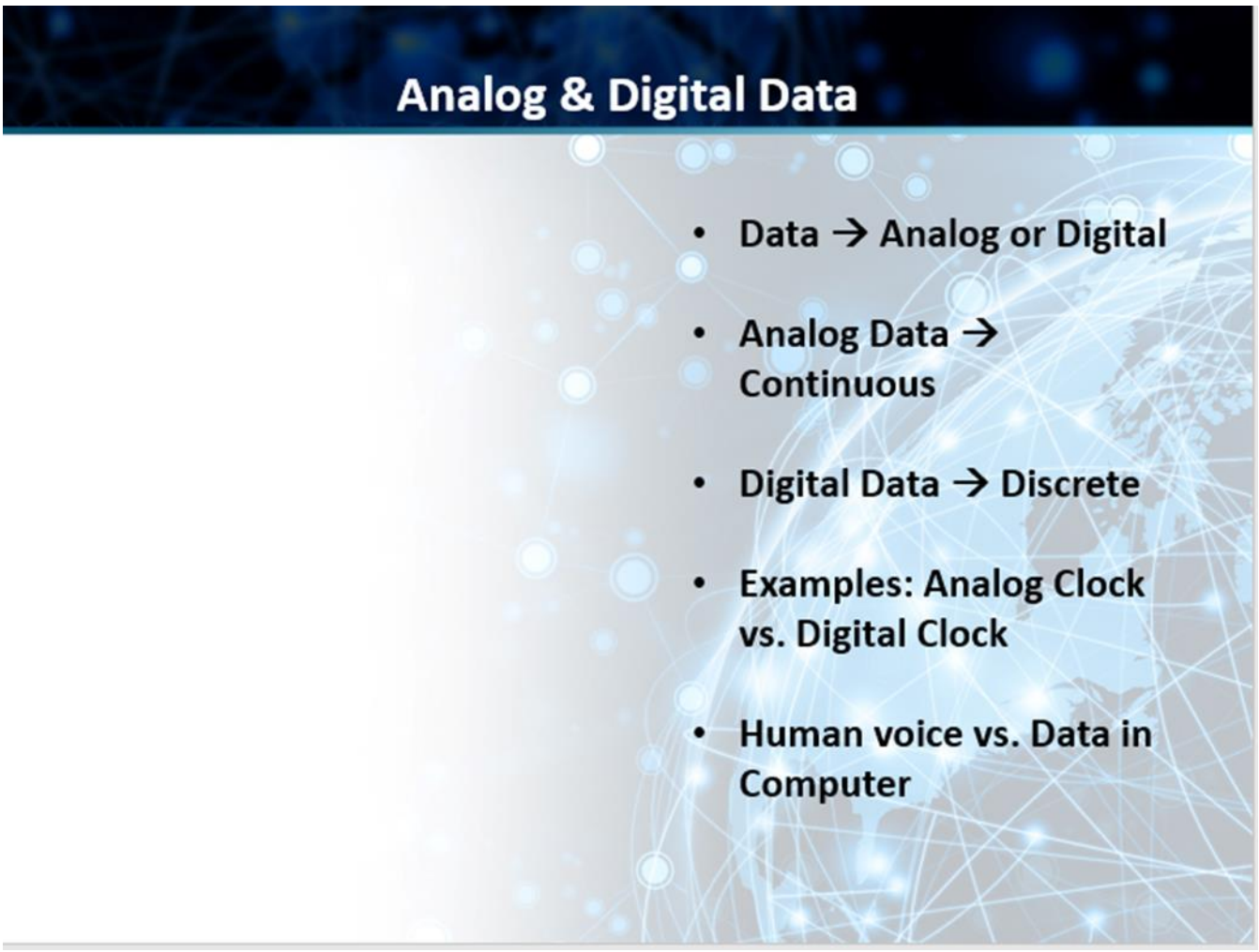
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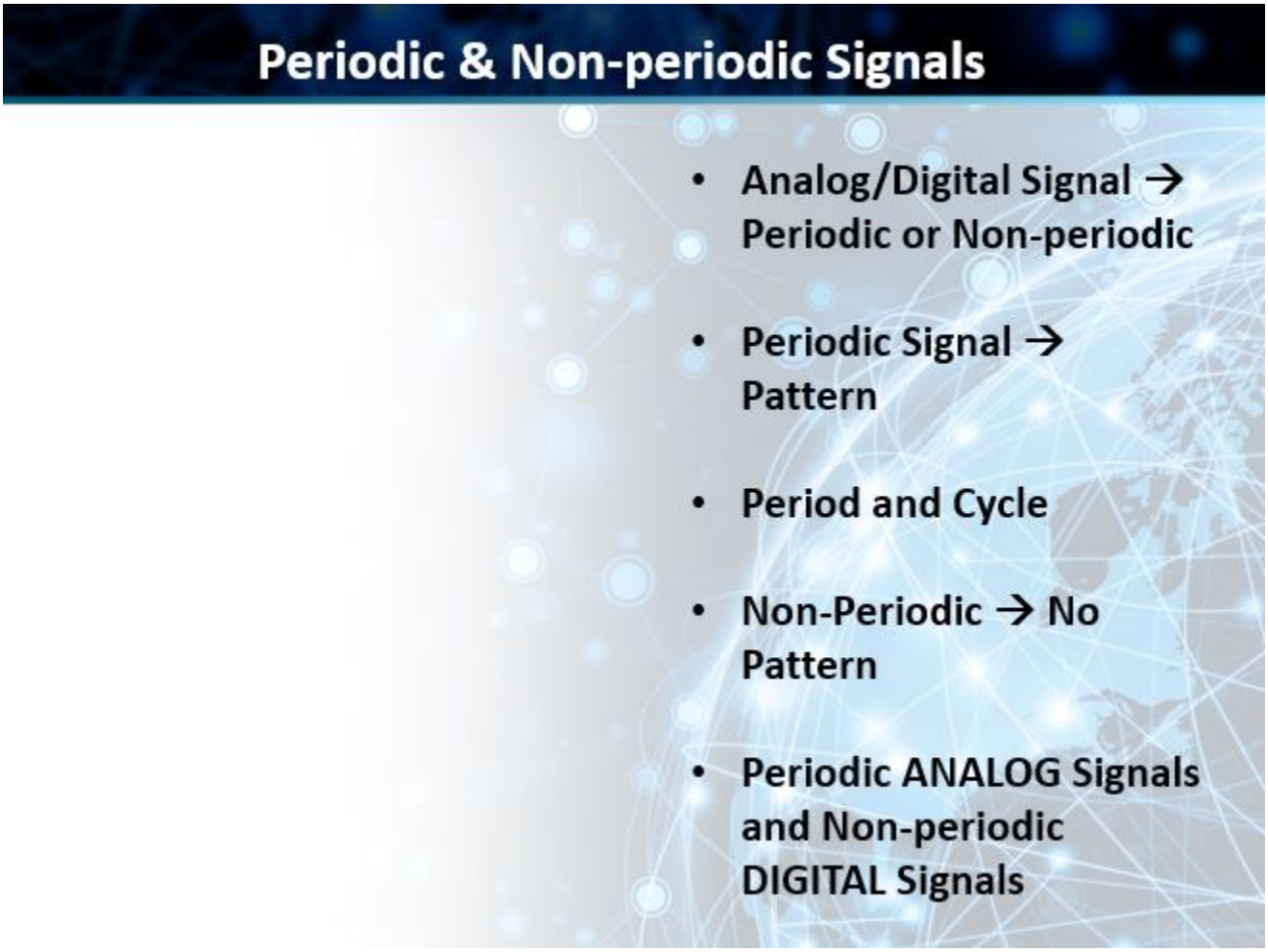
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## Periodic & Non-periodic Signals

A signal is said to be periodic signal if it has a definite pattern and repeats itself at a regular interval of time. Whereas, the signal which does not at the regular interval of time is known as an aperiodic signal or non-periodic signal

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## Periodic Signals

A Periodic Signals is a signal that complete pattern in a certain period of time. (Cycle).

**Simple Periodic Analog Signal** - Sine Wave

**Composite Periodic Analog Signal** - Multiple Sine Wave

## Non-Periodic Signals

A Non-Periodic Signals is a signal that changes its pattern in each cycle.

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Lecture Prepared By

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## A Simple Periodic Analog Signal, A Sine Wave

**A simple periodic analog signal**, a sine wave, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves. A sine wave can be represented by three parameters: the peak amplitude, the frequency, and the phase. These three parameters fully describe a sine wave.

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# Periodic Signal – Sine Wave

There are three main attribute of a Sine Wave

**1. Amplitude** **2. Period / Frequency** **3. Phase**

**1. Amplitude** (Key Attribute):

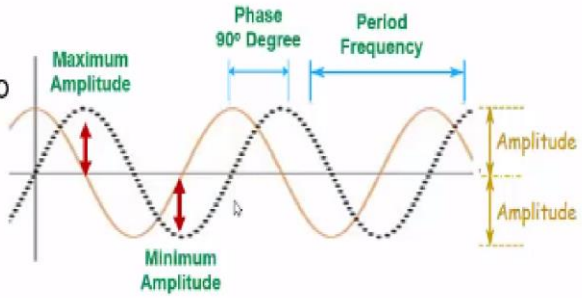
It is equal to the vertical distance from a given point on the wave form to the horizontal axis

**2. Period / Frequency** (Second Attribute):

**Period:** Amount of time (in seconds) a signal need to complete one cycle.

**Frequency:** Number of cycles completed in one second

**Unit of Period:** Period is expressed in seconds




Periodic Analog Signal

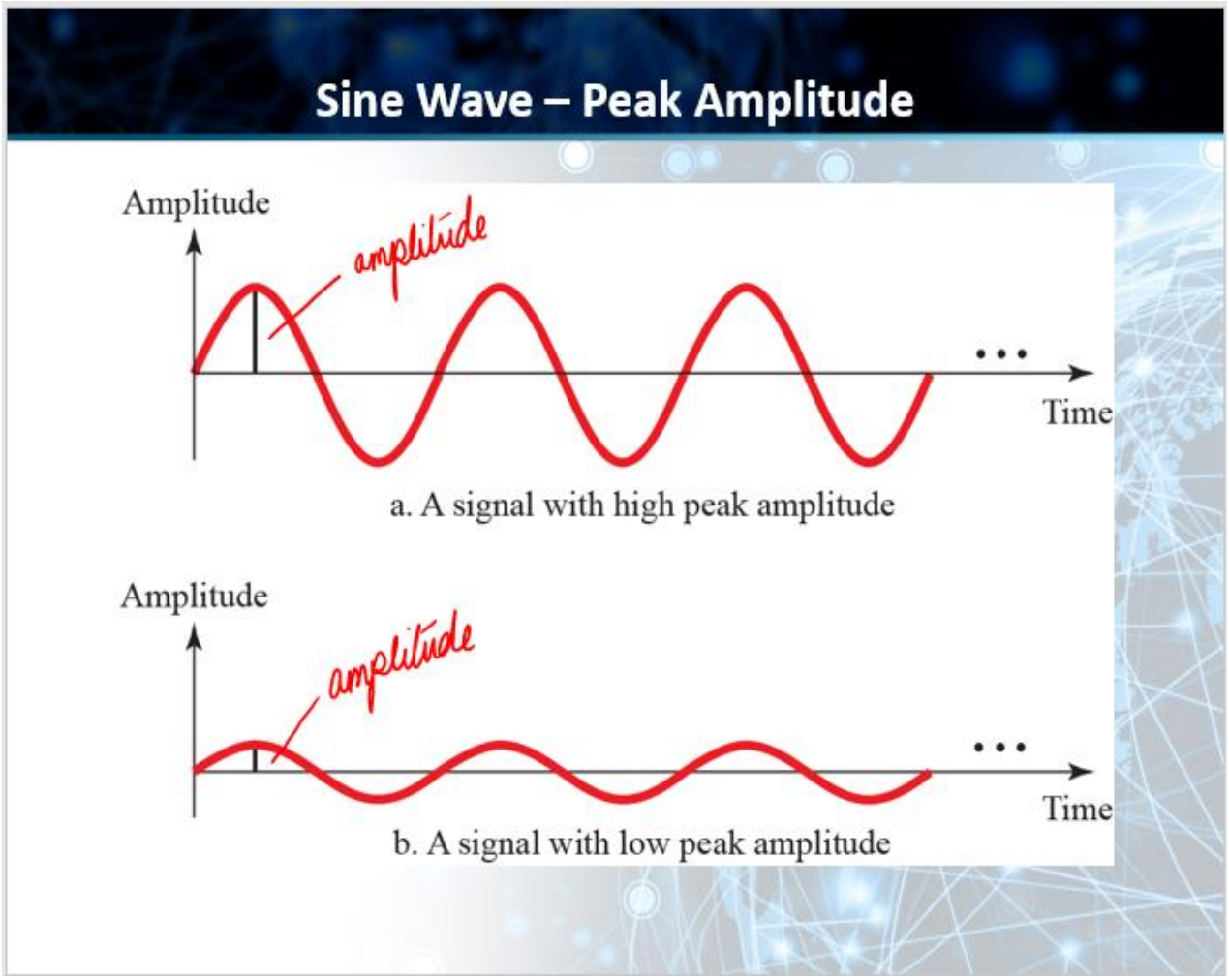
**3. Phase**

The Phase Shift is how far the function is shifted horizontally from the usual position.

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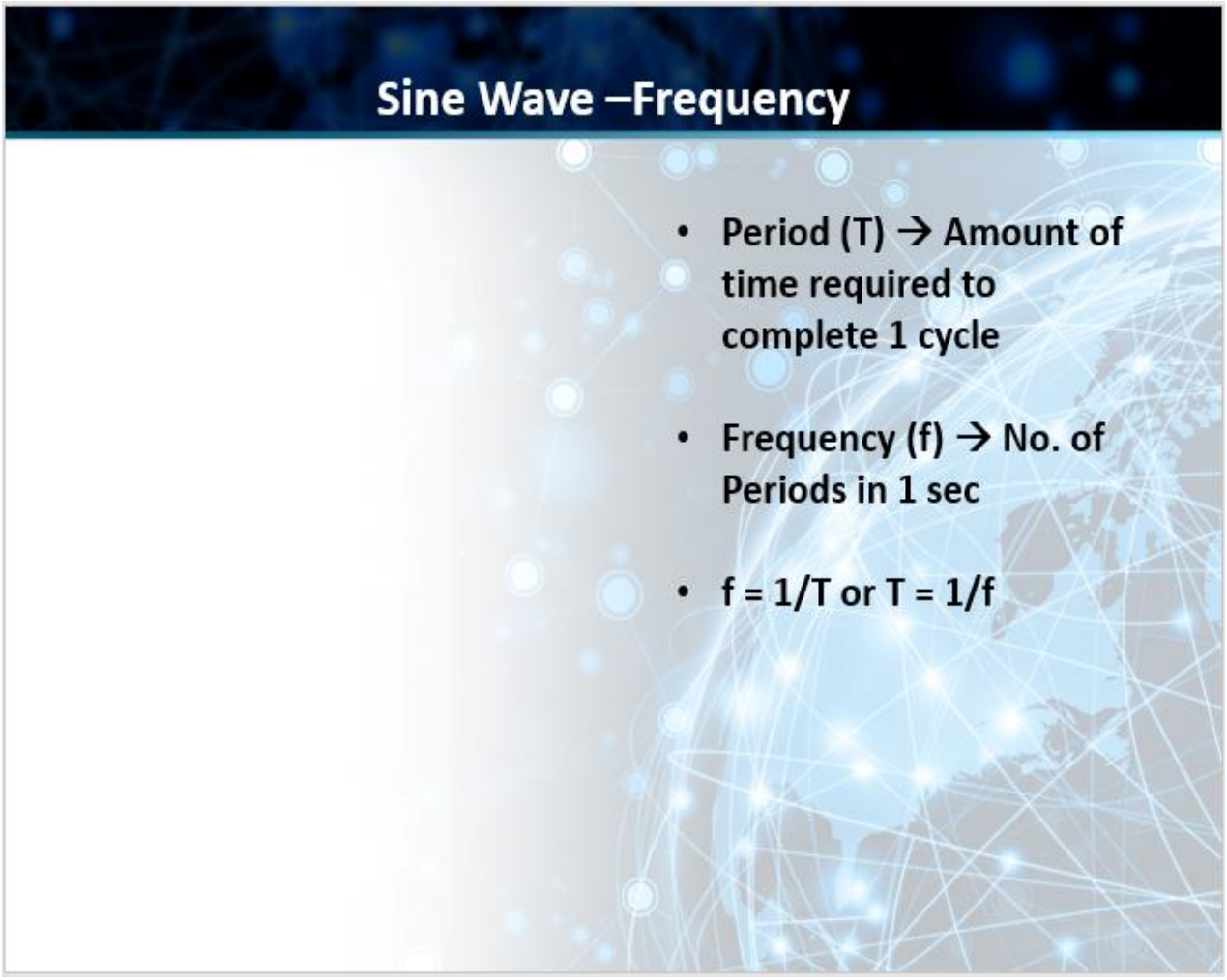
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## Period / Frequency Examples

- A Sine wave has a frequency of 6 Hz. What is its period?
- A Sine wave completes one cycle in 4 seconds. What is its frequency?
- **Period: Formula:** $T = 1 / f$
- **Frequency: Formula:** $f = 1 / T$

| *Period* | | *Frequency* | |
|---|---|---|---|
| *Unit* | *Equivalent* | *Unit* | *Equivalent* |
| Seconds (s) | 1 s | Hertz (Hz) | 1 Hz |
| Milliseconds (ms) | $10^{-3}$ s | Kilohertz (kHz) | $10^{3}$ Hz |
| Microseconds (μs) | $10^{-6}$ s | Megahertz (MHz) | $10^{6}$ Hz |
| Nanoseconds (ns) | $10^{-9}$ s | Gigahertz (GHz) | $10^{9}$ Hz |
| Picoseconds (ps) | $10^{-12}$ s | Terahertz (THz) | $10^{12}$ Hz |

Pause (k)

## SOLUTION

- Period: Formula: T = 1 / f
  = 1 / 6
  = 0.17 Second
- Frequency: Formula: f = 1 / T
  = 1 / 4
  = 0.25 Hz

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## Sine Wave –Frequency

| Period | | Frequency | |
|---|---|---|---|
| *Unit* | *Equivalent* | *Unit* | *Equivalent* |
| Seconds (s) | 1 s | Hertz (Hz) | 1 Hz |
| Milliseconds (ms) | $10^{-3}$ s | Kilohertz (kHz) | $10^{3}$ Hz |
| Microseconds (μs) | $10^{-6}$ s | Megahertz (MHz) | $10^{6}$ Hz |
| Nanoseconds (ns) | $10^{-9}$ s | Gigahertz (GHz) | $10^{9}$ Hz |
| Picoseconds (ps) | $10^{-12}$ s | Terahertz (THz) | $10^{12}$ Hz |

## Example

**The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:**

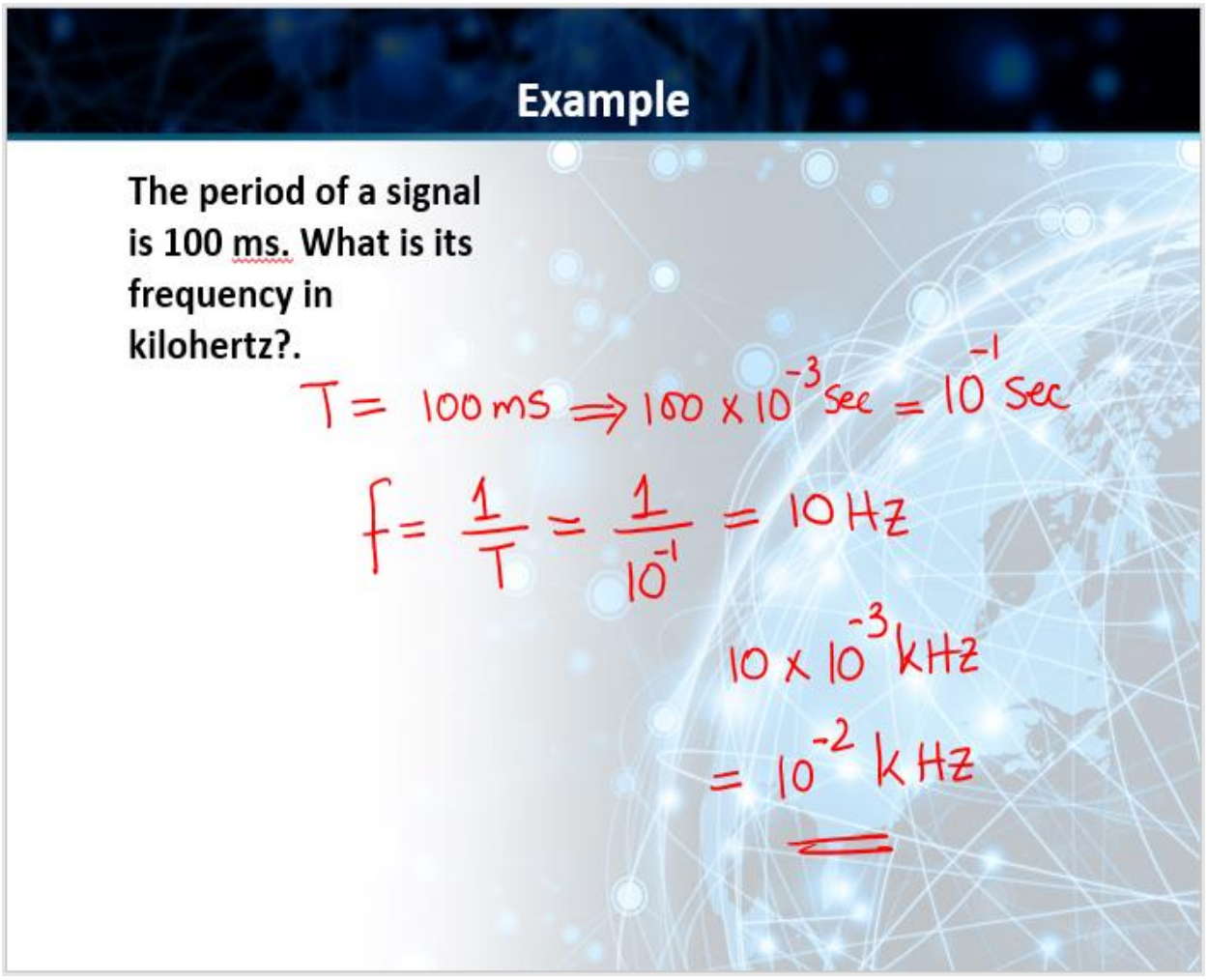
$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ sec}$$

$$= 16.6 \text{ msec}$$

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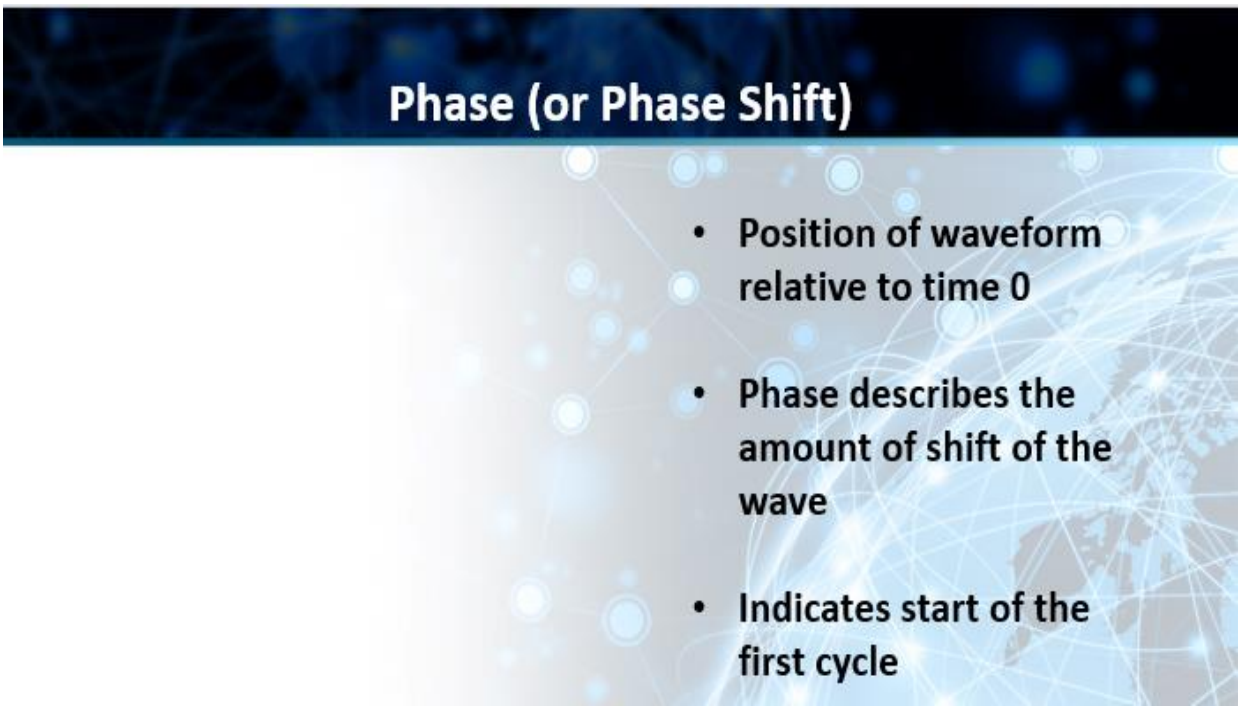
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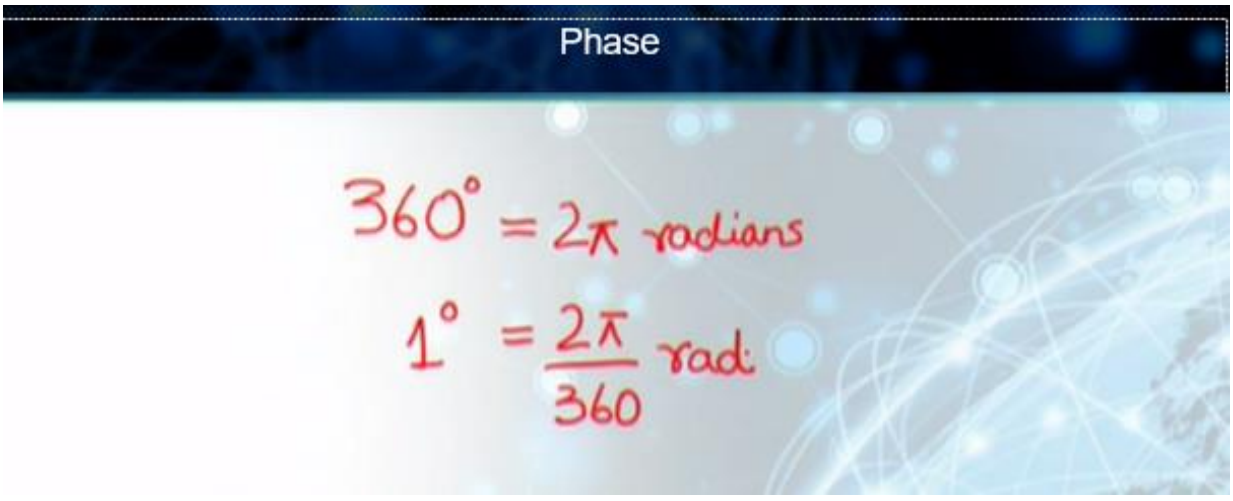


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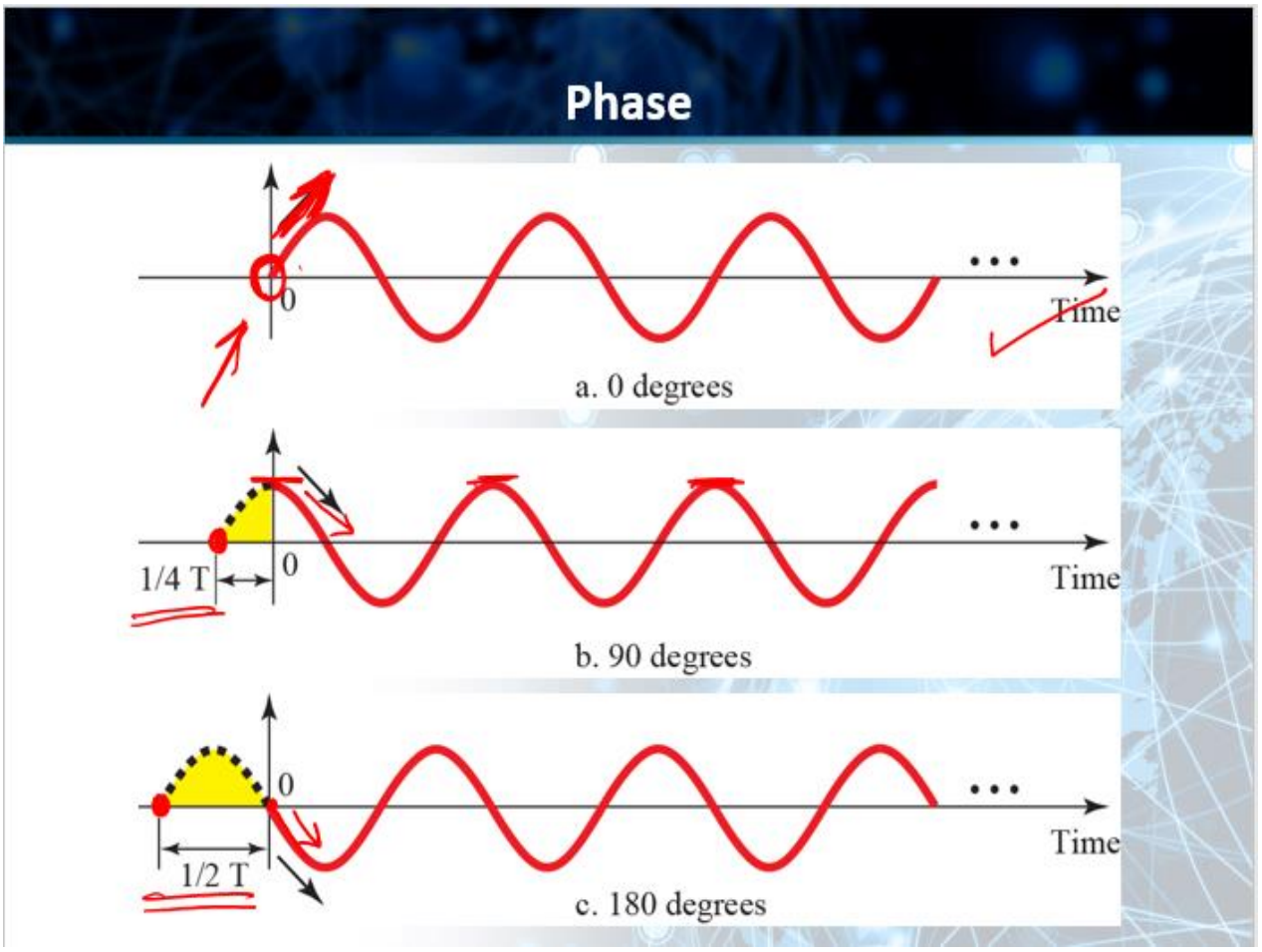


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## SOLUTION

- Phase:

= 1 / 6 x 360

= 360 / 6

= $60^0$ Degree

## Phase Example

A sine wave is offset 1/6 of a cycle with respect to time zero. What is its phase in radius?

- One Cycle = 360 Degrees

($1^0 = 2\pi / 360^0$ radios)

$= 60 \times 2\pi / 360$

$= 60 \times (2 \times (3.146 / 360))$

$= 60 \times (2 \times 0.0087)$

$= 60 \times 0.0174$

= 1.04 Radius

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## EXERECISE

- A Sine wave has a frequency of 60 Hz. What is its period?
- A Sine wave completes one cycle in 100 seconds. What is its frequency?
- A sine wave is offset 1/4 of a cycle with respect to time zero. What is its phase in radius?

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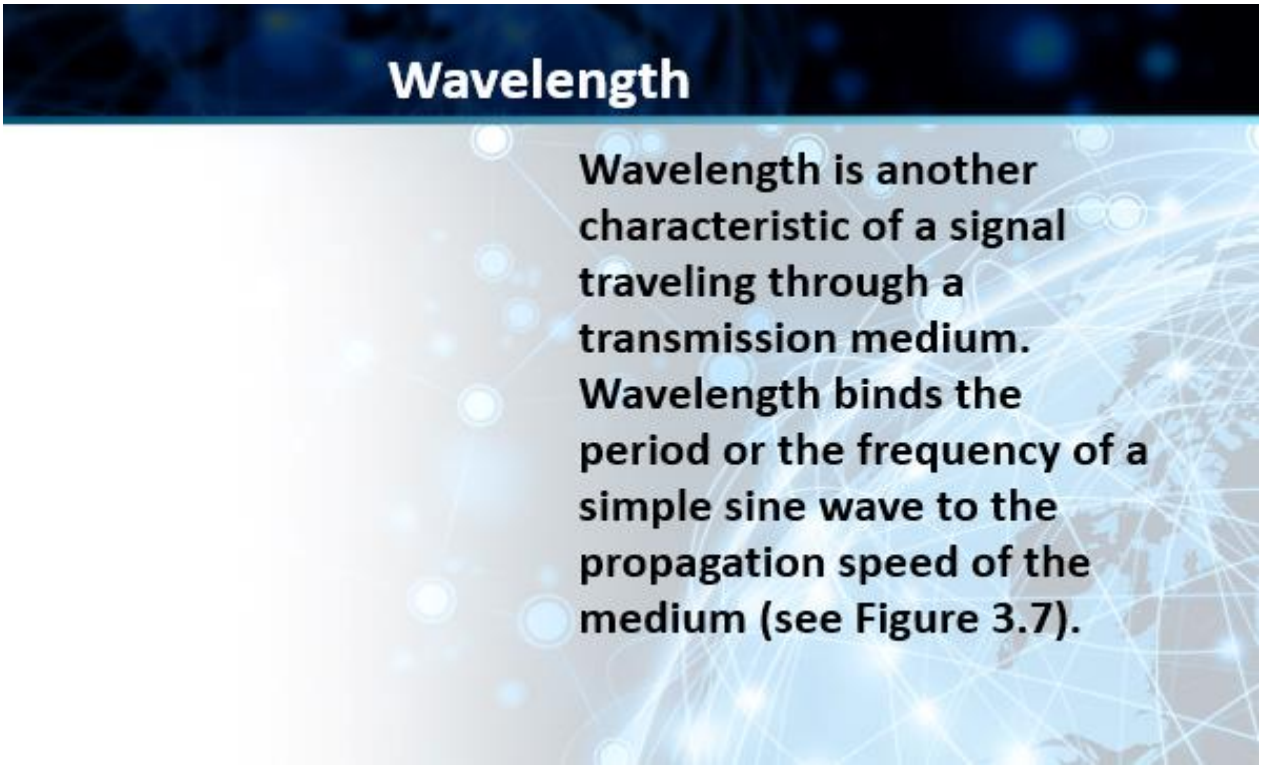
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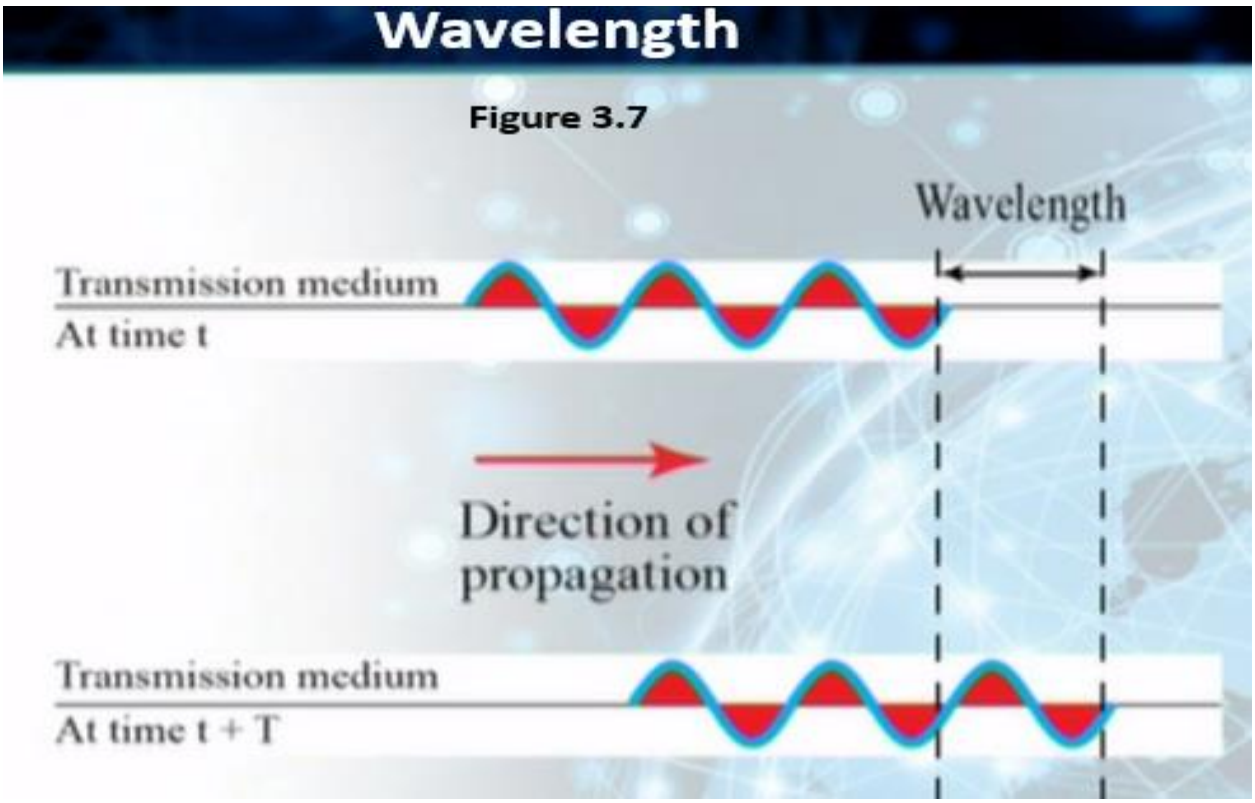
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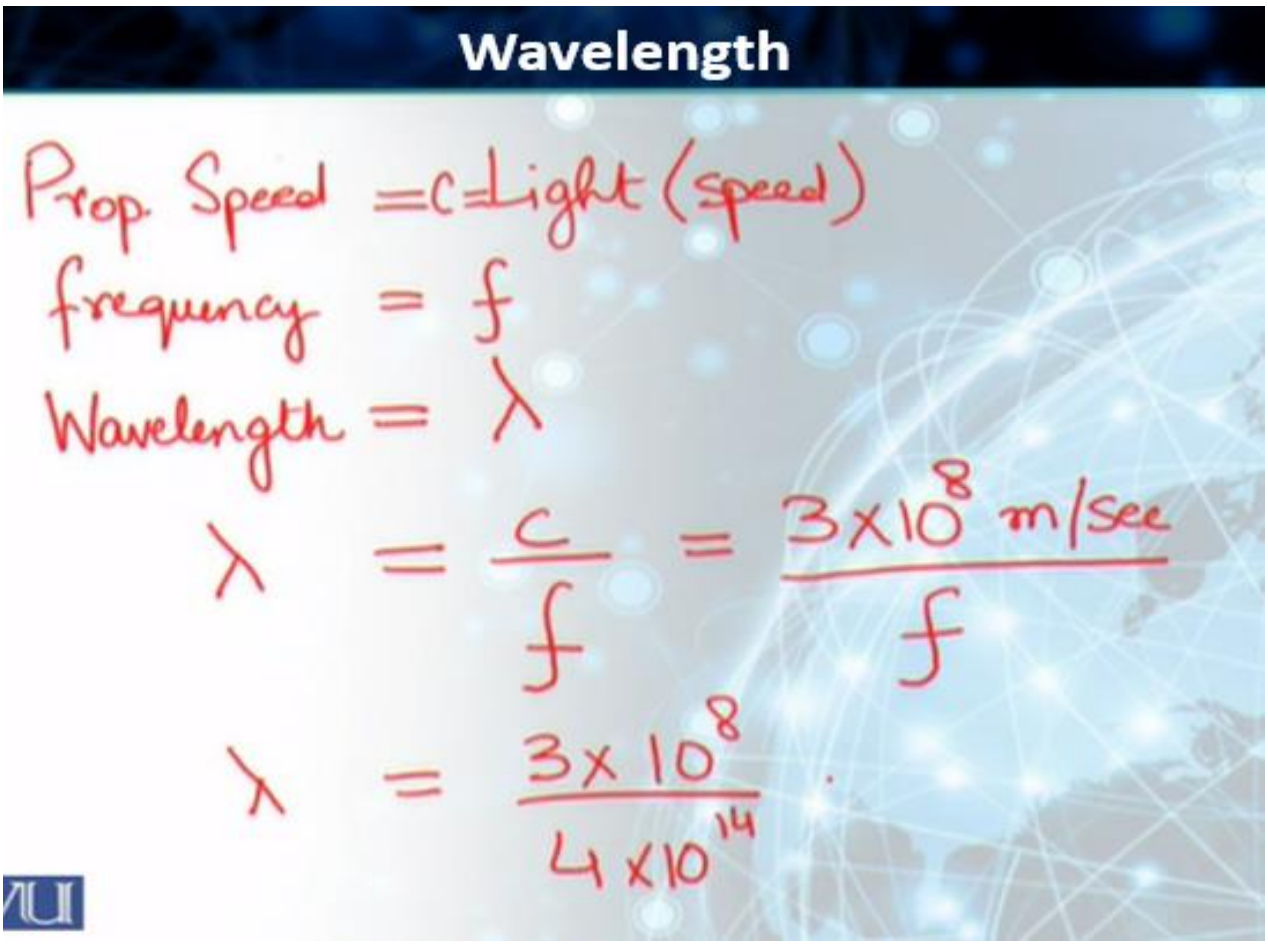
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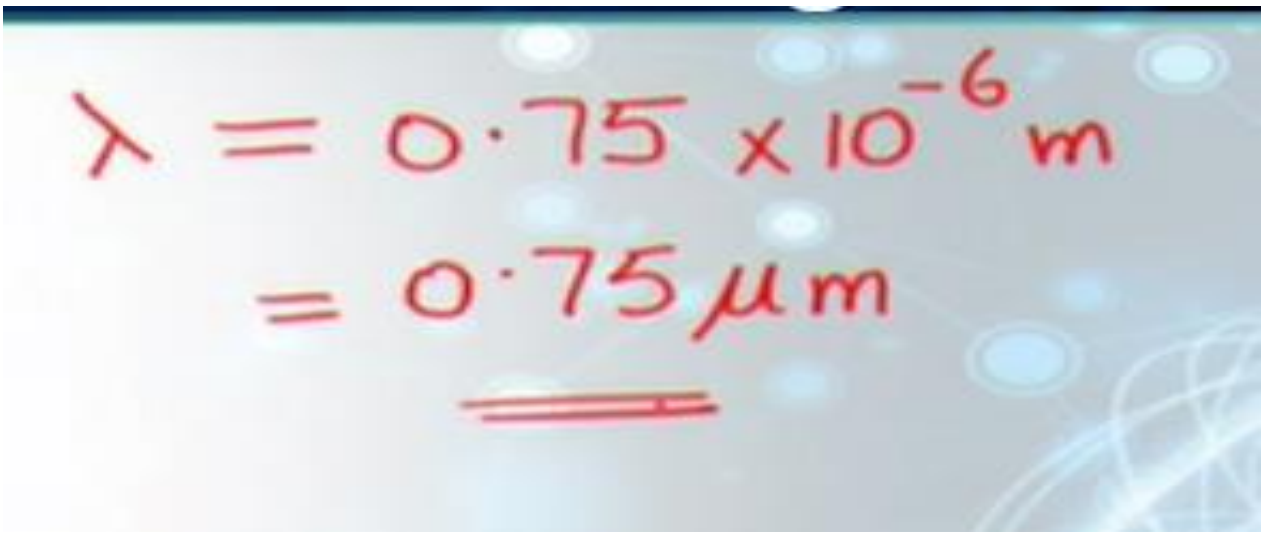
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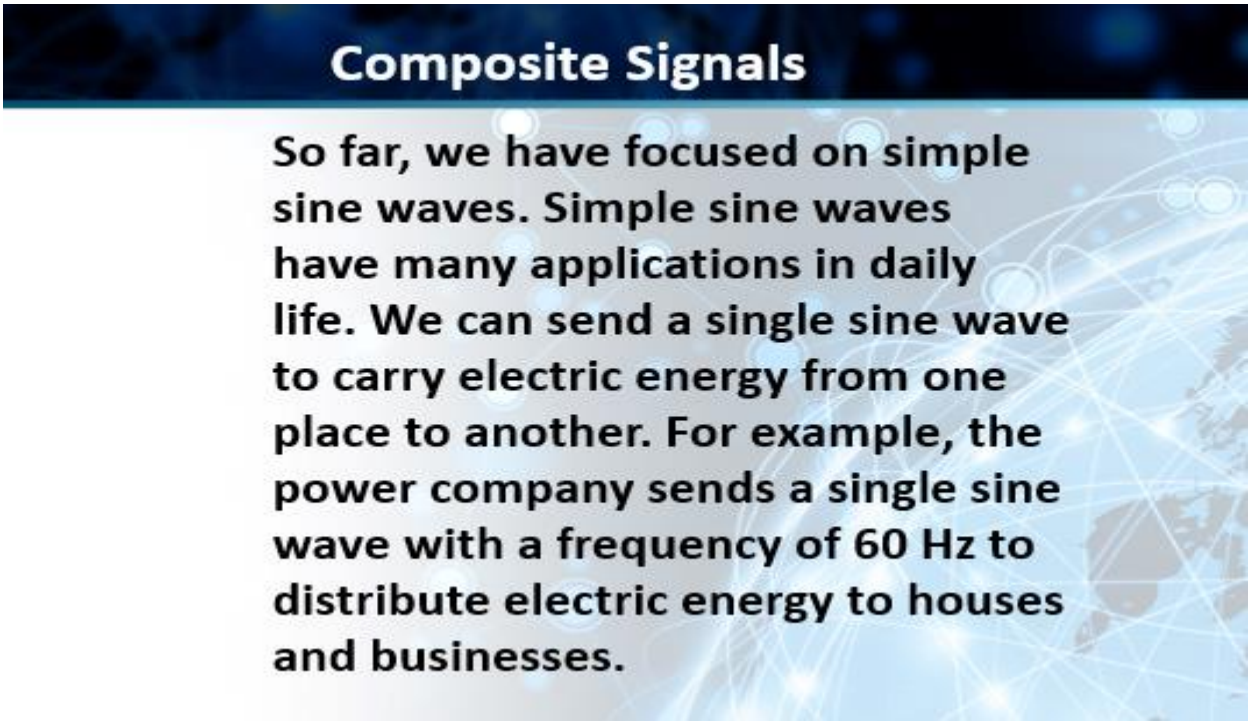


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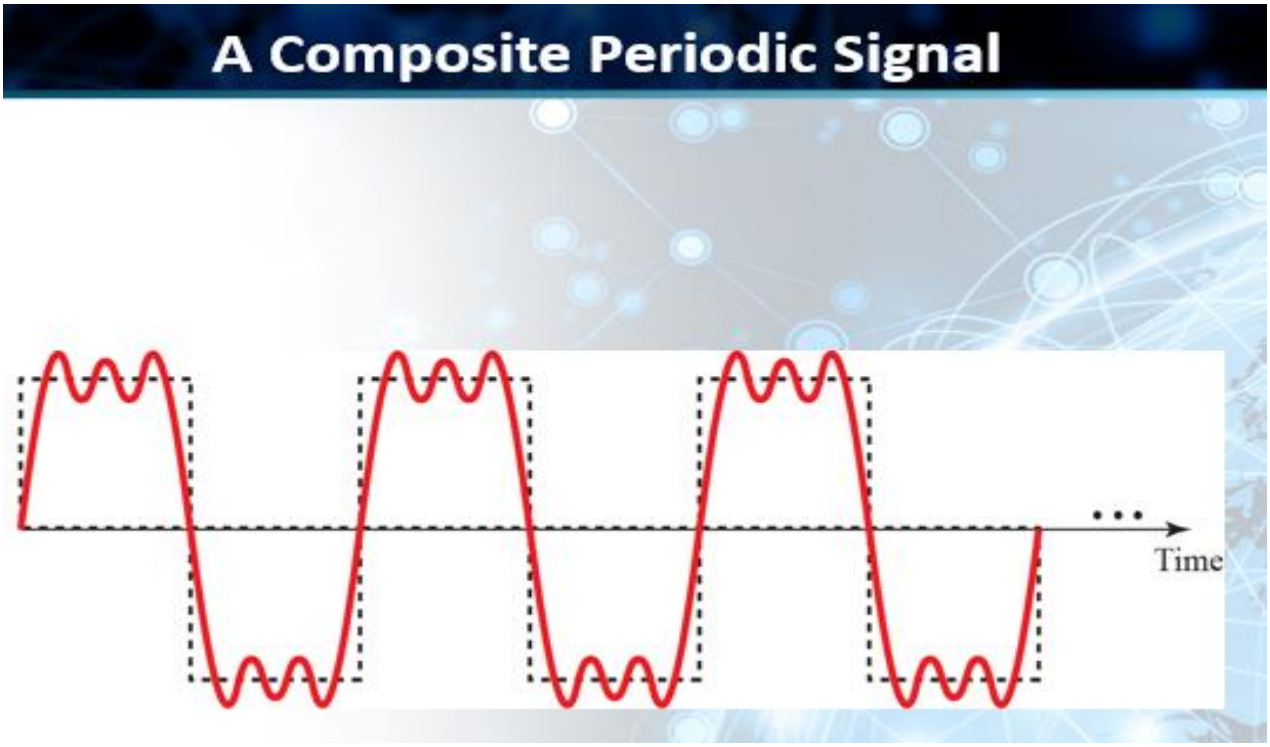
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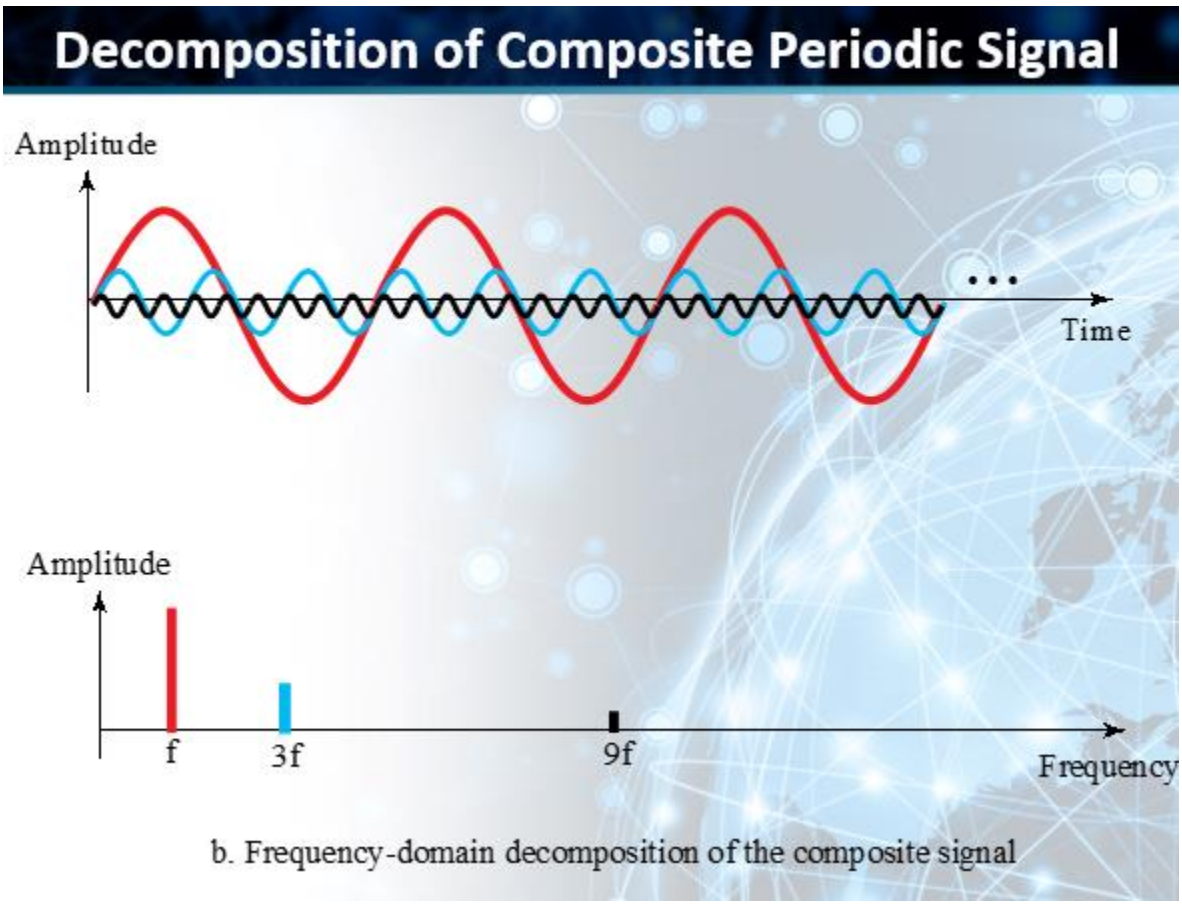
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## Bandwidth

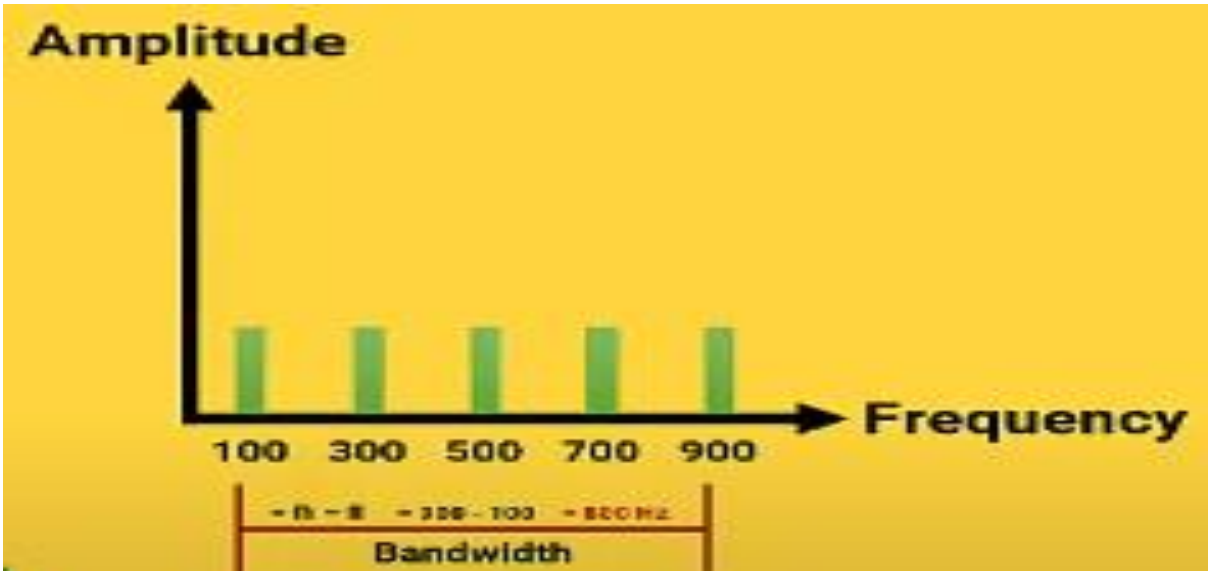
Bandwidth give us the range of frequencies contained in a composite signal. It give normally between two frequencies (High and Low).

**How to Calculate the Bandwidth:**

To calculate Bandwidth, subtract the lowest frequency from the highest frequency.

**Example:**

If a periodic signal is decomposed into five sine waves with frequencies 100, 300, 500, 700, and 900 Hz, what is the Bandwidth?



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## Example

**If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.**

$B = f_h - f_l$

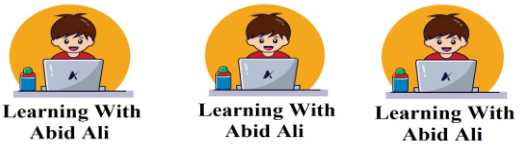
$= 900 - 100$

$B = 800\,Hz$



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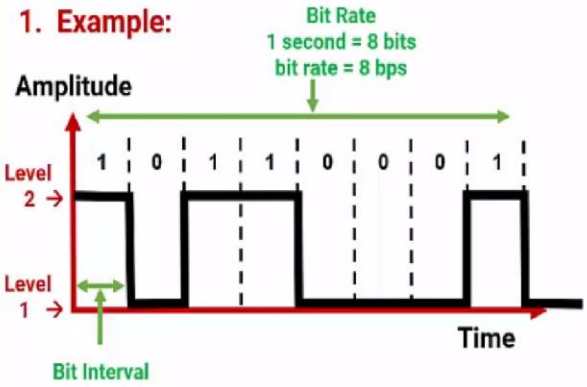



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## Digital Signal – Non-Periodic Signals

A signal in which the original information is converted into bits (0s Zero & 1s Positive) before being transmitted. And Digital Signal have more than 2 level.

**1. Example:**




**Bit Interval (Seconds):** The time required to send one signal bit.

**Bit Rate (bps):** Number of bits sent per second.

**Bit Length:** The distance one bit occupies on transmission medium.

**Signals:** It generate L level signals. If generate 1 level than $\log^2 2$. and if we have 4 level than $\log^2 4$.

**Example 1:** A digital signal has 8 Level. How many bits are needed?

**Formula:** No. of Bits = $\text{Log}_2 L$ = $\text{Log}_2 8$ = 3 bits

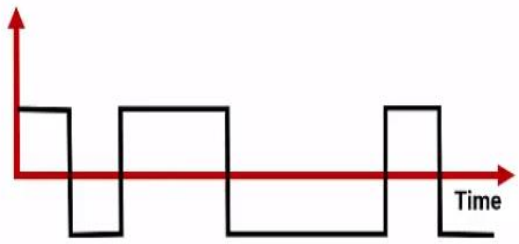
**Example 2:** Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel? 1 Page contained 24 line and 1 line contain 80 character and 1 character contain 8 bits.

**Formula:** No. of Pages x No. of Lines x No. of Character x No. of Bits

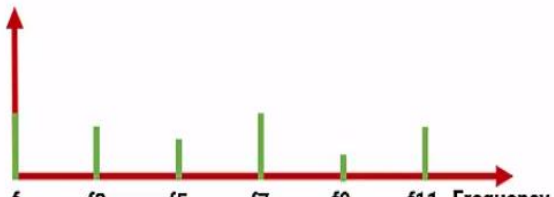
= 100 x 24 x 80 x 8 = 1.536 Mbps

## Digital Signals as Composite Analog Signal

- A method is used for converting the digital signals into composite analog signals that is Fourier Analysis. It has infinite Bandwidth.
- Vertical line in the time domain: Frequency of infinity
- Horizontal line in the time domain: Frequency of zero




Time Domain Plot




Frequency Domain Plot

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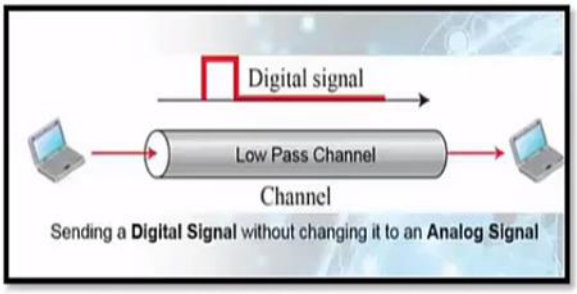
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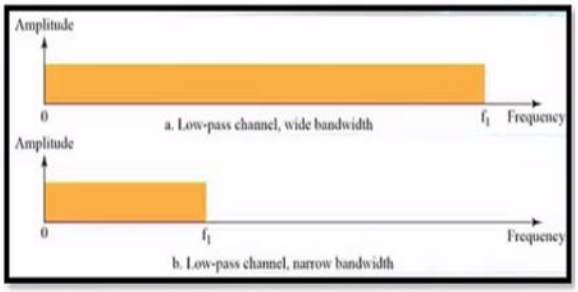
# Transmission of Digital Signals

We have already study that when the digital signal seen as composite signal it has infinite bandwidth and it has frequency from ZERO to ONE. To Transmit such signals we need two methods.

- **Baseband Transmission Digital Signal to Digital Signals**

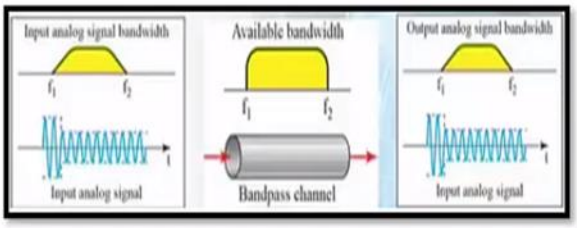


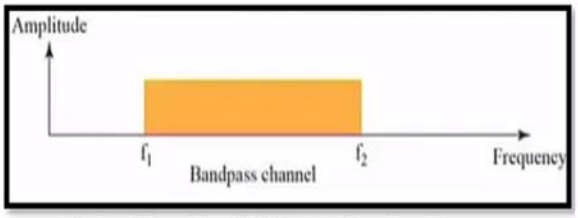

The output of the digital signal is NOT changed.




The low Pass Chanel started from Zero to Infinite Frequency
For this Transmission we need dedicated link (**Bus Topology**)

- **Broadband Transmission (Modulation) Digital Signal to Analog Signals**



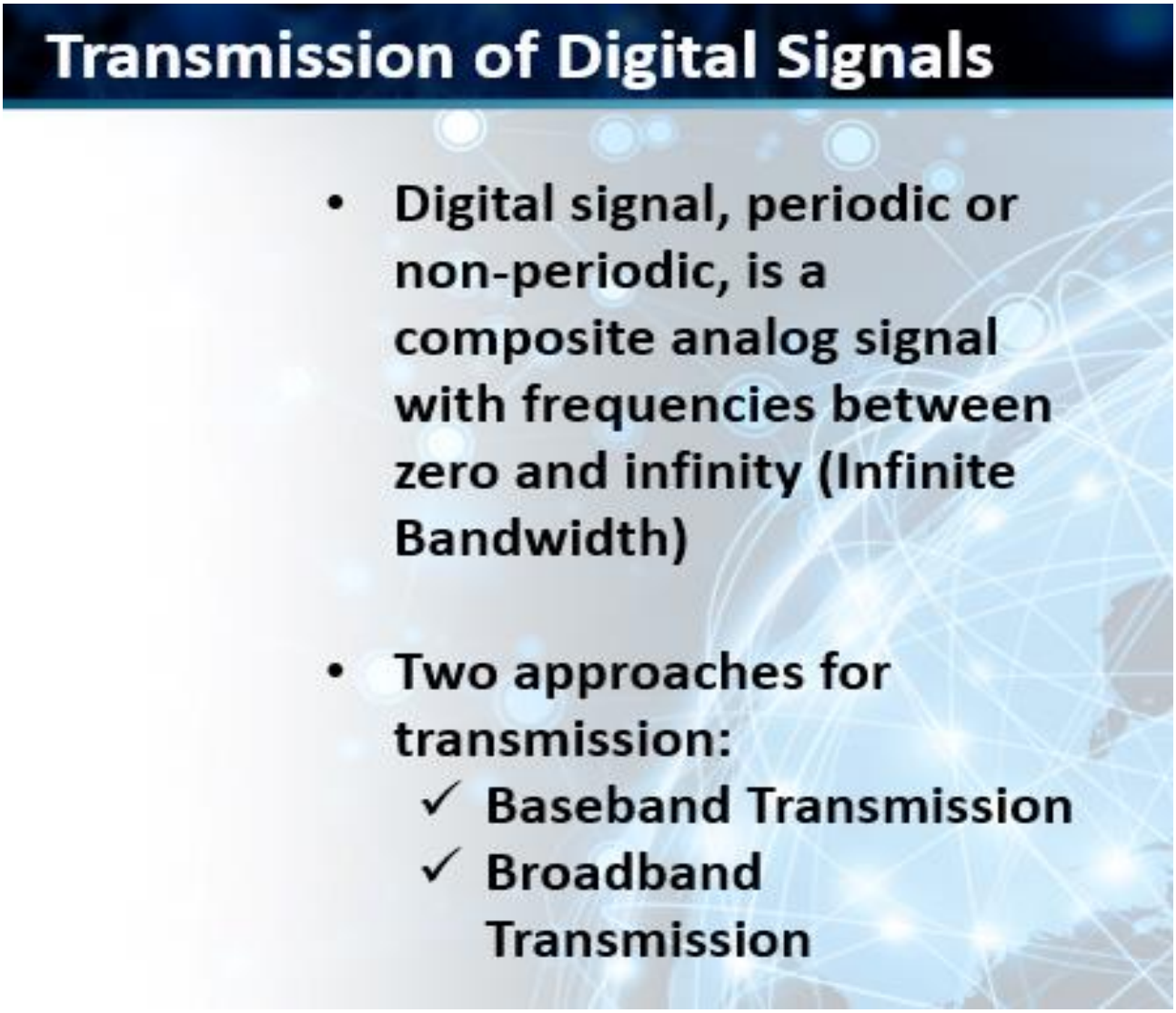




The band Pass Chanel NOT started from Zero Frequency

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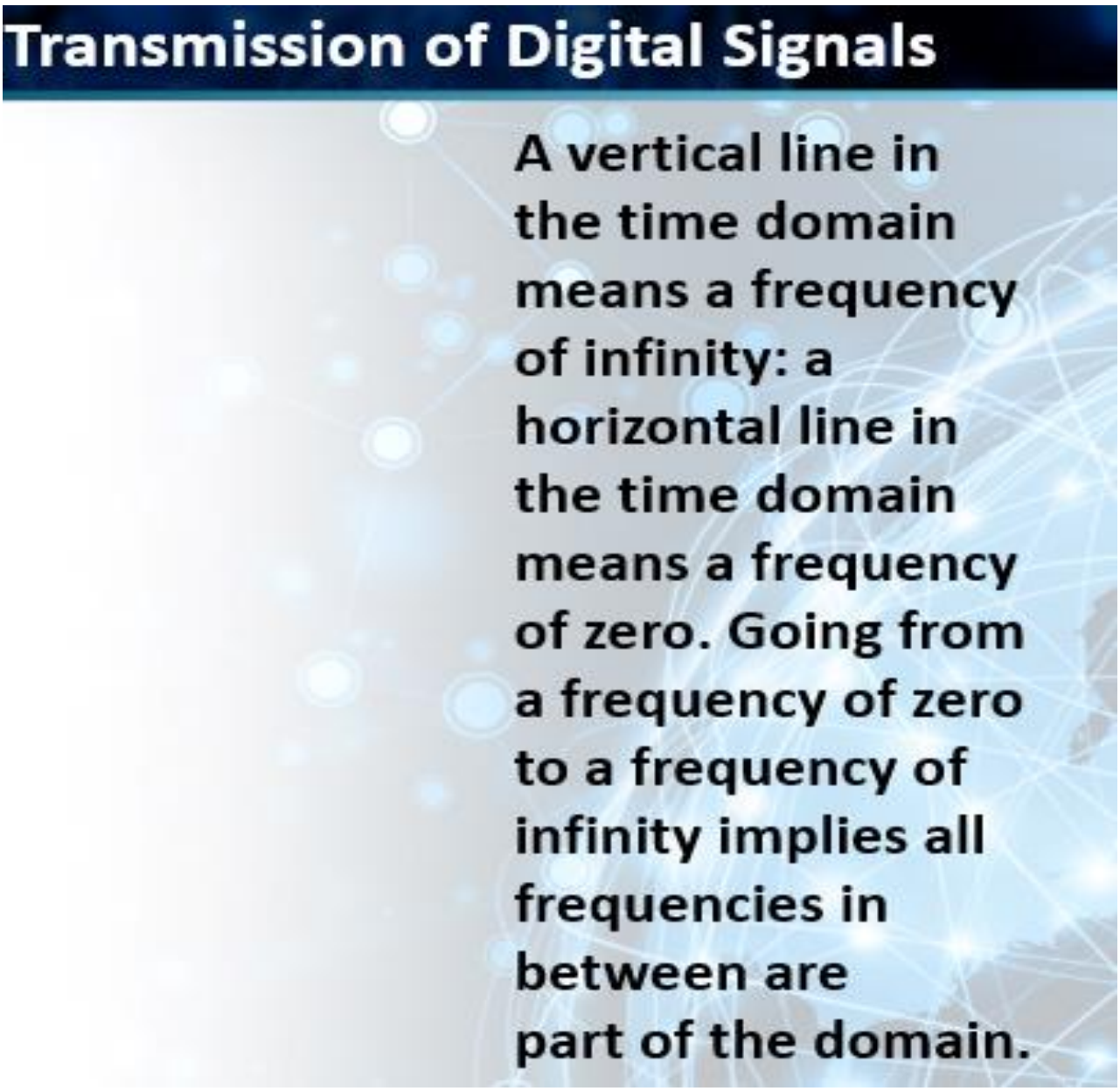
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# Transmission Impairments

When we are sending signals from one point to another point the signals has lost some thing. These Losses can be heat, external factors etc. These in perfect signals are known as Transmission Impartments.

## Causes of Transmission Impairments

Following factors affect on Transmission Impairments.

1. Attenuation
2. Distortion
3. Noise

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## Attenuation | Distortion | Noise

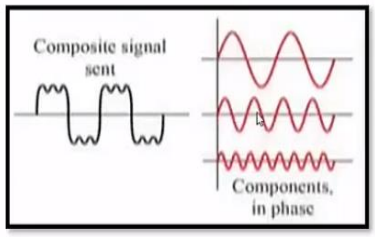
**1. Attenuation** – Loss of Energy. It cannot be recovered. (Heat)

The signals Strength can measured in terms of **Decible (dB).** We can measure it at two Pints (Point 1 and Point 2).

**Formula:** 10 $\text{Log}_{10}$ P2 / P1

If Decible is Negative its mean Signals has Attenuated and Lost Strength.

**2. Distortion** – is change in the form or shape of signal. (Delay)




### Attenuation Example

Suppose a signal travels through a transmission medium and its power is reduced to one half. Calculate the attenuation?

**Formula: 10 $\text{Log}_{10}$ P2 / P1**

= 10 $\text{Log}_{10}$ P2 / P1

= 10 $\text{Log}_{10}$ (0.5 P1) / P1

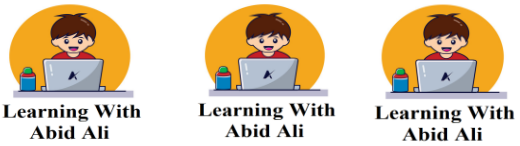
= 10 $\text{Log}_{10}$ (0.5)

**= -3 dB**

= Signal has lost one half of the Power.

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## Noise Example

The signal power is 10 mW (Milli Wats) and noise power is 1 uW (Micro Wats). What is SNR?

**Formula = Avg. Signal Power / Avg. Noise Power**

**SNR = 10 mW / 1 uW**

= 10,000 Milli Wats

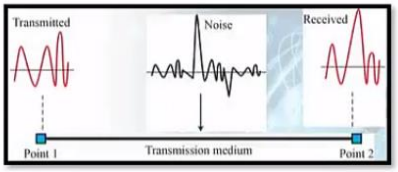
**$SNR_{dB}$ = 10 $Log_{10}$ 10,000**

= $40_{dB}$

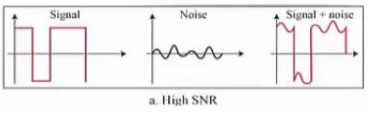
## Attenuation | Distortion | Noise

**3. Noise** – Noise Corrupt Signal

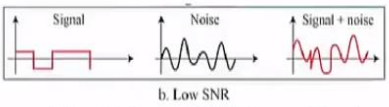
1. Thermal Noise 2. Induced Noise 3. Crosstalk Noise




## Signal to Noise Ratio (SNR)




1. High SNR mean Low Corrupt Signal




2. Low SNR mean High Corrupt Signal

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**Both Formulas are used to measure Data Rate Limits. Shannon's formula gives us the upper limit while the Nyquist formula gives us the signal levels.**

## Data Rate Limits

- Data Rate Limits mean how much data can be transfer over transmission medium or channel. It can be depend on three factors:
  - Band Width Availability
  - Level of Signals
  - Level of Noise (Quality of Channel)
- The Data Rate Limits can be calculate by two methods:

**1. One is provided by Nyquist (Noise = 0)**

**Formula:** Bit Rate = 2 x Bandwidth x $\text{Log}_2$ L

**Example:** A Noiseless channel with a bandwidth of 3,000 Hz transmitting a signal with four signal levels. What is the maximum bit rate of the said channel?

Bit Rate = 2 x 3,000 x $\text{Log}_2$ 4

= 6,000 x $\text{Log}_2$ 4

= 6,000 x 2

Bit Rate = 12,000 bps

- **Second is provided by Shanon (Noise > 0)**

**Formula:** Capacity = Bandwidth x $\text{Log}_2$ (1 + SNR)

**Example:** Theoretical highest bit rate of a Telephone line Bandwidth of 3000 Hz assigned for data communication. SNR is usually 3162. What is capacity?

Capacity = Bandwidth x $\text{Log}_2$ (1 + SNR)

= 3,000 x $\text{Log}_2$ (1+3162)

= 3,000 x $\text{Log}_2$ (3,163)

Capacity = 34,881 bps

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## Network Performance

The network performance is based on three factors. Which includes:

- **Bandwidth**

Bandwidth can be measure in two Terms:

1. Bandwidth in Herts (Range of Frequencies in a Composite Signals)
2. Bandwidth in Bits Per Second (No. of Bits Transmits on Channel or Network)

- **Throughput**

The throughput is how fast we can actually send data through a network.

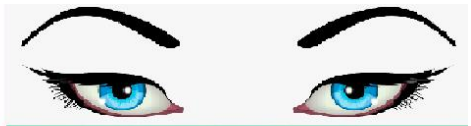
1. Throughput T bps is always less than Bandwidth B.

- **Latency - Delay**

The time period to sent a signal from source to destination. Its depend on

Propagation Time + Transmission Time + Queuing Time + Processing Time

$$\frac{\text{Distance}}{\text{Propagation Speed}} \quad \frac{\text{Message Size}}{\text{Bandwidth}} \quad \text{Message Time Hold}$$



### Throughput Example

A network with bandwidth of 8 Mbps can pass only an average of 10,000 frames per minute with each frame carrying an average of 8,000 bits. What is the throughput of this network?

**Formula: No. of Frames in per Minute x No. of Bits per Second**

Throughput = 10,000 x 8,000

= 80,000,000 / 60

**= 1,333,333.33 bit per second**

**Throughput = 1.33 Mbps**

**Bandwidth = 8 Mbps**

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# Example Propagation Time

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be $2.4 \times 10^8$ m/s in cable.

## SOLUTION

Propagation Time + Transmission Time + Queuing Time + Processing Time

$$\frac{\text{Distance}}{\text{Propagation Speed}} \quad \frac{\text{Message Size}}{\text{Bandwidth}} \quad \text{Message Time Hold}$$

**Propagation Time = Distance / Propagation Speed**

$= 12{,}000 \times 10{,}000 / 2.4 \times 10^8$

$= 120{,}000{,}000 / 2.4 \times 100{,}000{,}000$

$= 120{,}000{,}000 / 240{,}000{,}000$

$= 0.5$

**= 50 m/s**

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## Exercise

**Question No. 1:**

A network with bandwidth of 15 Mbps can pass only an average of 20,000 frames per minute with each frame carrying an average of 14,000 bits. What is the throughput of this network?

**Question No. 2:**

What is the propagation time if the distance between the two points is 15,000 km? Assume the propagation speed to be $3.4 \times 10^{10}$ m/s in cable.

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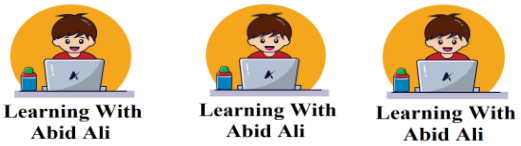
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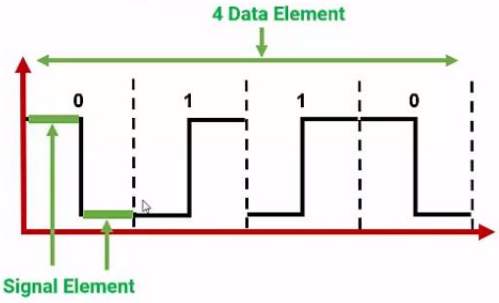

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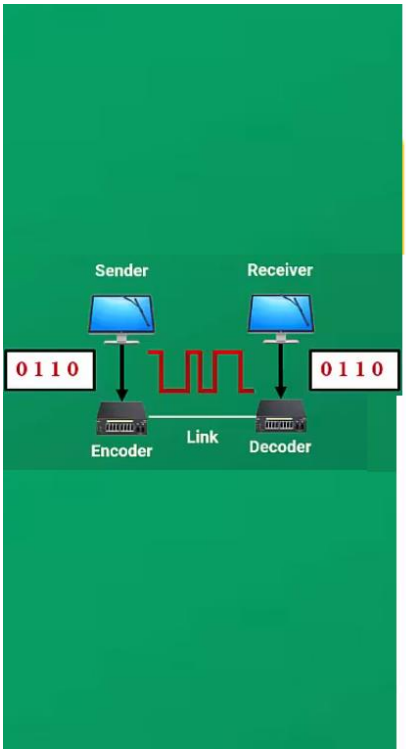
## Digital Transmission

- There are two types of signals that can be transmitted. These transmitted signals are known as Digital Transmission. These two digital transmission are:
- Digital to Digital Conversion
- Analog to Digital Conversion
- First of all we discussed Digital to Digital Conversion. Three methods are used for this conversion:
- Line Coding — Always Needed
- Block Coding — May NOT needed
- Scrambling — May NOT needed

**Data Element:** A smallest unit which carry some piece of Information (Bits) → Carried

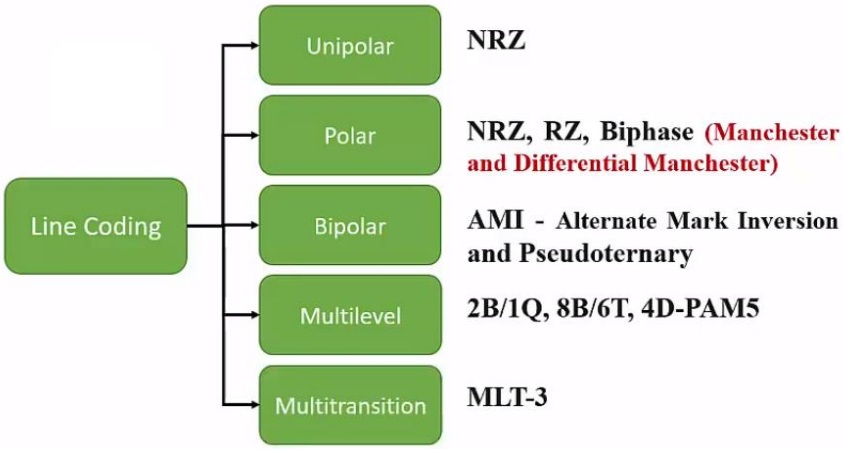
**Signal Element:** A shortest unit in terms of time of the digital signal. → Carriers




## Line Coding

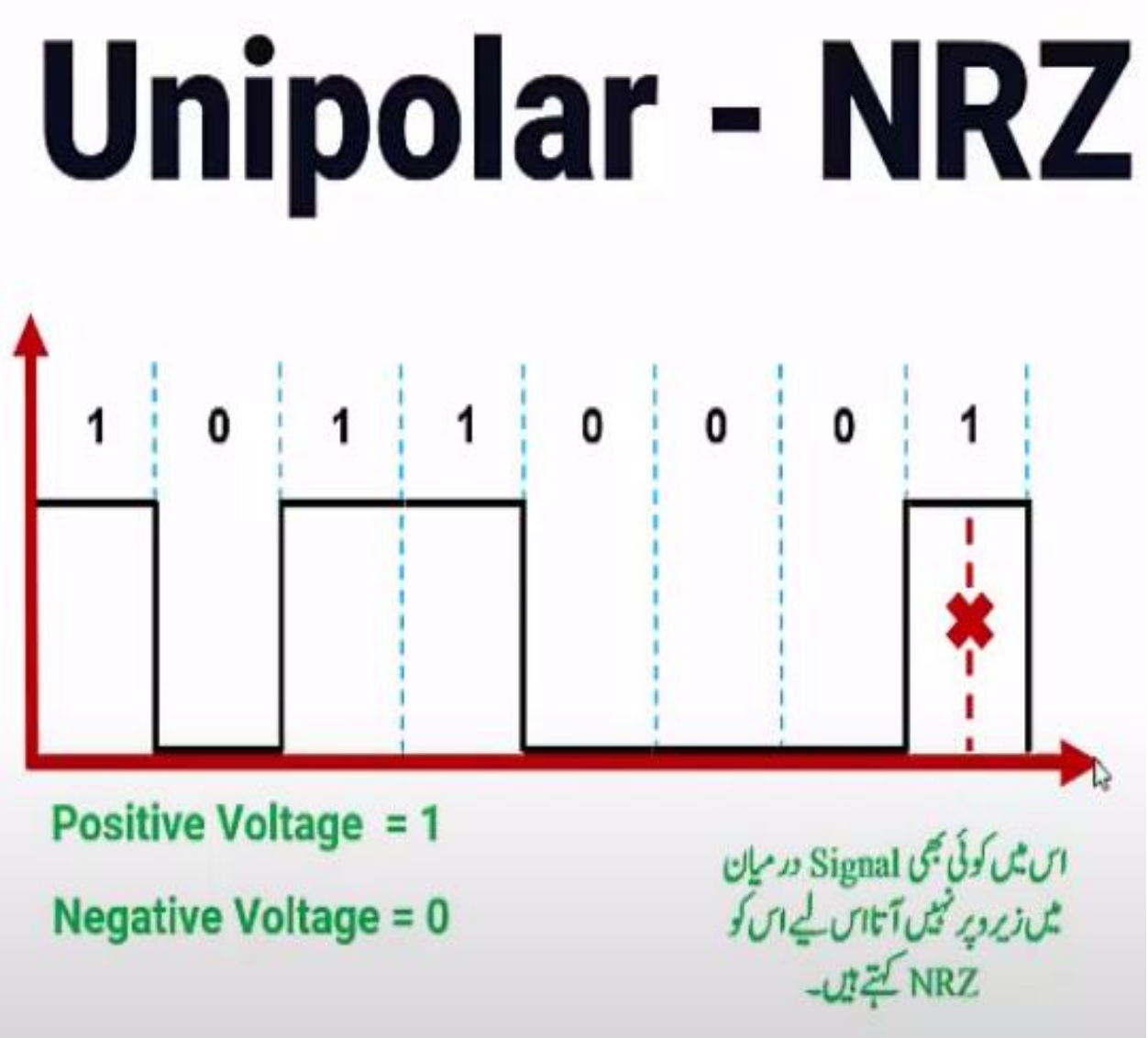
Data can be Text, Numbers, Images, Audio or Video store into the computer in forms of bits in a sequence. In Line Coding these bits converted into Digital Signals. There are multiple techniques for line coding these are five in numbers.



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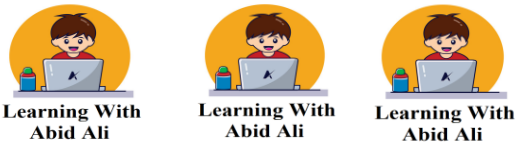
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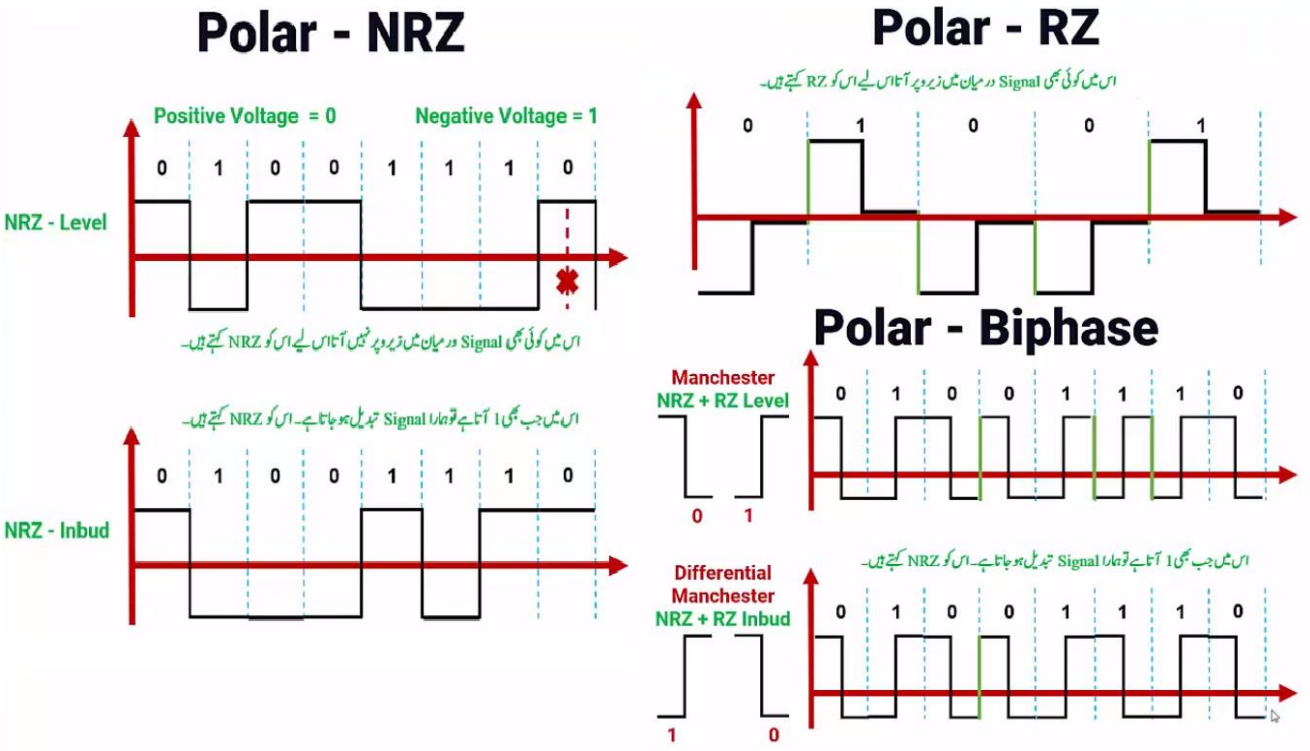


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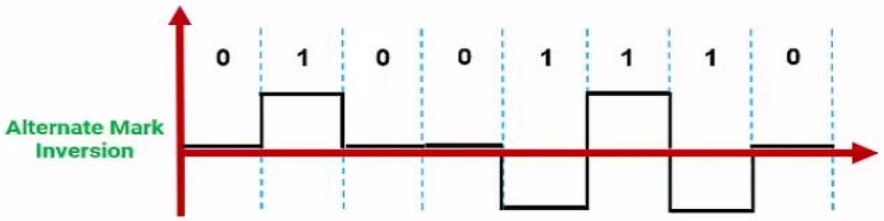


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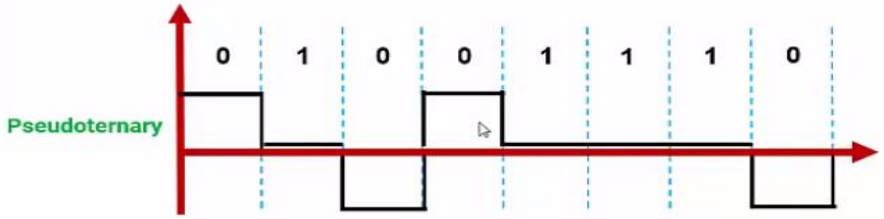
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## Bipolar - AMI




اس میں جب 1 آتا ہے تو ایک بار Positive ہوتا ہے اور ایک بار Negative ہوتا ہے۔

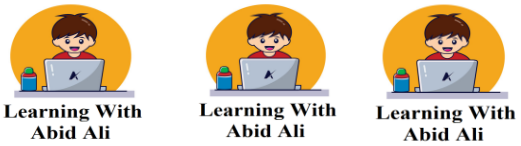
## Bipolar - Pseudoternary




اس میں جب 0 آتا ہے تو ایک بار Positive ہوتا ہے اور ایک بار Negative ہوتا ہے۔

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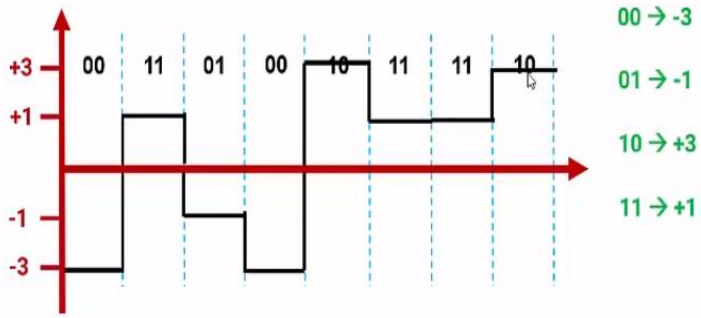



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# Multilevel - 2B/1Q (Two Binary and One Quaternary)

The objective of this method is to increase the data rate and decrease the required bandwidth.

اس Technique میں ڈیٹا کی 2 Bit ٹرانسفر ہوتی ہیں۔



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# Today's Topic

1. Block Coding
    a) Division
    b) Substitution
    c) Combination
2. Scrambling
    a) Types of Scrambling
        i) B8ZS
        ii) High Density HDB3

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## Block Coding (Error Detection)

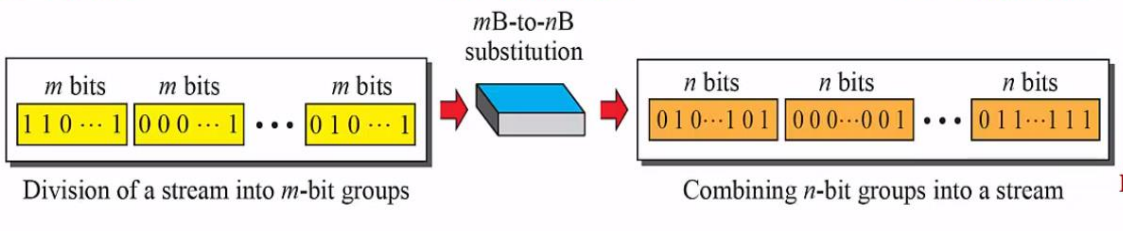
**Block coding refers to the technique of adding extra bits to a digital word in order to improve the reliability of transmission.**

When we are using Line Coding we not know where to start long ZEROs "0s" and where from start long Ones "1s". This problem is known as synchronization. To over the said problem we use second technique is known as Block Coding.

In this method we convert the "m" bits into "n" bits. These n bits is less than the m n < m. Its tell about where is long series of ZEROs "0s" and where from start long series of Ones "1s". We add some extra bit with original bits known as redundancy bits.

Three Steps of Block Coding:

- Division
- Substitution
- Combination




Division of a stream into *m*-bit groups

Combining *n*-bit groups into a stream

Redundancy
n > m

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## Block Coding with Line Coding

We have select 4B / 5B scheme.

**Example:**

**We need to send data at a 1 Mbps rate. What is the minimum required bandwidth, using a combination of 4B/5B and NRZ-I or Manchester coding?**

**Block Code Resolve:**

Synchronization → ✓

Bandwidth (NRZ-I) < Bandwidth (Manchester) ✓

DC Components → X

## SOLUTION

M = 4

N = 5

n > m

Data Rate

4 Bit = 1Mbps

5 Bit = 1.25 Mbps

↑ Signal Rate Grow Up

NRZ-I (Minimum Bandwidth is half of the Data Rate)

Formula = N / 2

= 1.25 / 2 Mbps

= 62.5 Mbps

= 625 KHZ

8B / 10B Scheme

Manchester (1 MHZ)

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# DC components

- When voltage level in a digital signal is constant for a while, spectrum creates very low frequencies (around zero), called DC *components*
- *Creates problems for a* system that cannot pass low frequencies
- a telephone line cannot pass frequencies below 200 Hz.

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# Scrambling

- ✓ In **Scrambling** we increase the **No. of Pulses** instead of **No. of Bits**. Because in Block Coding when we increase the No. of Bits it will increase the data rate as well as it increase bandwidth. But in Scrambling when we increase Pulses it will increase data rate but not increase the bandwidth.
  - ✓ Biphase schemes suitable for Dedicated Link (LAN) but not for Long Distance
  - ✓ Block Coding + NRZ-I solves Synchronization issue but has DC Component
  - ✓ Bipolar AMI has a narrow bandwidth (no DC Component) but synch issue (long series of 0s)

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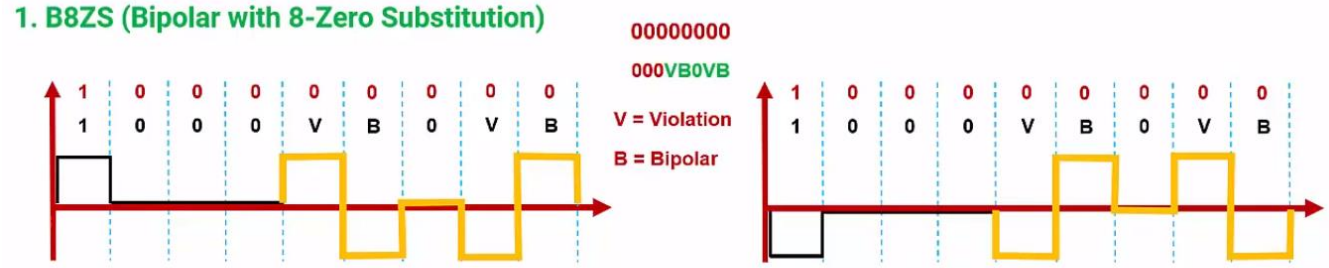
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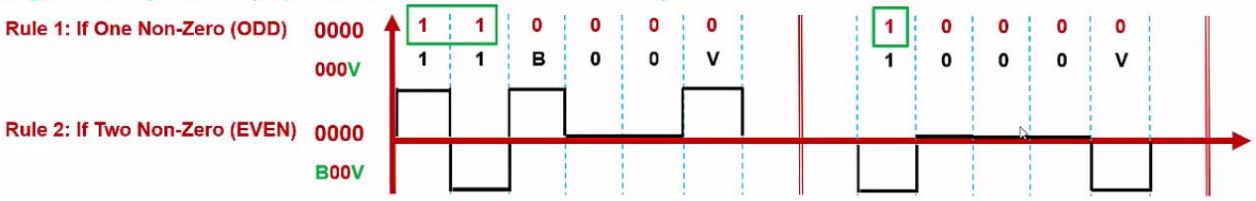

https://www.youtube.com/watch?v=-0kr2sG8JUQ&list=PLAUocDaOe9K6nDHAxKa-7fHsIFAKBPE8q&index=9

## Types of Scrambling Techniques

### 1. B8ZS (Bipolar with 8-Zero Substitution)




### 2. High Density HDB3 (Bipolar with 3-Zero Substitution)



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# Today's Topic

## Analog to Digital Conversion

1. (PCM) - Pulse Code Modulation
    - i) Sampling
    - ii) Quantization
    - iii) Encoding
2. (DM) - Delta Modulation

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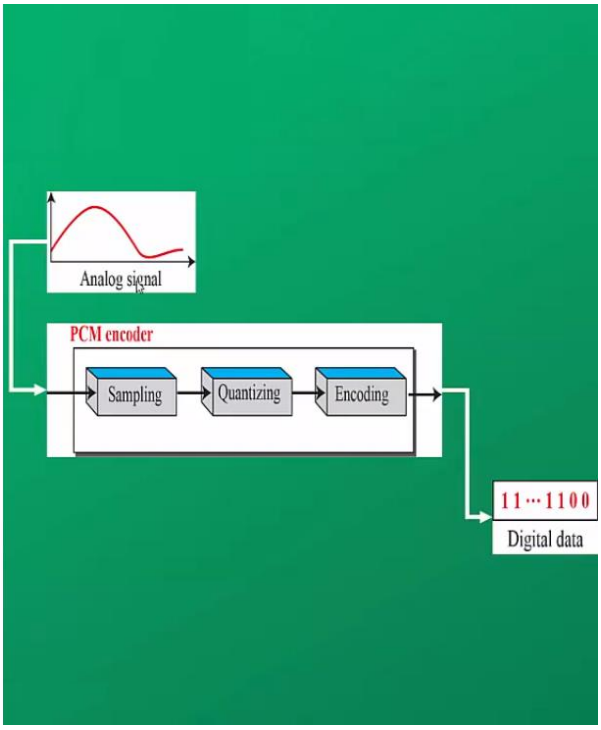



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## Analog to Digital Conversion

In this method we convert the analog signals to the digital signals. In this method first of all we convert the data into Digitization than convert it into Digital Signals. Its mean that two process are involved in it.

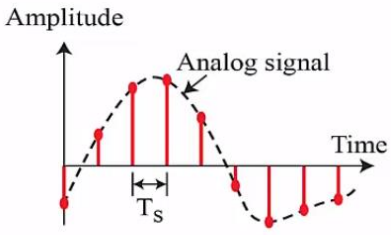
- PCM consists of three steps to digitize an analog signal:
  1. (PCM) - Pulse Code Modulation
     -- Sampling (Pulse Amplitude Modulation)
     -- Quantization
     -- Encoding
  2. (DM) - Delta Modulation



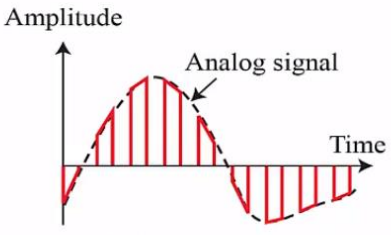

## Sampling

Series of pulses with amplitude values between min and max signal amplitude.

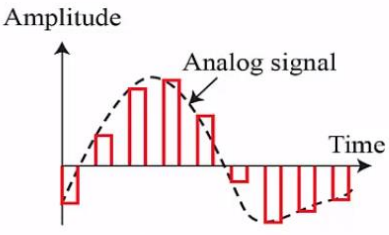
## Three Different Sampling Methods for PCM




a. Ideal sampling




b. Natural sampling




c. Flat-top sampling

TS → Sampled Interval or Sample Period

It has different pluses at different Amplitude. It is very difficult to deploy in Hardware.

It has high speed Switch. It has very small time interval

It is also called sample and hold technique.

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Sample is a piece of data taken from the whole data which is continuous in the time domain. When a source generates an analog signal and if that has to be digitized, having 1s and 0s i.e., High or Low, the signal has to be discretized in time. This discretization of analog signal is called as Sampling.

نمونہ پورے ڈیٹا سے لیا گیا ڈیٹا کا ایک ٹکڑا ہے جو ٹائم ڈومین میں مسلسل ہوتا ہے۔ جب کوئی ذریعہ ایک یعنی ہائی یا لو ہو، تو سگنل کو 1s اور 0s اینالاگ سگنل تیار کرتا ہے اور اگر اسے ڈیجیٹائز کرنا ہو، جس میں وقت کے ساتھ الگ کرنا ہوگا۔ ینالاگ سگنل کی اس ڈسکریٹائزیشن کو سیمپلنگ کہا جاتا ہے۔

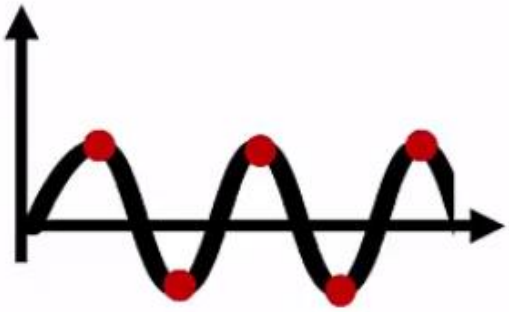
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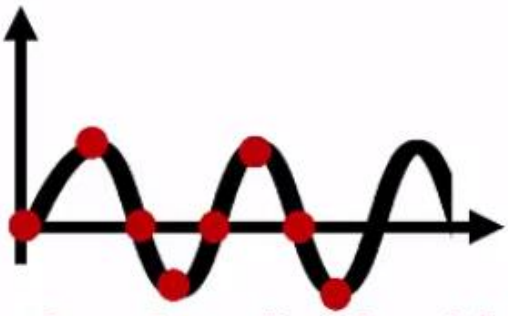
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# Nyquist Sampling Rate

Nyquist → fs (Frequency of Sampling) = 2fh (Twice Heights Frequency)



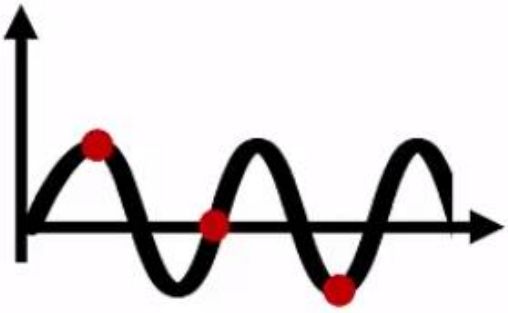
Ideal Sampling $f_s = 2\,f$



Over Sampling $f_s = 4\,f$

(Double Nyquist Rate)

Extra on Line Sample has no Value



Under Sampling $f_s = f$

(Half Nyquist Rate)

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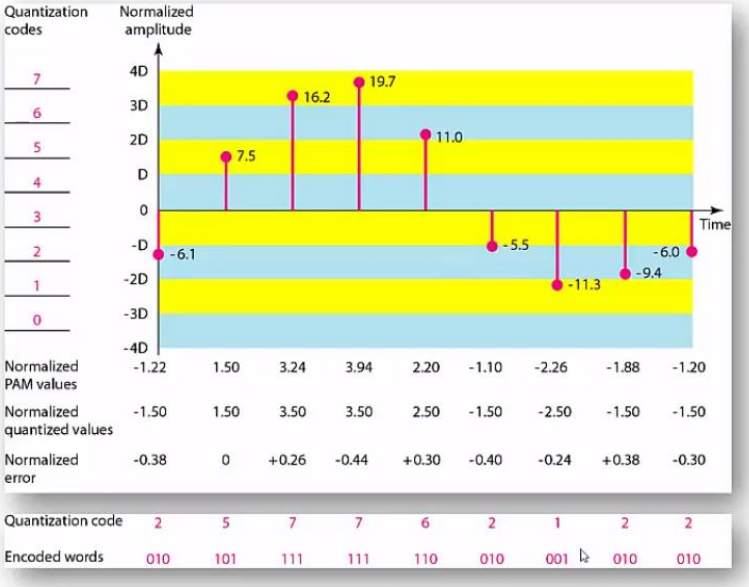

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## Quantization

Quantization is normalization of sample data according to certain level. The no. of level chose on two factors.

-- Range of Amplitude

-- Accuracy (If we acquire higher accuracy than we need more Quantization Level and Voice Versa).




## Encoding

The digitization of the analog signal is done by the encoder. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n bit code. Encoding also minimizes the bandwidth used.

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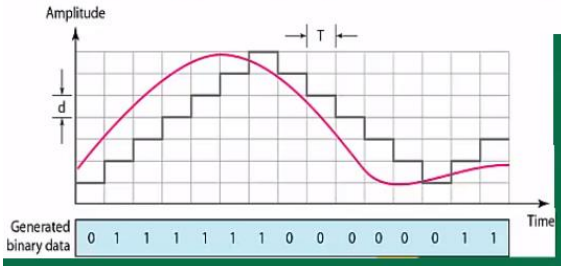



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## Delta Modulation

This scheme sends only the difference between pulses, if the pulse at time $t_{n+1}$ is higher in amplitude value than the pulse at time $t_n$,

- Then a single bit, say a "1", is used to indicate the positive value.
- If the pulse is lower in value, resulting in a negative value, a "0" is used.
- This scheme works well for small changes in signal values between samples.
- If changes in amplitude are large, this will result in large errors.




## Delta Modulation

- Delta modulation is a simpler technique
- DM finds the change from the previous sample
- No code words

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## 1. Transmission Mode

a) Parallel Transmission

b) Serial Transmission

i) Asynchronous Transmission

ii) Synchronous Transmission

iii) Isochronous Transmission

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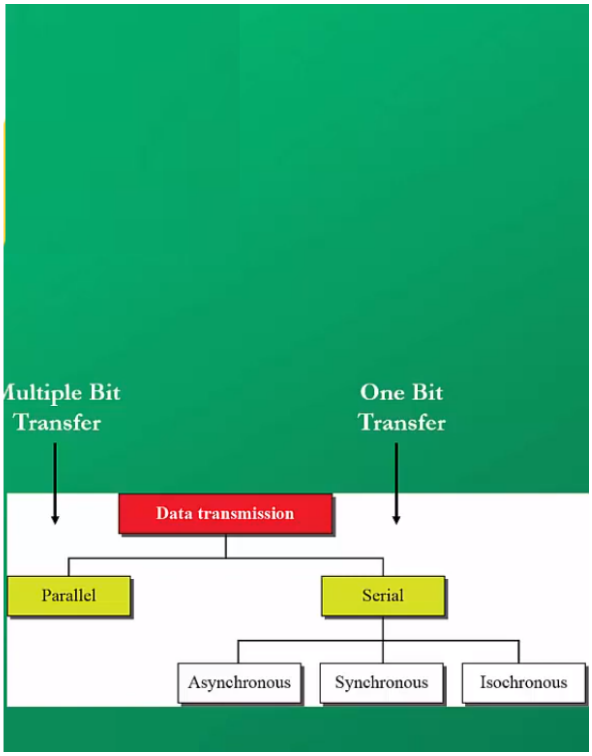



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## Transmission Mode

We have convert the data into digital data. Now, the process is to transmission that from source to destination. This data is transmitted through Transmission Mode. One problem is arises before transmitting data that is **"WIRING"**. We have used different types of wires to transmit the data. Its depends on no. of bits to transmit. We have one bit or multiple bits to transmit.

- We have two method to transmit data.
  1. Parallel Transmission
  2. Serial Transmission

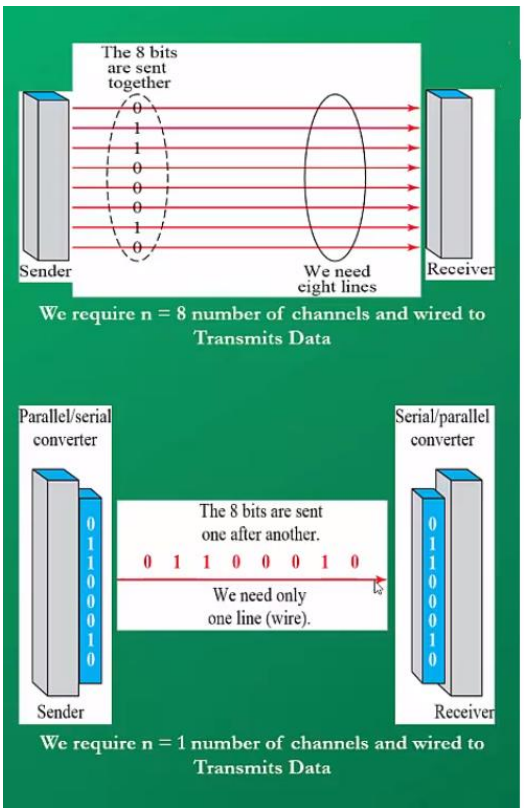


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## Parallel Transmission

In this type of Transmission Mode we have Transmit Multiple Bit to the destination at the same time.

**Advantages** → Speed (Data Transmit Quickly)

**Disadvantages** → Cost (We Require n[8] Wire)

## Serial Transmission

In this type of Transmission Mode we have Transmit Only One Bit to the destination.

**Advantages** → Cost (We Require n[1] Wire)

**Disadvantages** → Speed (Data Transmit Slowly)

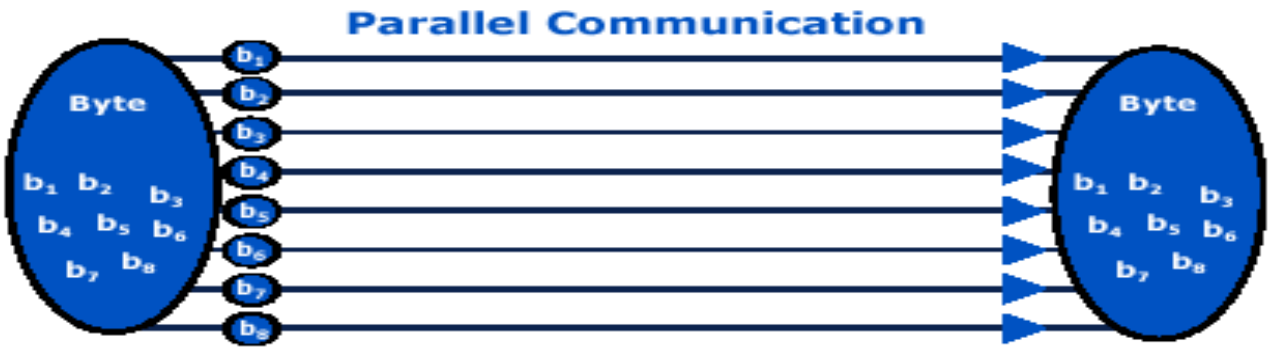
Serial Transmission: In this type of transmission a single link is used to transmit data bits and only one bit is transferred at a time. Parallel Transmission: In this transmission, data bits are transmitted simultaneously through multiple links, which are placed parallel to each other.

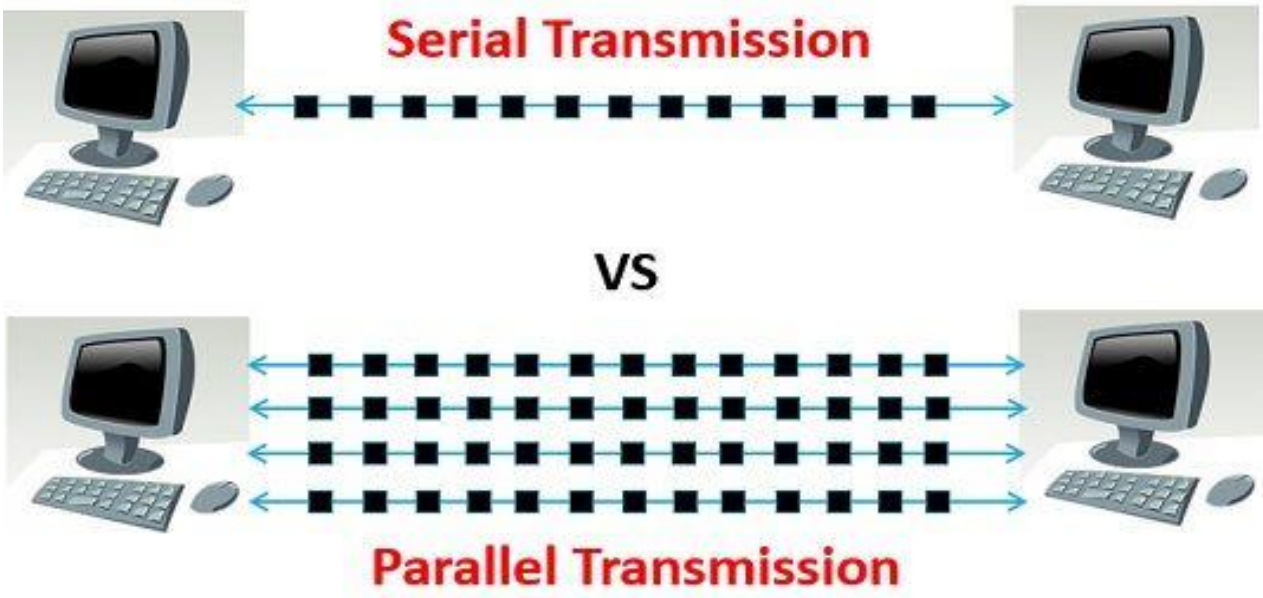
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## Asynchronous Transmission

In this type of Transmission Mode we have Transmit Bits on a specific pattern.

Advantages → Cost (Cheap)

Disadvantages → Speed (Slow)

Example → Keyboard input to PC

## Synchronous Transmission

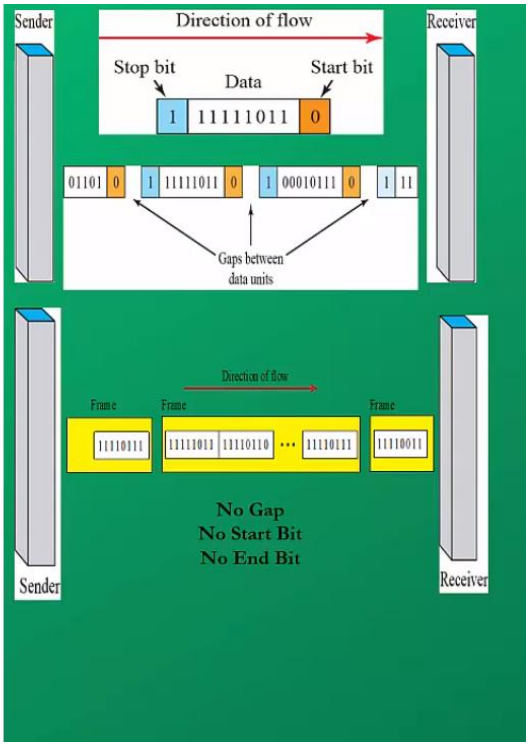
In this type of Transmission Mode we have Transmit data into large units called frame.

Advantages → Speed (No Gap)

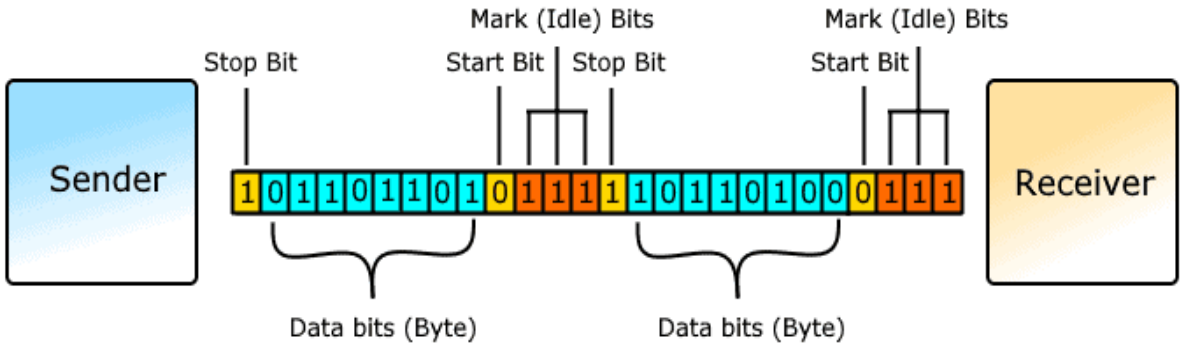
Example → High Speed Application

## Isochronous Transmission

In this type of Transmission Mode we have synchronized Entire bit of stream and then Transmit it.




The Isochronous Transmission Format is also known as **START-STOP mode or CHARACTER mode**. Each character or byte is framed as a separate and independent unit of DATA that may be transmitted and received at irregular and independent time intervals.



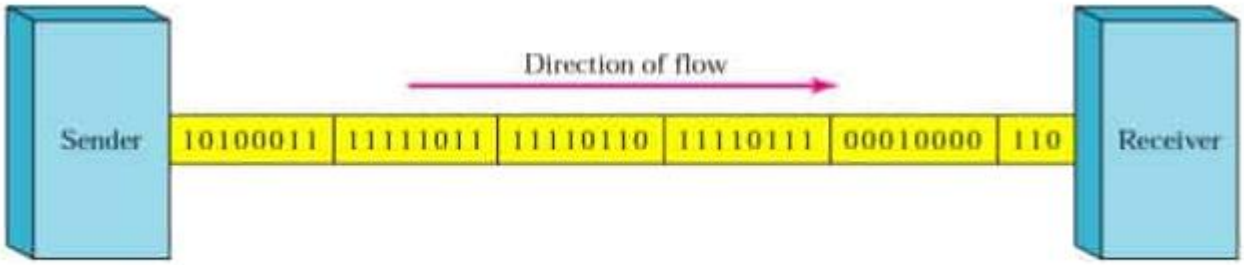

Asynchronous Transmission

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## Synchronous Transmission




| Synchronous Transmission | Asynchronous Transmission |
|---|---|
| Transmits data in the form of chunks or frames or blocks | Transmits 1 byte or character at a time |
| Transmission speed is faster | Transmission speed is slower |
| Cost is expensive | Cost is economical |
| Time interval is constant | Time interval is random |
| There are gaps between the data | There are no gaps between the data |
| Used in Chat Rooms, Telephonic Conversations, Video Conferencing | Used in Email, Forums, Letters |

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# Today's Topic

1. Transmission Mode of Analog Data

   a) Amplitude Shift Keying(ASK)

   b) Frequency Shift Keying(FSK)

   c) Phase Shift Keying (PSK)

      i) Bit Rate

      ii) Data Rate

      iii) Baud Rate

      iv) Signal Rate

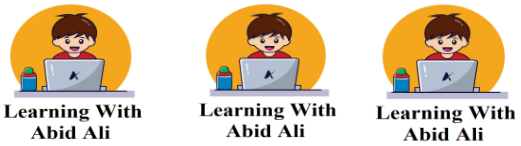
      v) Carrier Signal

      vi) No. of Data Element

      vii) High Frequency Signals

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## Digital-To-Analog Conversion

Digital-To-Analog Conversion (DAC), **Process by which digital signals (which have a binary state) are converted to analog signals** (which theoretically have an infinite number of states). For example, a modem converts computer digital data to analog audio-frequency signals that can be transmitted over telephone lines.

A sine wave is defined by 3 characteristics:

Amplitude

Frequency

Phase

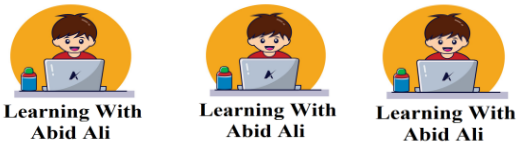
How do your data convert from digital-to-analog?

**There are three kinds of digital-to-analog conversions:**

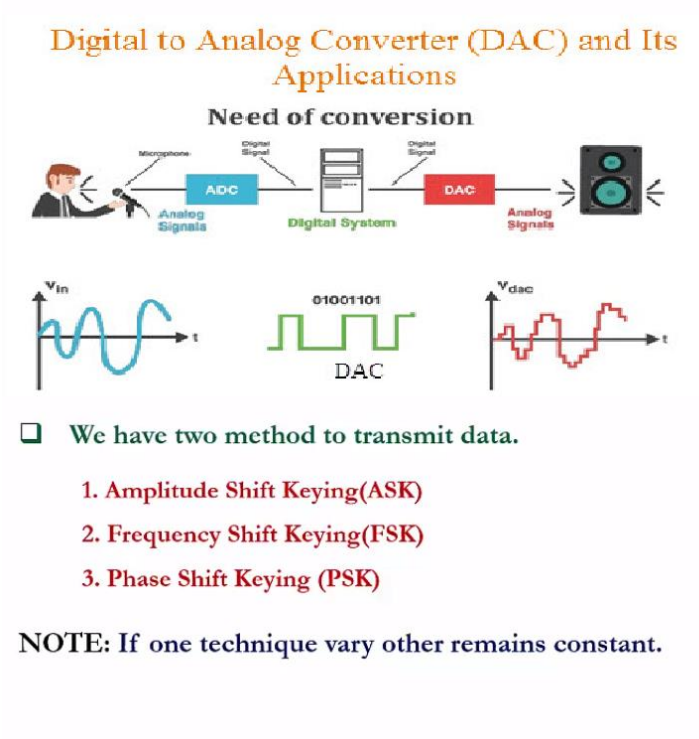
1. Amplitude Shift Keying. In this conversion technique, the amplitude of analog carrier signal is modified to reflect binary data. ...
2. Frequency Shift Keying. ...
3. Phase Shift Keying. ...
4. Quadrature Phase Shift Keying.

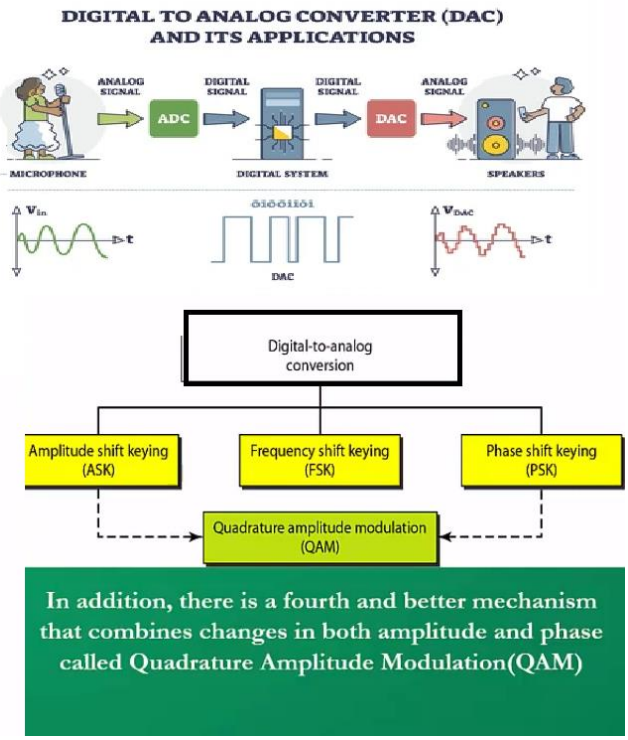
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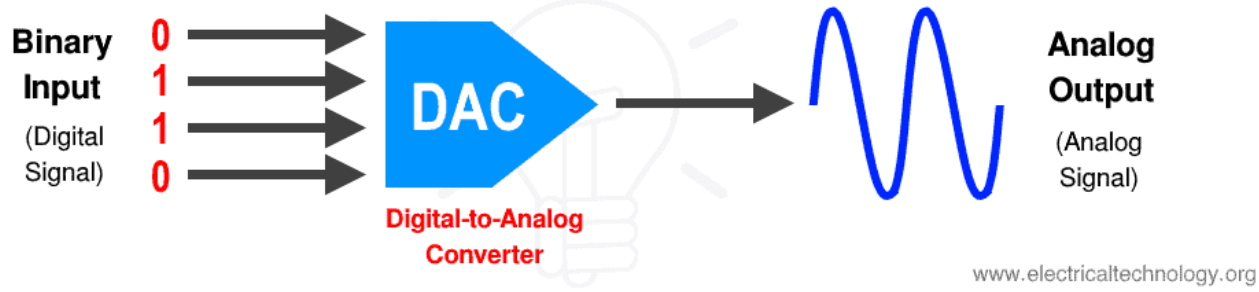
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# DAC

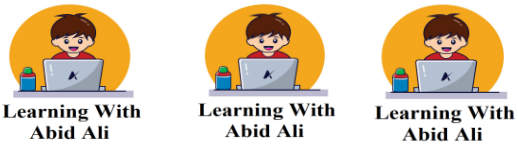
## DIGITAL TO ANALOG CONVERTER

Types, Working, Block Diagram & Applications



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An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate?

**SOLUTION**

**Formula: S = N / r**

S = 1000 R = 4 N = ?

S = N / r N = r x S N = 4 x 1000 **4000 pbs**

An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

**SOLUTION**

**Formula: S = N / r**

N = 8000 S = 1000 L = ? r = ?

S = N / r r = N / S r = 8000/1000 **8 bits**

$L = 2^r$ $L = 2^8$ **256 bits**

## Digital to Analog Conversion

In conversion from digital to analog signals we must review two issues.

1. Bit Rate → Data Rate
   No of Bits Transmitted in One Second
2. Baud Rate → Signal Rate
   No of Signal Units Per Second

} Baud Rate is less than Bit Rate

## Calculation of Bit & Baud Rate

**(S = N / r) Signal Rate = (Data Rate / No. of Data Element)**

$R = \log_2 L$ L = No. of Different Signal Elements

**Bandwidth Required → Its depend on Signal Rate as more signal required more bandwidth & vice versa. FSK**

**3. The Carrier Signal →**

When we transmit the signals from sender to receiver. Than we transmit **High Frequency Signals**, these are called **Carrier Signal**.

# Exercise

**Example # 01**

An analog signal carries 8 bits per signal element. If 2000 signal elements are sent per second, find the bit rate?

**Example # 02**

An analog signal has a bit rate of 16000 bps and a baud rate of 2000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

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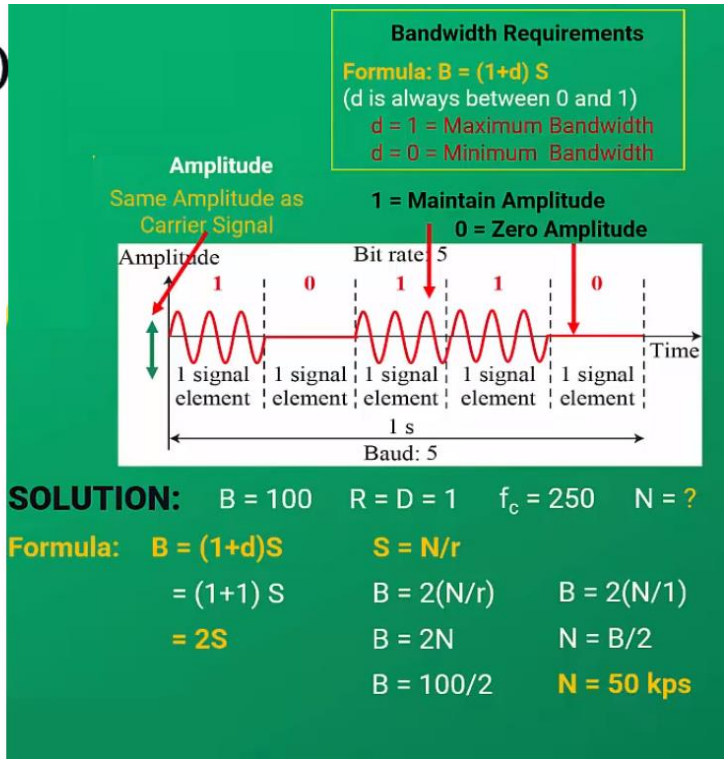

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## Amplitude Shift Keying (ASK)

In this technique the amplitude is vary and it vary with the carrier signals as the carrier signal change the amplitude changed.

- This technique is also known as (OOK - On Off Key) or Binary Technique (0's and 1's)
- When Amplitude change than Frequency and Phase remain (Constant).

**Example:** We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with d = 1?

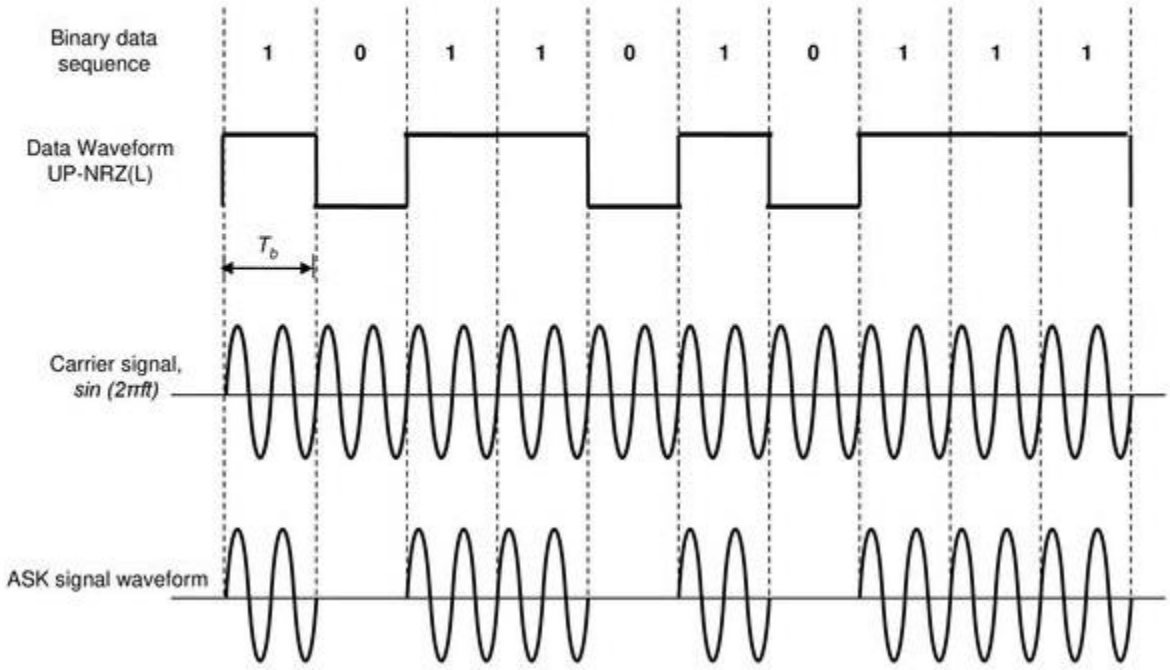


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## 3.1.1 Amplitude Shift Keying (ASK)

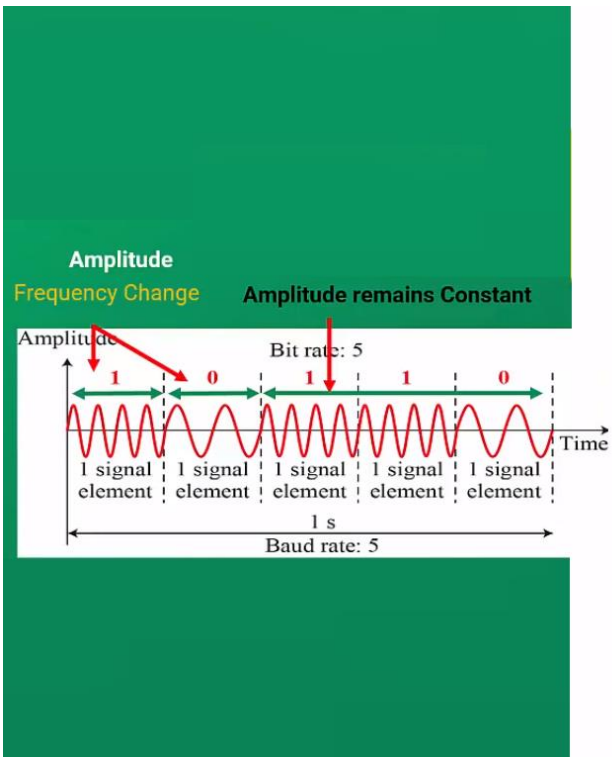


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## Frequency Shift Keying (FSK)

In this technique in which the frequency vary (change) but the Amplitude and Phase remain constant.

- Formula: $B = (1+d)\,S + 2\,\Delta f$

**Example:** We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with d = 1?

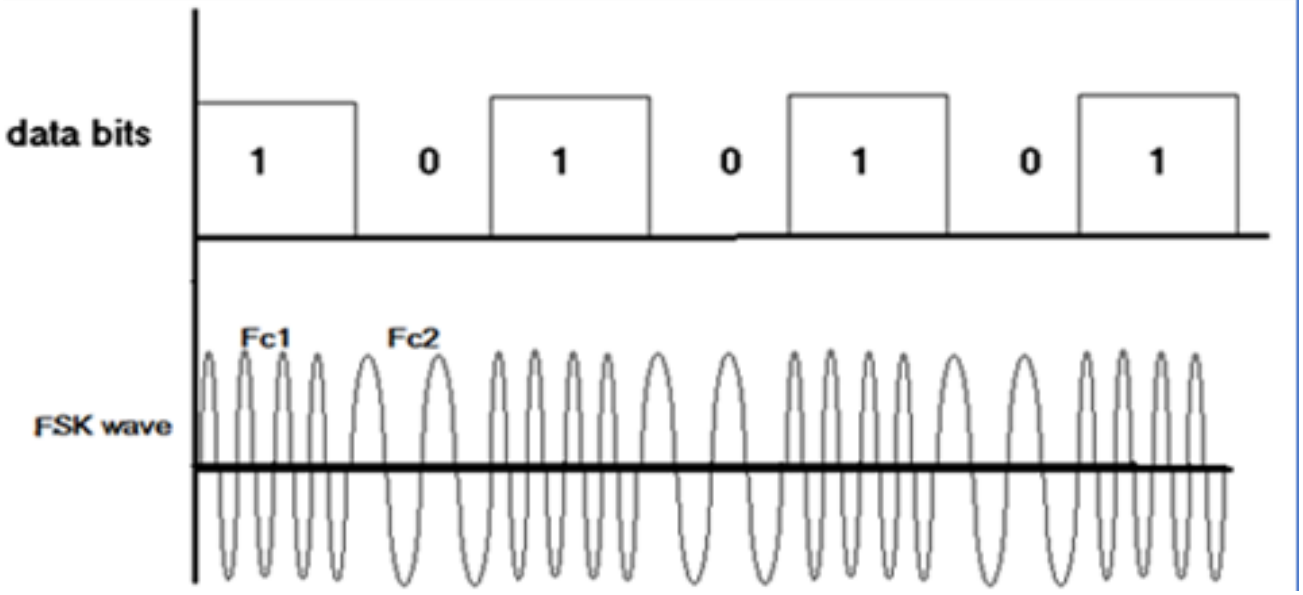
**SOLUTION:** B = 100 R = D = 1 $\Delta f$ = 50 N = ?

| | |
|---|---|
| **$B = (1+d)S + 2\Delta f$** | $100 = 2S + 50$ |
| $B = (1+1)\,S + 50$ | $100 - 50 = 2S$ |
| $B = 2S + 50$ | $2S = 50$ |
| $100 = 2S + 50$ | $S = 25$ kHz |

Activate Windows
Go to Settings to activate Windows.

**N = 25 kHz**
Because Data Rate = 1

## FSK Modulation and Demodulation



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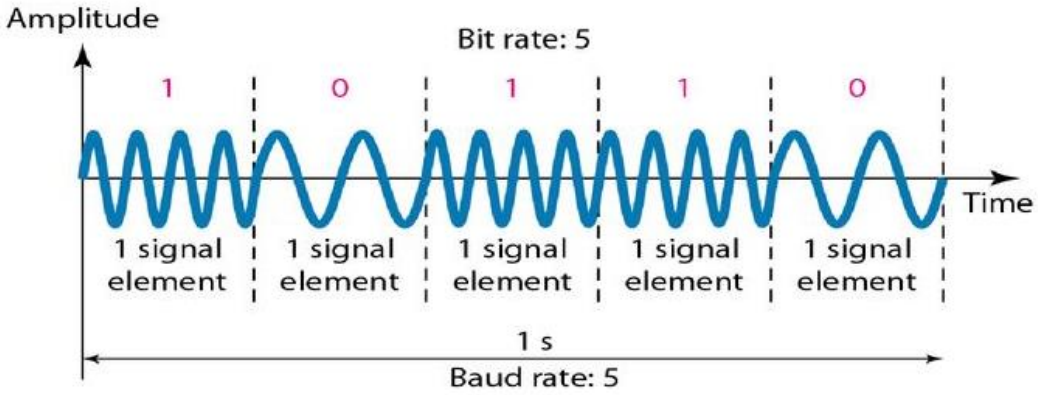





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## FSK (Frequency Shift Keying)

- The frequency of the carrier signal is varied to represent binary 1 or 0.
- Both peak amplitude and phase remain constant while the frequency changes.
- The frequency of the signal during each bit duration is constant, and its value depends on the bit (0 or 1).



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## Phase Shift Keying (PSK)

In this technique in which the Phase vary (change, but the Amplitude and Frequency remain constant. PSK is relatively common than ASK or FSK

**Example:** Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of d = 0.

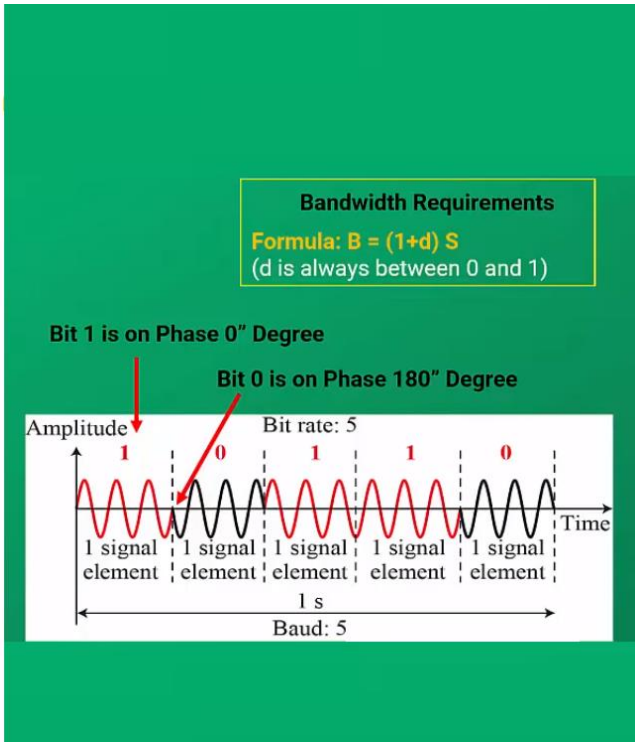
**SOLUTION:**

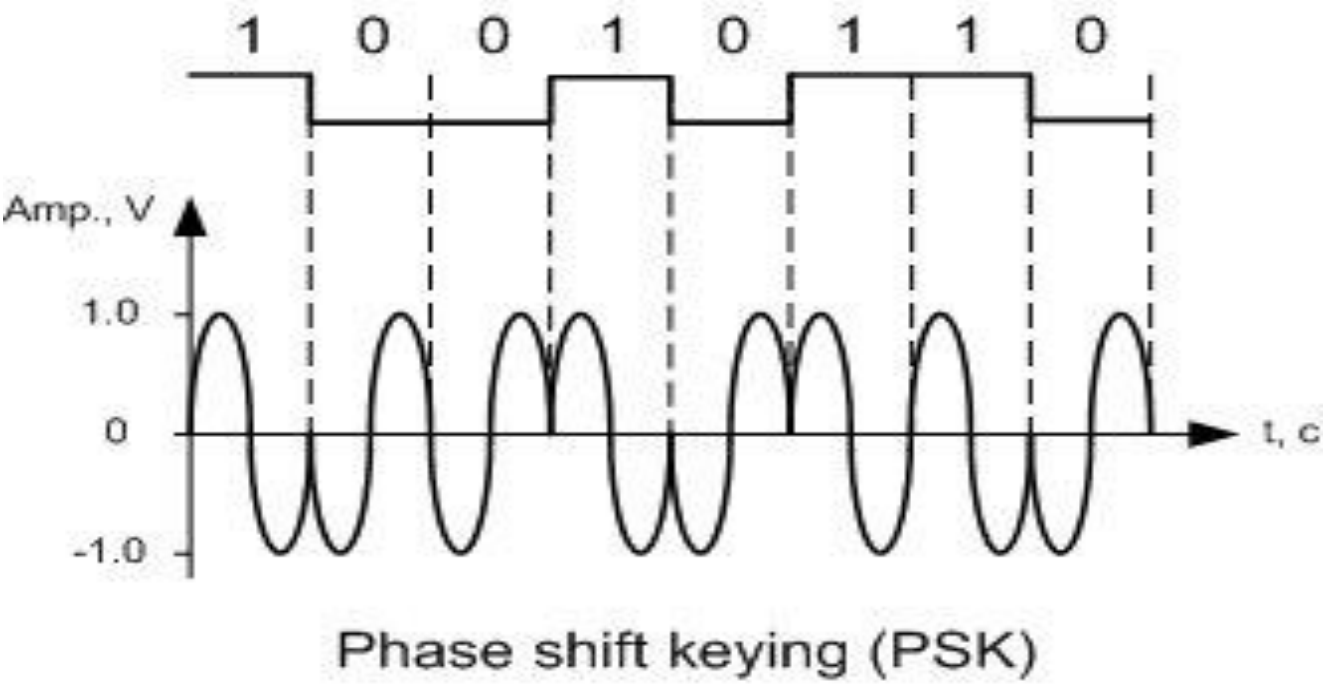
N = 12Mbps R = 2 = 1 S = ?

**S = N/r**

= 12/2

**S = 6 Mbps**




Phase shift keying (PSK)

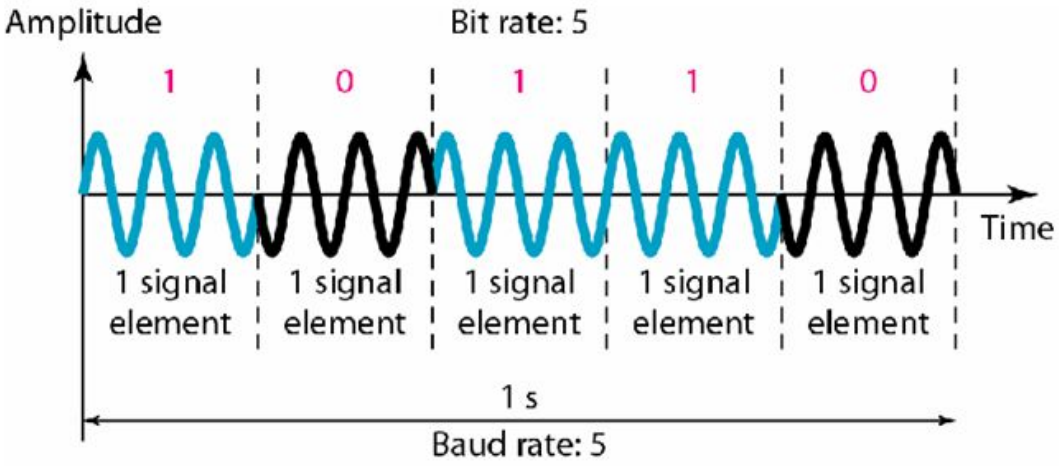
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# Phase Shift Keying (PSK)

- In PSK, the phase of the carrier is shifted to represent data.
- Two-Level PSK(Binary PSK) - BPSK
  - Uses two phases (0 and 180°) to represent the two binary digits.
  - The resulting transmitted signal for one bit time is:



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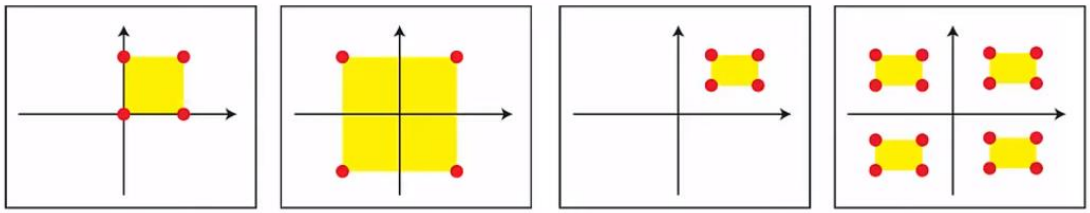
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# (QAM)
# Quadrature Amplitude Modulation

We have studies about ASK, FSK, and PSK. In these three techniques the **PSK is most effective** technique.

We have **combine two ASK and PSK** to make more effective that become QAM. It has **four variation**.



a. 4-QAM b. 4-QAM c. 4-QAM d. 16-QAM

**Bandwidth Requirements:**

The QAM require Bandwidth as ASK and PSK requires.

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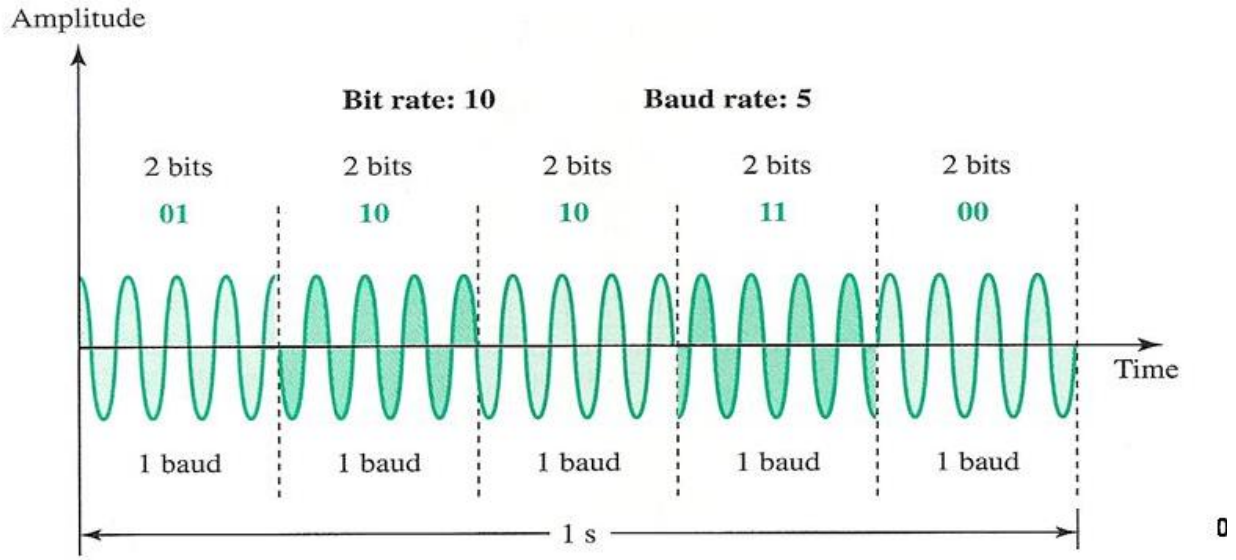





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# Quadrature PSK (QPSK)

- The term "quadrature" implies that there are four possible phases (4-PSK) which the carrier can have at a given time.
- The pair of bits represented by each phase is called **dibit**.
- The rate of change (baud) in this signal determines the signal bandwidth.
- BUT the throughput or bit rate for QPSK is twice the baud rate.



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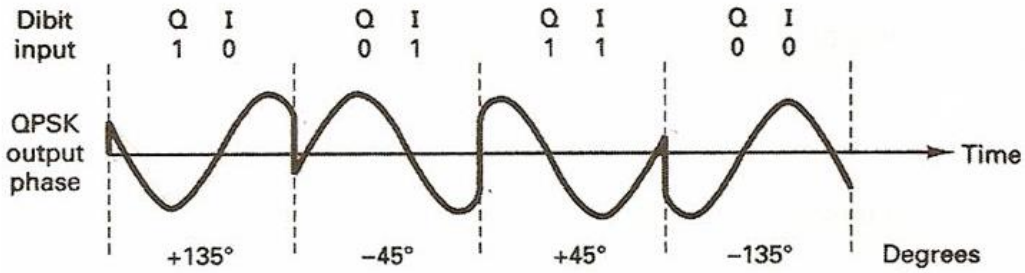





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## QPSK = 4-PSK

**PSK that uses phase shifts of 90°=π/2 rad ⇒ 4 different signals are generated, each representing 2 bits**




Assumption

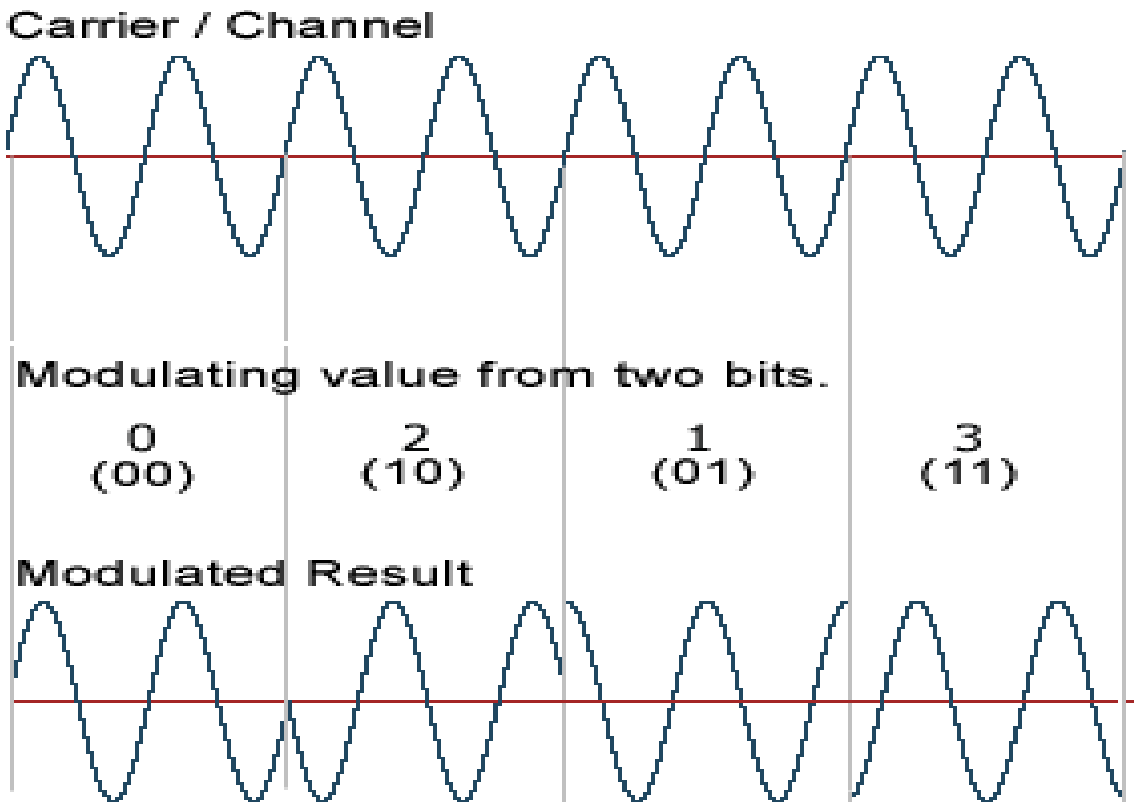
| Binary input | | QPSK output phase |
|---|---|---|
| Q | I | |
| 0 | 0 | −135° |
| 0 | 1 | −45° |
| 1 | 0 | +135° |
| 1 | 1 | +45° |

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## QPSK

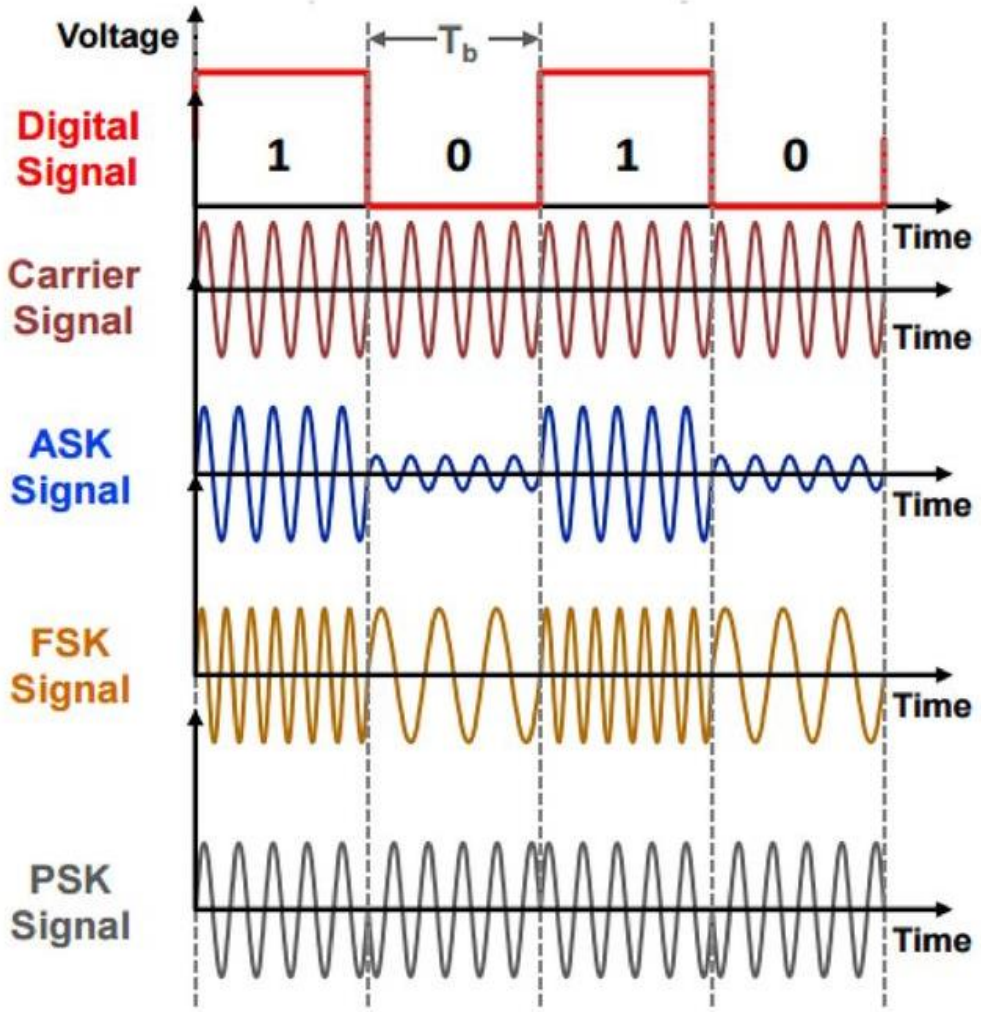


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## Comparison between ASK, FSK and PSK



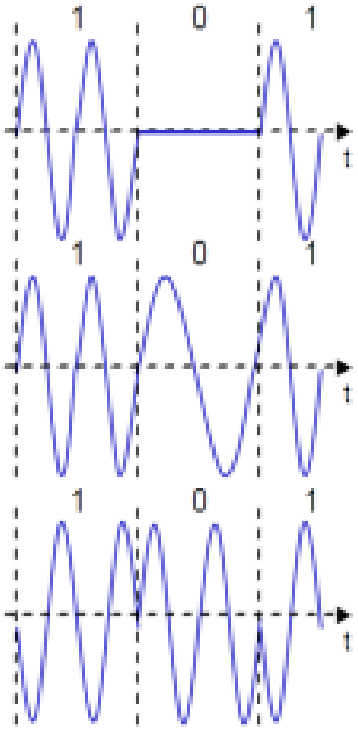
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- Amplitude Shift Keying (ASK):
  - change amplitude with each symbol – frequency constant
  - low bandwidth requirements
  - very susceptible to interference
- Frequency Shift Keying (FSK):
  - change frequency with each symbol
  - needs larger bandwidth
- Phase Shift Keying (PSK):
  - Change phase with each symbol
  - More complex
  - robust against interference



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## Data Communication CS601 Lecture No 18

Lecture Prepared By

Muhammad Abid Ali

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## CS601 Lecture No 18 Topic 86 to 89

## Analog to Analog Conversion

Analog-to-analog conversion, or analog modulation, is the representation of analog information by an analog signal. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.

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An example is radio. The government assigns a narrow bandwidth to each radio station. The analog signal produced by each station is a low-pass signal, all in the same range. To be able to listen to different stations, the low-pass signals need to be shifted, each to a different range.

## Analog To Analog Conversion



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| Que. | The example of an analog to analog conversion is |
|---|---|
| a. | radio |
| b. | Video |
| c. | Television |
| d. | internet |
| | **Answer:** radio |

## What are the 3 ways of conversion analog to analog?

**Analog to Analog conversion can be done in three ways:**

- Amplitude Modulation.
- Frequency Modulation.
- Phase Modulation.

## What is difference between modulated Signal Carrier Signal and modulating signal?

We take the help of such high frequency signal which is called as a carrier signal to transmit our message signal. Such a process is simply called as Modulation. Modulation is the process of changing the parameters of the carrier signal, in accordance with the instantaneous values of the modulating signal.

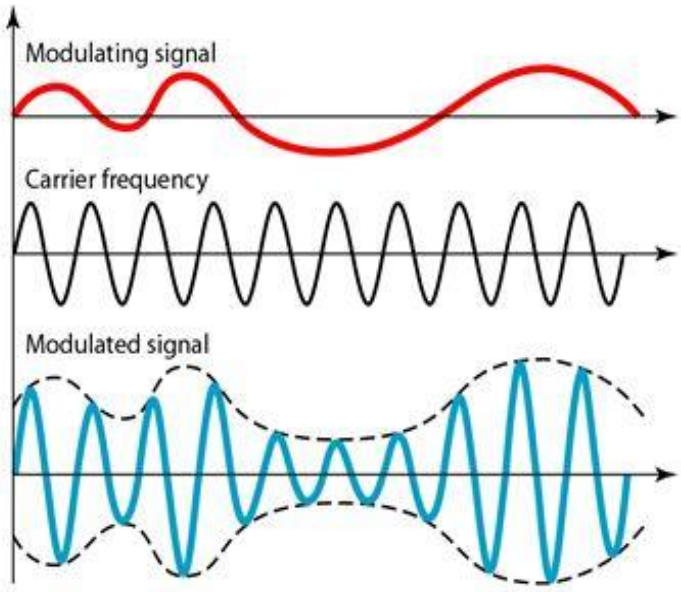
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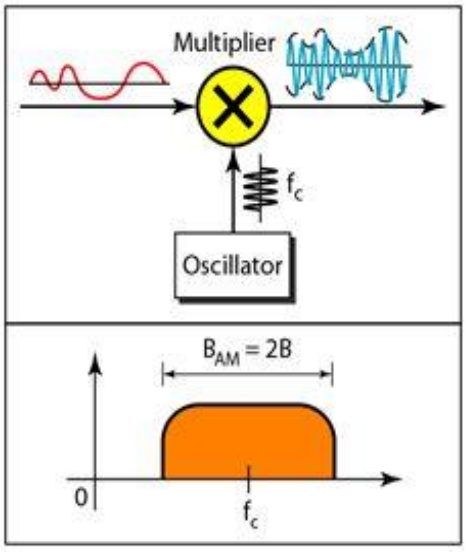
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## 1. Amplitude Modulation:

In AM transmission, the carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal. The frequency and phase of the carrier remain the same. Only the amplitude changes to follow variations in the information. The following figure shows how this concept works. The modulating signal is the envelope of the carrier.



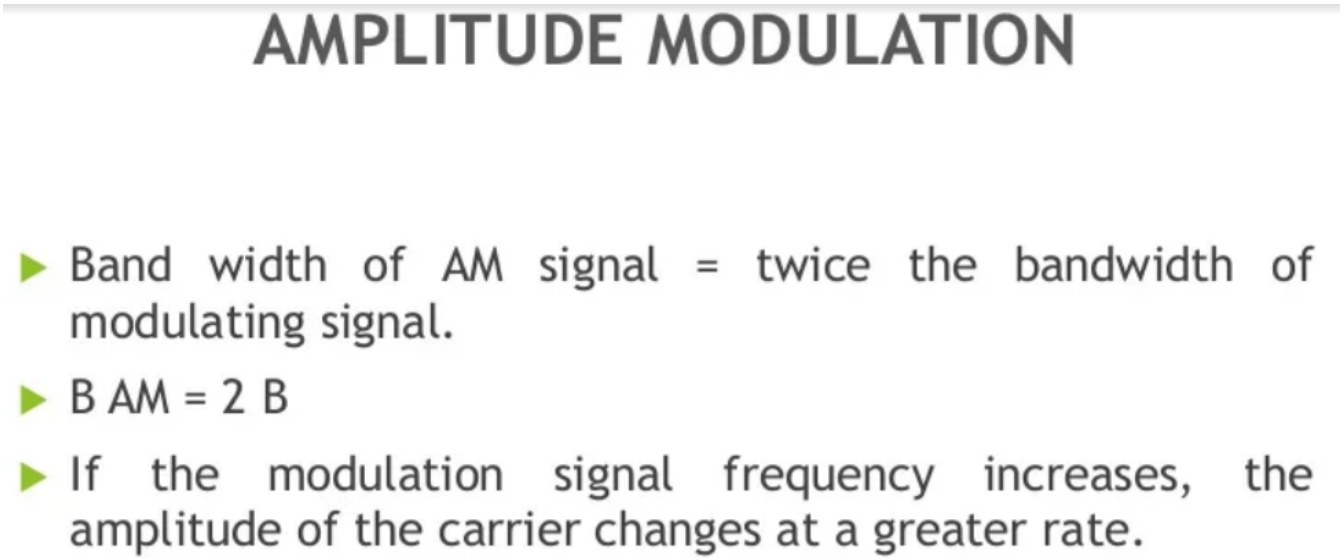



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AM is normally implemented by using a simple multiplier because the amplitude of the carrier signal needs to be changed according to the amplitude of the modulating signal.

## 2. Frequency Modulation

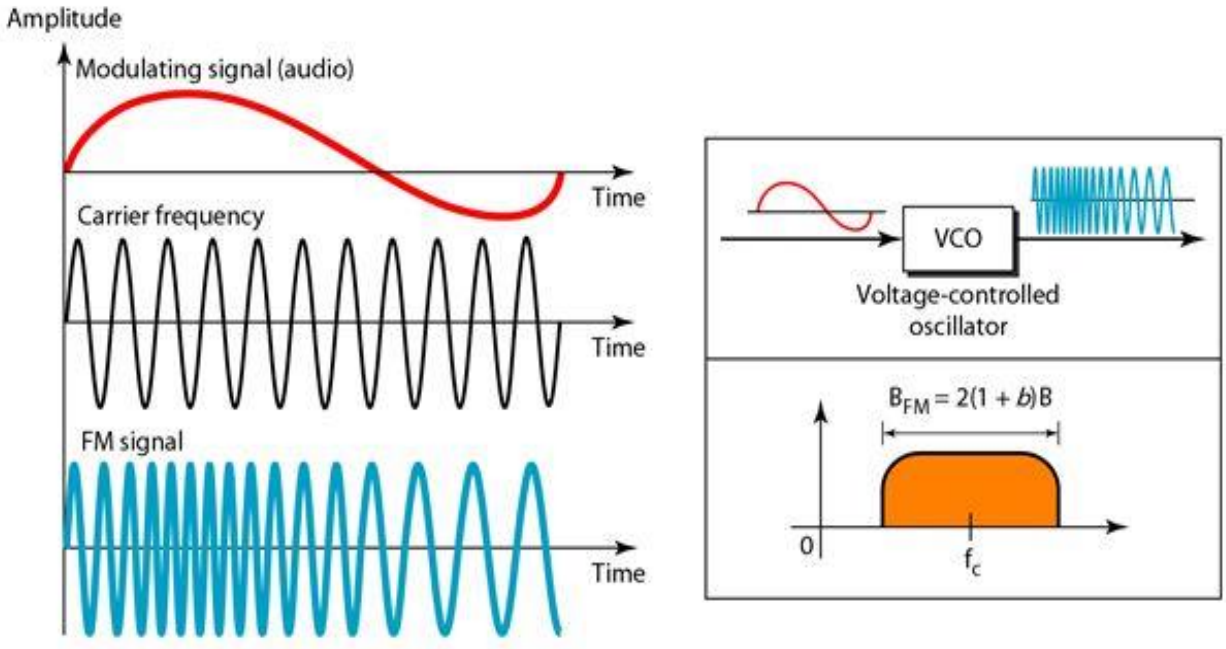
In FM transmission, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and phase of the carrier signal remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly.

The following figure shows the relationships of the modulating signal, the carrier signal, and the resultant FM signal. FM is normally implemented by using a voltage-controlled oscillator as with FSK. The frequency of the oscillator changes according to the input voltage which is the amplitude of the modulating signal.

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## 3. Phase Modulation:

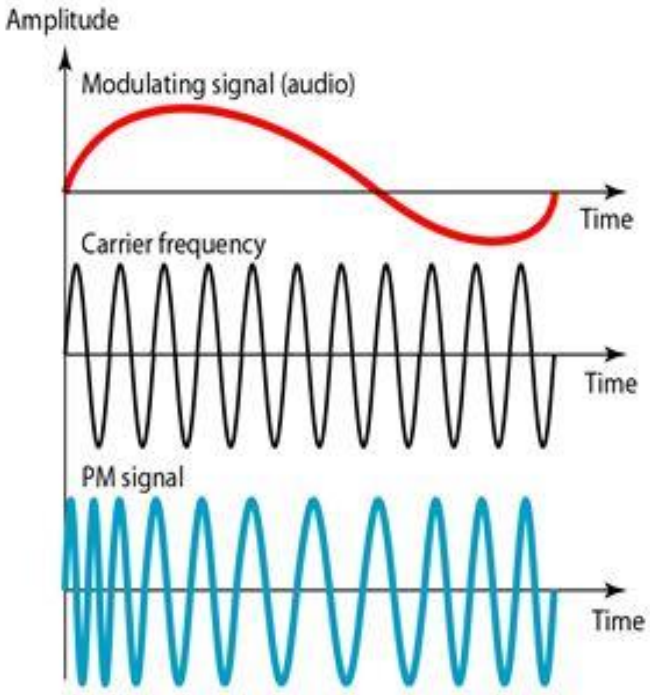
In PM transmission, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly. It is proved mathematically that PM is the same as FM with one difference.

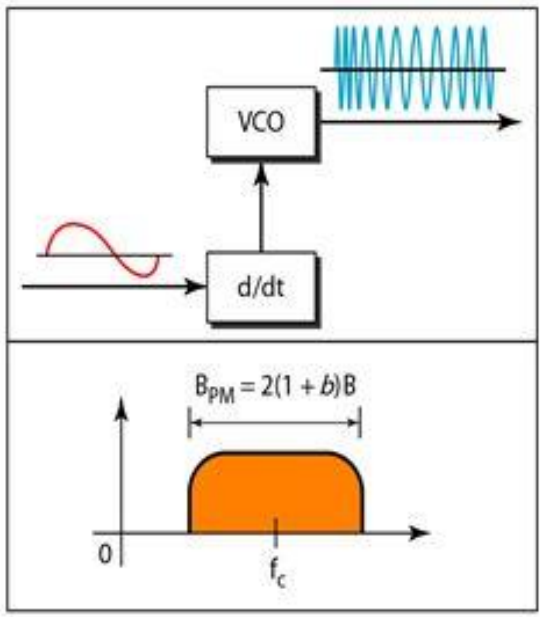
In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal; in PM the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal. The following figure shows the relationships of the modulating signal, the carrier signal, and the resultant PM signal.

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## CS601 Lecture No 19 Topic 89 to 100

### Multiplexing

Simultaneous transmission of multiple signals across a single data link MCQS

In telecommunications and computer networking, multiplexing (sometimes contracted to muxing) is a method by which multiple analog or digital signals are combined into one signal over a shared medium.

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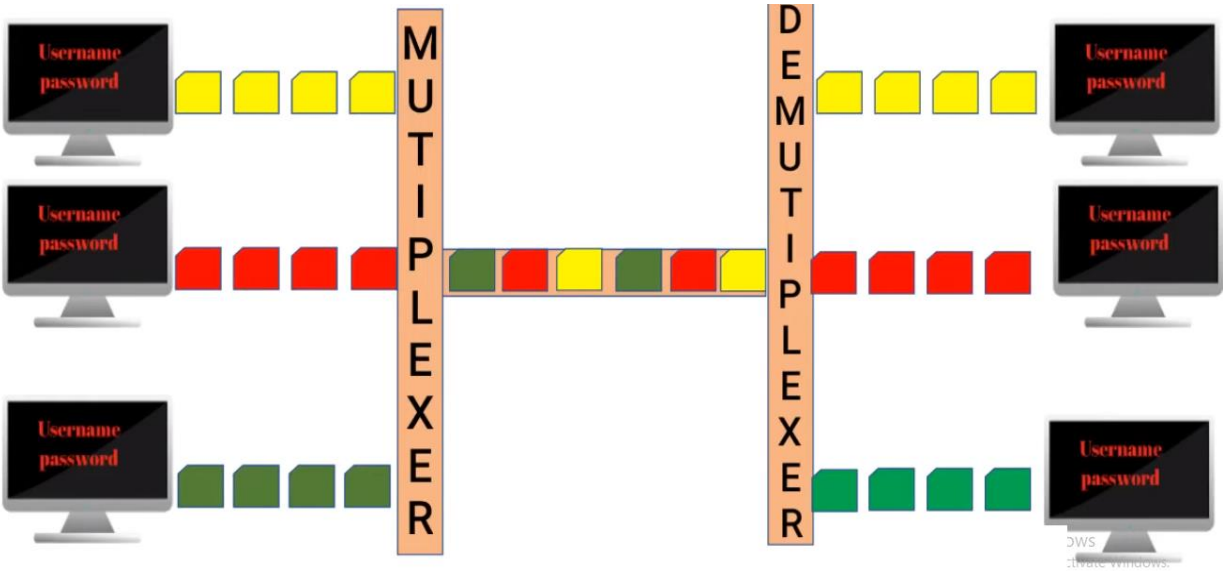
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The aim is to share a scarce resource. For example, in telecommunications, several telephone calls may be carried using one wire. Multiplexing originated in telegraphy in the 1870s, and is now widely applied in communications. In telephony, George Owen Squier is credited with the development of telephone carrier multiplexing in 1910.

The multiplexed signal is transmitted over a communication channel such as a cable. The multiplexing divides the capacity of the communication channel into several logical channels, one for each message signal or data stream to be transferred. A reverse process, known as demultiplexing, extracts the original channels on the receiver end.

A device that performs the multiplexing is called a multiplexer (MUX), and a device that performs the reverse process is called a demultiplexer (DEMUX or DMX).

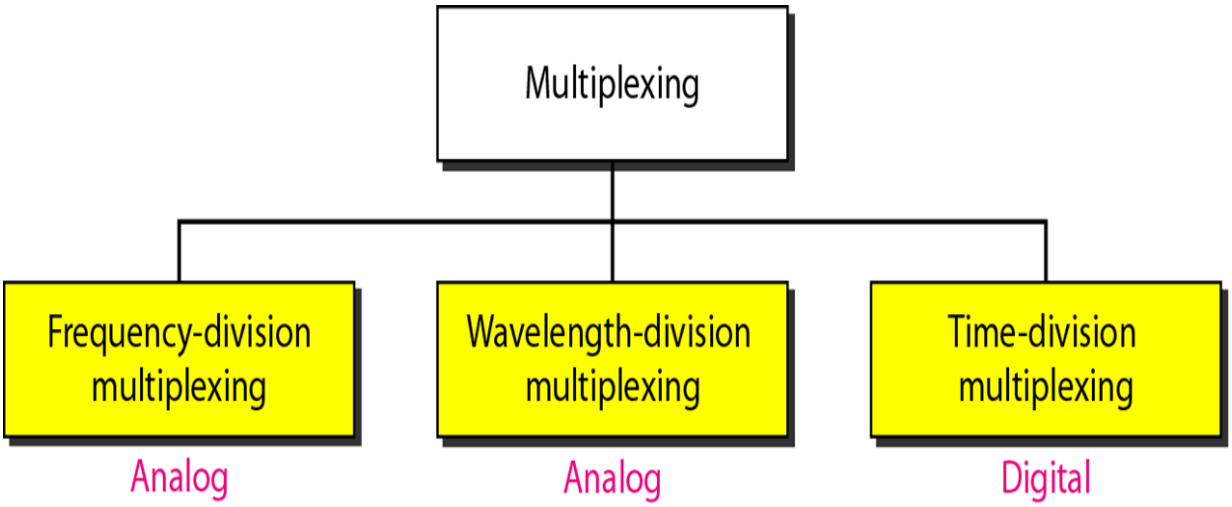
Inverse multiplexing (IMUX) has the opposite aim as multiplexing, namely to break one data stream into several streams, transfer them simultaneously over several communication channels, and recreate the original data stream.



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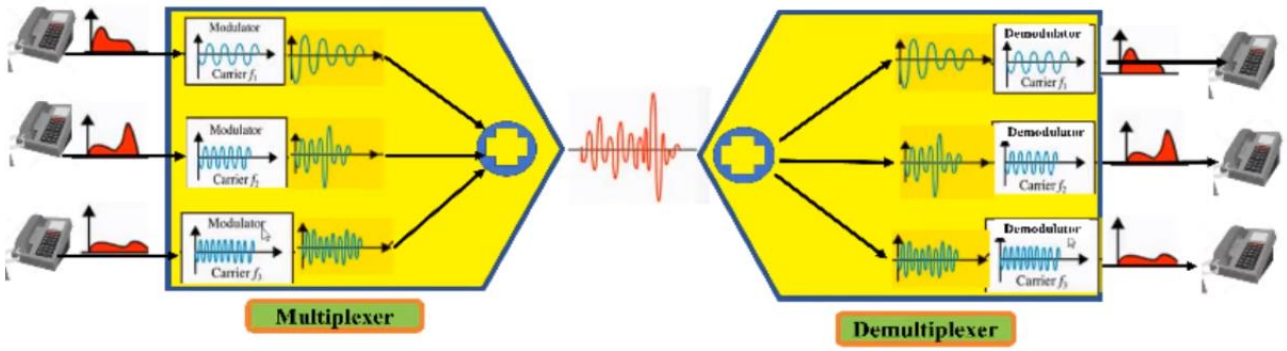
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## Frequency-Division Multiplexing

An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.

**Frequency-division multiplexing**

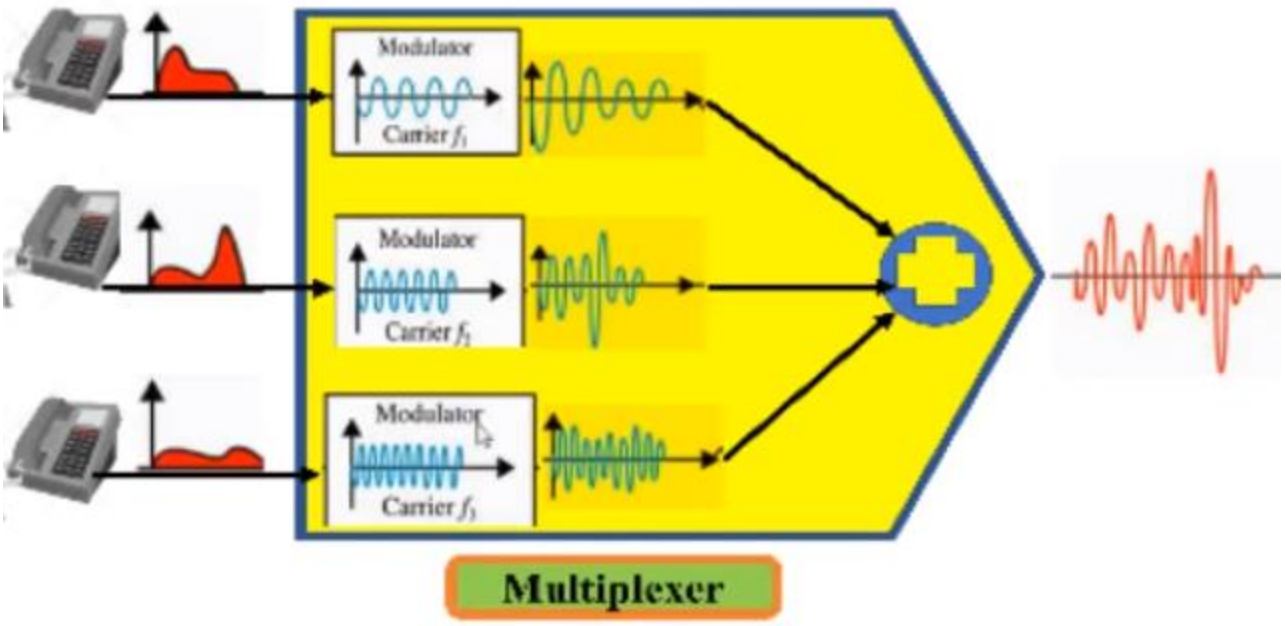
is an analog technology that supports the transmission of multiple communications over a single carrier by shifting each communication to a unique frequency range. A related technology for fiber channels is known as wavelength division multiplexing. This technology requires that each communication be transmitted as unique wavelength (i.e. color) of light along the shared fiber optic channel.



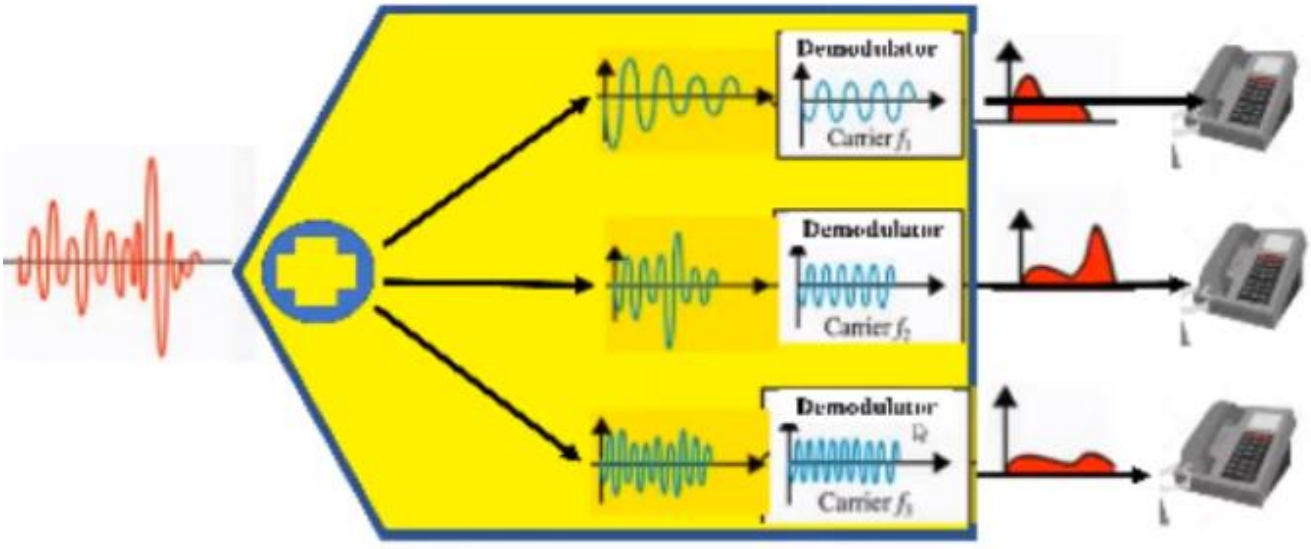
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Multiplexer




Demultiplexer

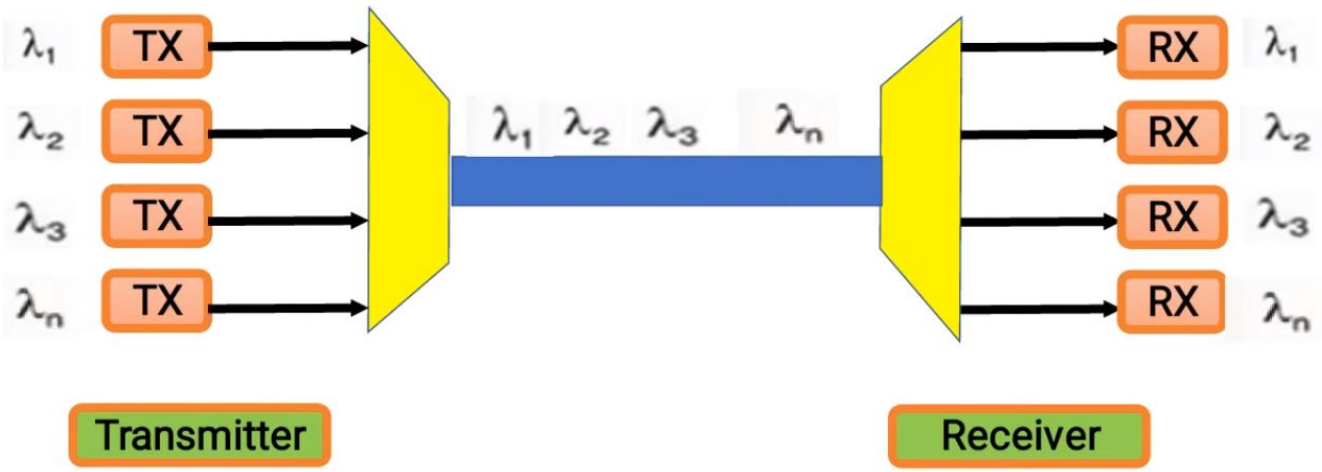
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## Wavelength-Division Multiplexing

Wavelength division multiplexing (WDM) is a technique of multiplexing multiple optical carrier signals through a single optical fiber channel by varying the wavelengths of laser lights. WDM allows communication in both the directions in the fiber cable.



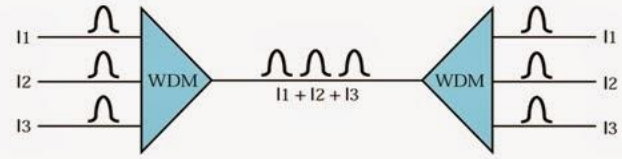
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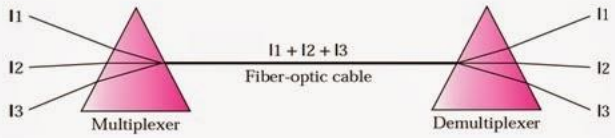
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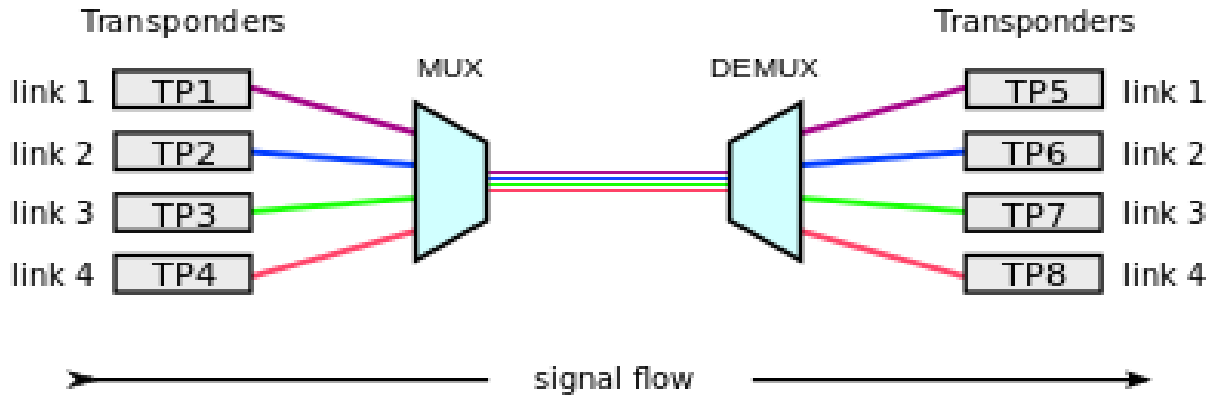
## WAVE DIVISION MULTIPLEXING (WDM)

- An analog multiplexing technique to combine optical signals
- Multiple beams of light at different frequency
- Carried by optical fibber
- A form of FDM
- Each color of light (wavelength) carries separate data channel
- Commercial systems of 160 channels of 10 Gbps now available




## wavelength-division multiplexing (WDM)



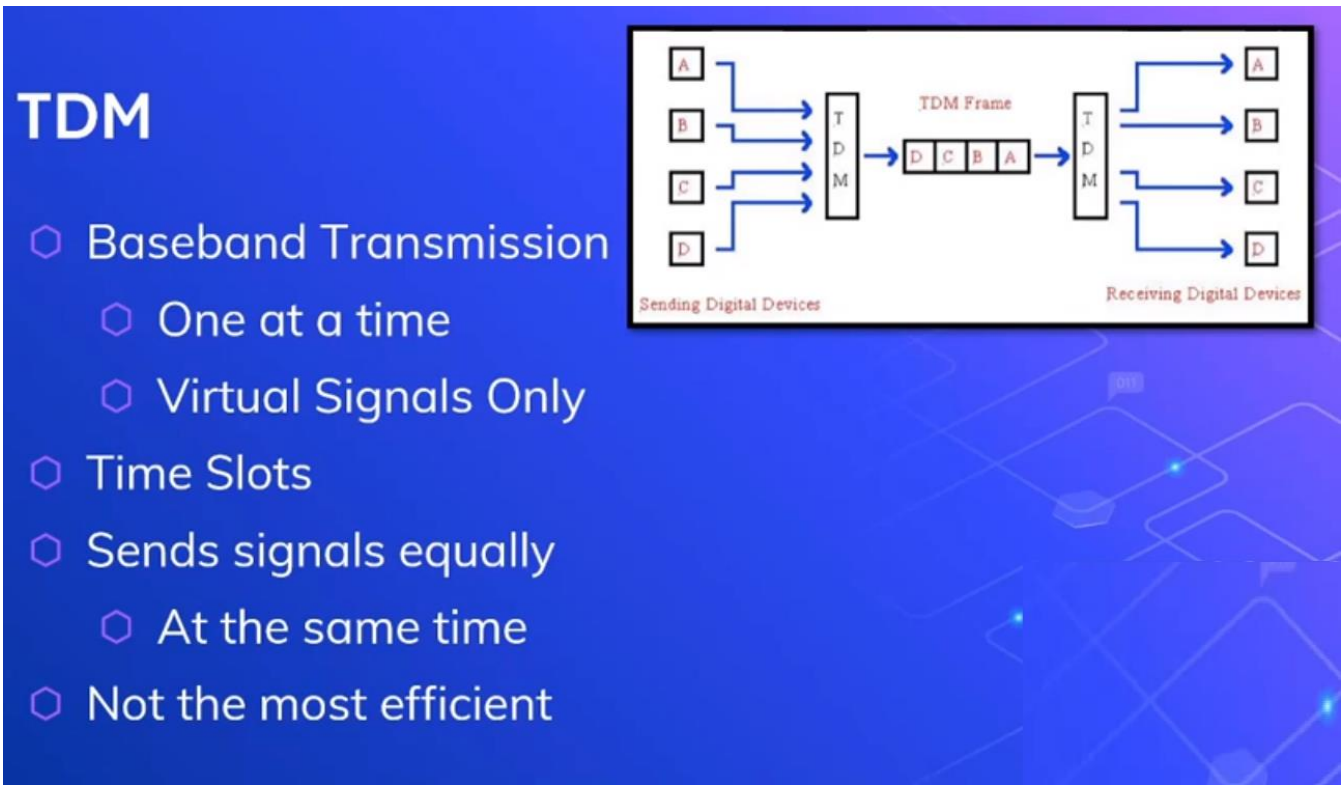
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## Time-Division Multiplexing

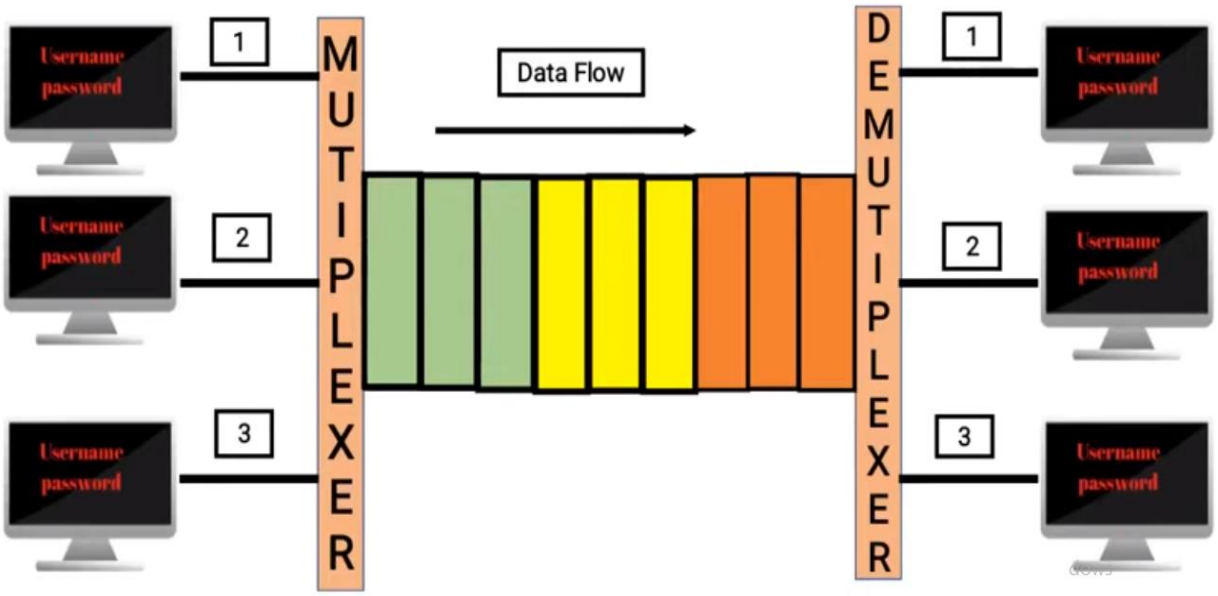
Digital process that allows several connections to share the high bandwidth of a link

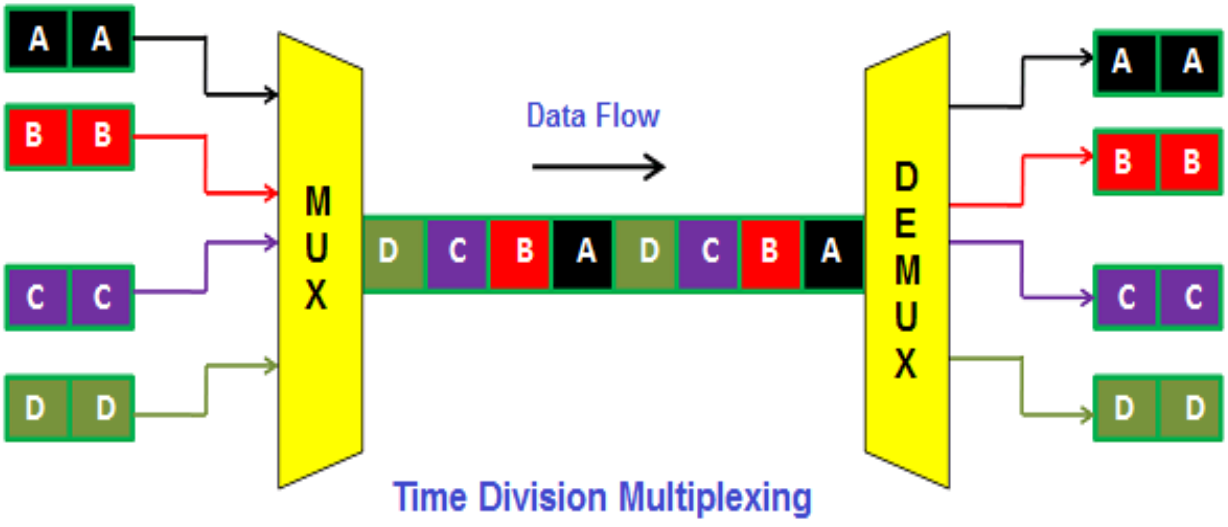


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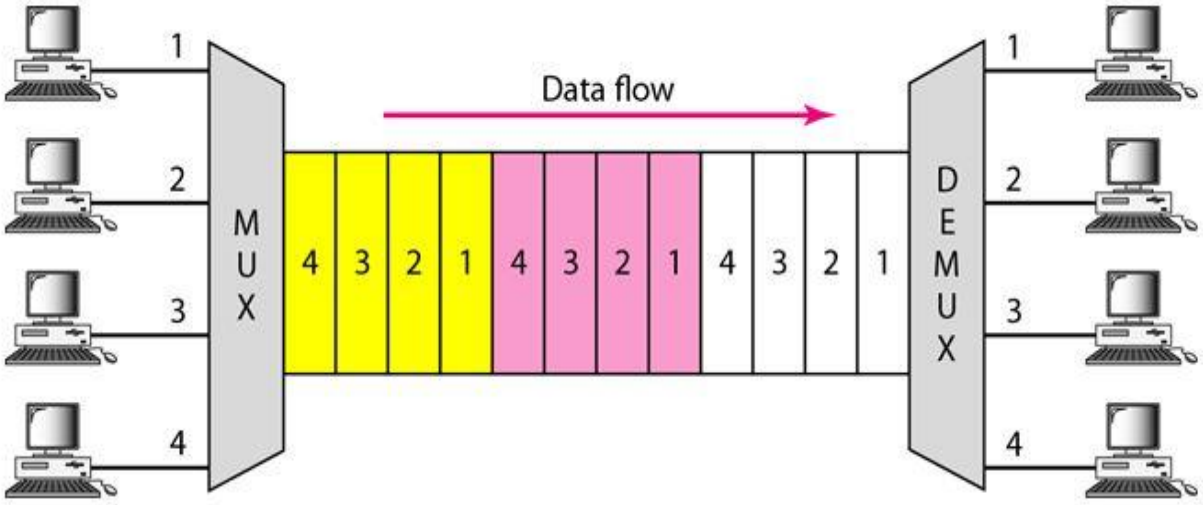




Time Division Multiplexing

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## Time-Division Multiplexing

- **Synchronous TDM**
- **Statistical TDM**

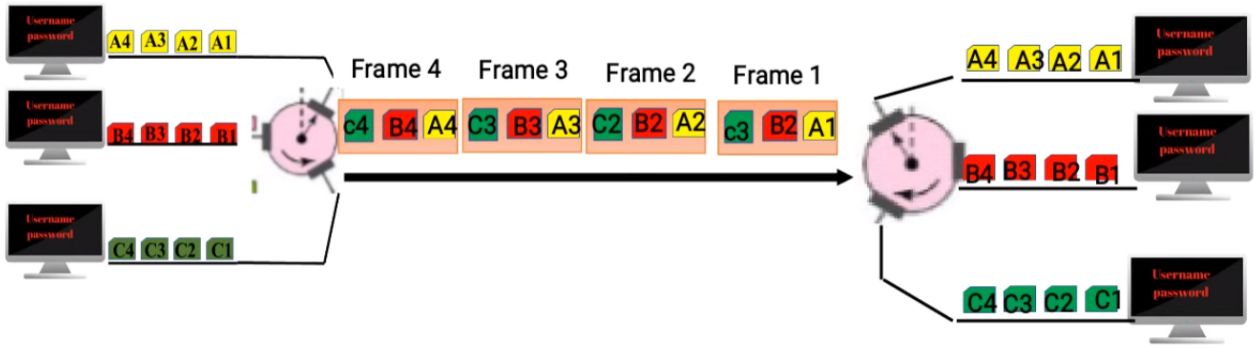
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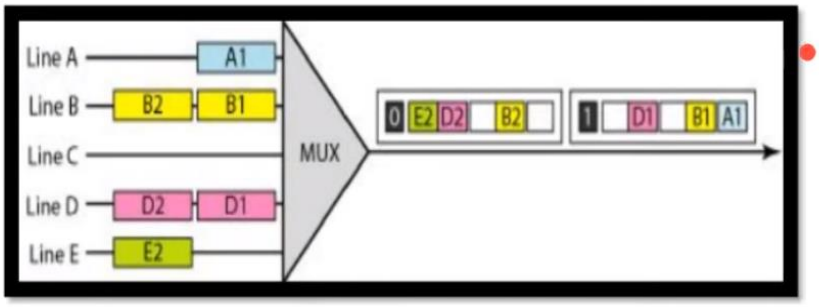

https://www.youtube.com/watch?v=-0kr2sG8JUQ&list=PLAUocDaOe9K6nDHAxKa-7fHsIFAKBPE8q&index=9

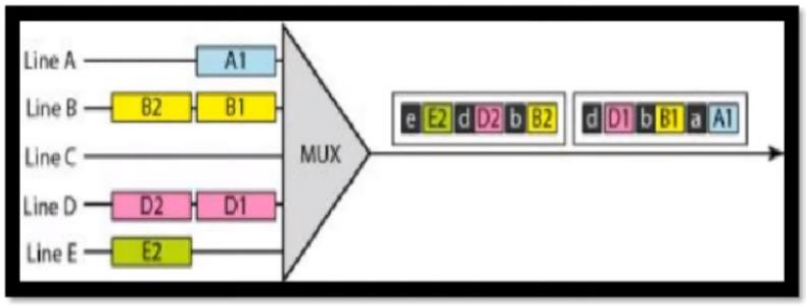
## Synchronous TDM




Empty slots If a source does not have data to send, the corresponding slot in the output frame is empty.

## Synchronous TDM




## Statistical TDM

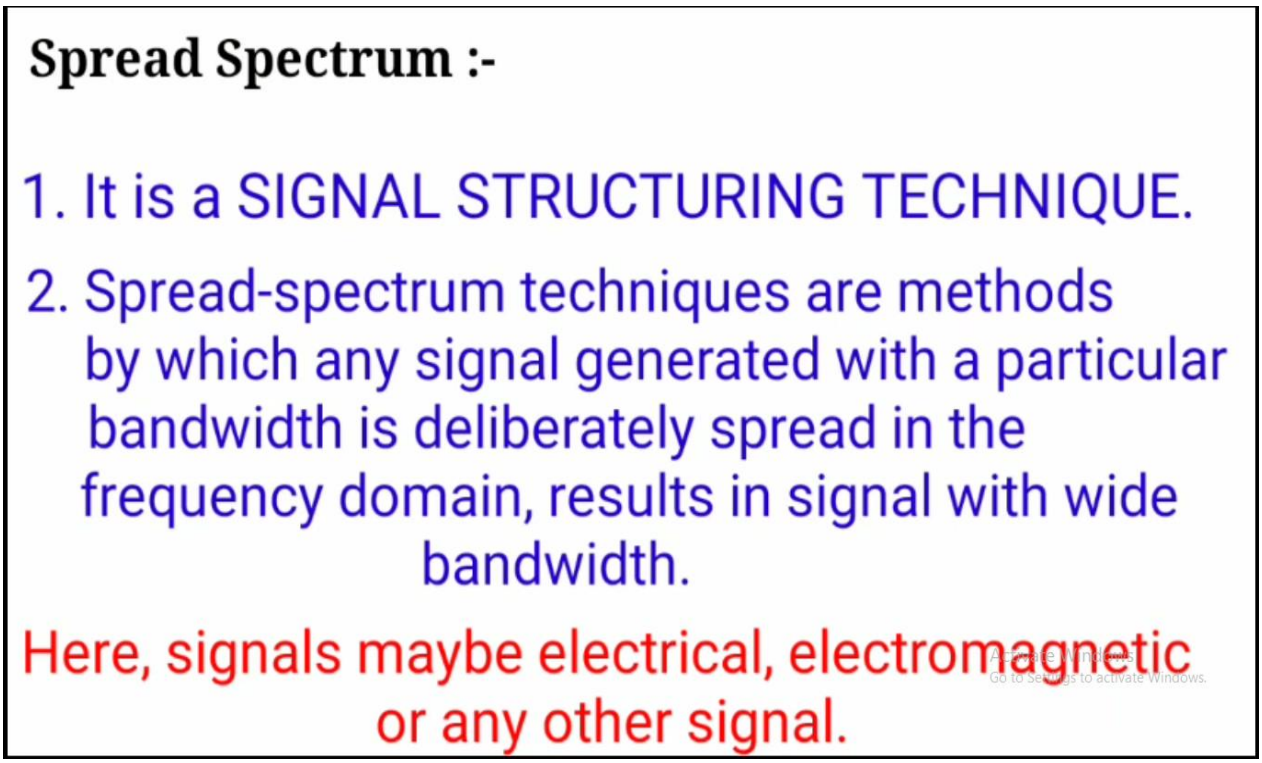
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## SPREAD SPECTRUM

Spread spectrum is **a form of wireless communications in which the frequency of the transmitted signal is deliberately varied**. This results in a much greater bandwidth than the signal would have if its frequency were not varied.




## What is spread spectrum & Types?

Spread Spectrum can be categorized into two types:

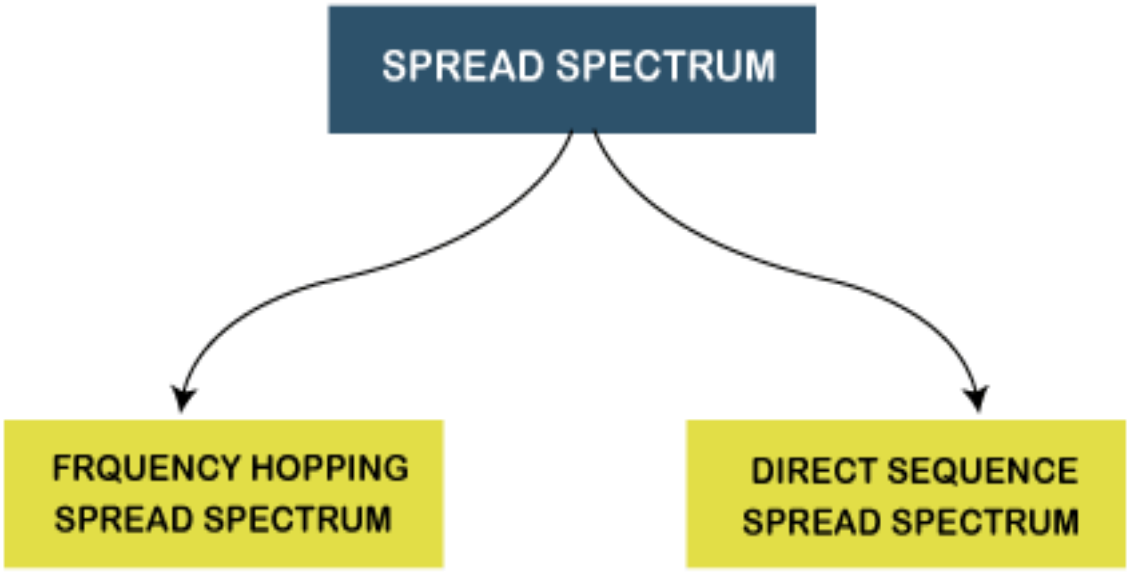
Frequency Hopping Spread Spectrum (FHSS)

Direct Sequence Spread Spectrum (DSSS)

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## DSSS

- **DSSS also expands the bandwidth of the original signal, but the process is different**
- **We replace each data bit with 'n' bits using a spreading code**
- **Each bit is assigned a code of 'n' bits, called chips, where the chip rate is 'n' times that of the data bit**

## What is the difference between FHSS and DSSS?

FHSS and DSSS, which stand for Frequency Hopping Spread Spectrum and Direct Sequence Spread Spectrum, are two spread spectrum techniques. The main difference is in how they spread the data into the wider bandwidth. **FHSS utilizes frequency hopping while DSSS utilizes pseudo noise to modify the phase of the signal**.

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